

Multiproduct Firms and Production Networks

Renata Abbott¹ and Marco Rojas²

¹UC San Diego

²Central Bank of Chile

August 25, 2026

PRELIMINARY DRAFT

Abstract

How do multi-product firms behave inside production networks? Using the universe of firm-to-firm transaction data from Chile, we show that supplier growth operates along two distinct but connected extensive margins: acquiring buyers and expanding product scope. We document three model-facing facts. First, the largest growth episodes are driven disproportionately by the simultaneous addition of new buyers and new products. Second, the composition of sales growth varies systematically with initial supplier size: smaller suppliers grow relatively more through new buyers and new products, while larger suppliers rely more on existing buyer–product relationships and on forming new buyer links. Third, buyer acquisition and product introduction are complements: even conditional on sales and supplier fixed effects, a 10% increase in buyers is associated with 0.2 new product introductions. We also provide direct evidence against cannibalization, meaning that expansion along one margin does not come at the expense of existing sales. Guided by these facts, we build a model that embeds the rank-based multiproduct structure of [Arkolakis et al. \(2021\)](#) into a production network. A firm-level product-adoption cost, paid once and amortized across buyer relationships, generates two-way complementarity between buyer acquisition and product scope, while idiosyncratic product–buyer matches generate the partial alignment between firm-level and within-link product importance observed in the data. The model reproduces the empirical amortization elasticity, and because each fixed cost affects both margins, lower relationship-formation costs also stimulate product expansion. Finally, shutting down one margin at a time shows that the buyer margin is quantitatively more important: restricting each firm to a single product reduces sales dispersion by 10%, whereas collapsing the buyer base to a single representative buyer reduces it by 37%. The buyer margin alone reproduces about 90% of firm-size dispersion, compared with 63% for the product margin. Together, these results decompose firm-size dispersion into its buyer and product components and show that policies aimed at product innovation may operate not only directly, but also indirectly through buyer–supplier network formation.

1 Introduction

How do suppliers choose the scope of products they produce and sell, and to whom do they sell these products within production networks? How do product and buyer scope contribute to the firm size distribution? This paper studies these questions in an environment in which supplier growth is shaped by two distinct and interacting margins: expanding product scope and forming buyer relationships. The central insight is that these margins are not only separately important but become complementary through the structure of firms' fixed costs. We do not impose complementarity directly. Instead, it emerges because a firm-level product-adoption cost can be amortized across buyer relationships, while a broader product portfolio raises the expected surplus generated by a potential buyer relationship. A larger buyer base raises the return to introducing additional products because firm-level product-adoption costs can be amortized across more relationships. A broader product portfolio, in turn, raises the value of forming additional buyer relationships. Understanding the firm-size distribution and growth within production networks therefore requires modeling these two margins jointly rather than collapsing them into a single product or match margin.

These two margins are quantitatively important but are typically studied separately. Multi-product firms¹ account for a large share of production, trade, and employment (Arkolakis et al., 2021), and a large literature emphasizes that product introduction and product switching are important for firm performance and resource reallocation (Bernard et al., 2010, 2011; Mayer et al., 2014; Argente et al., 2018; Copeland and Shapiro, 2016; Hottman et al., 2016). Because these models focus primarily on firm and product dynamics, however, they generally abstract from heterogeneity across individual supplier–buyer relationships.

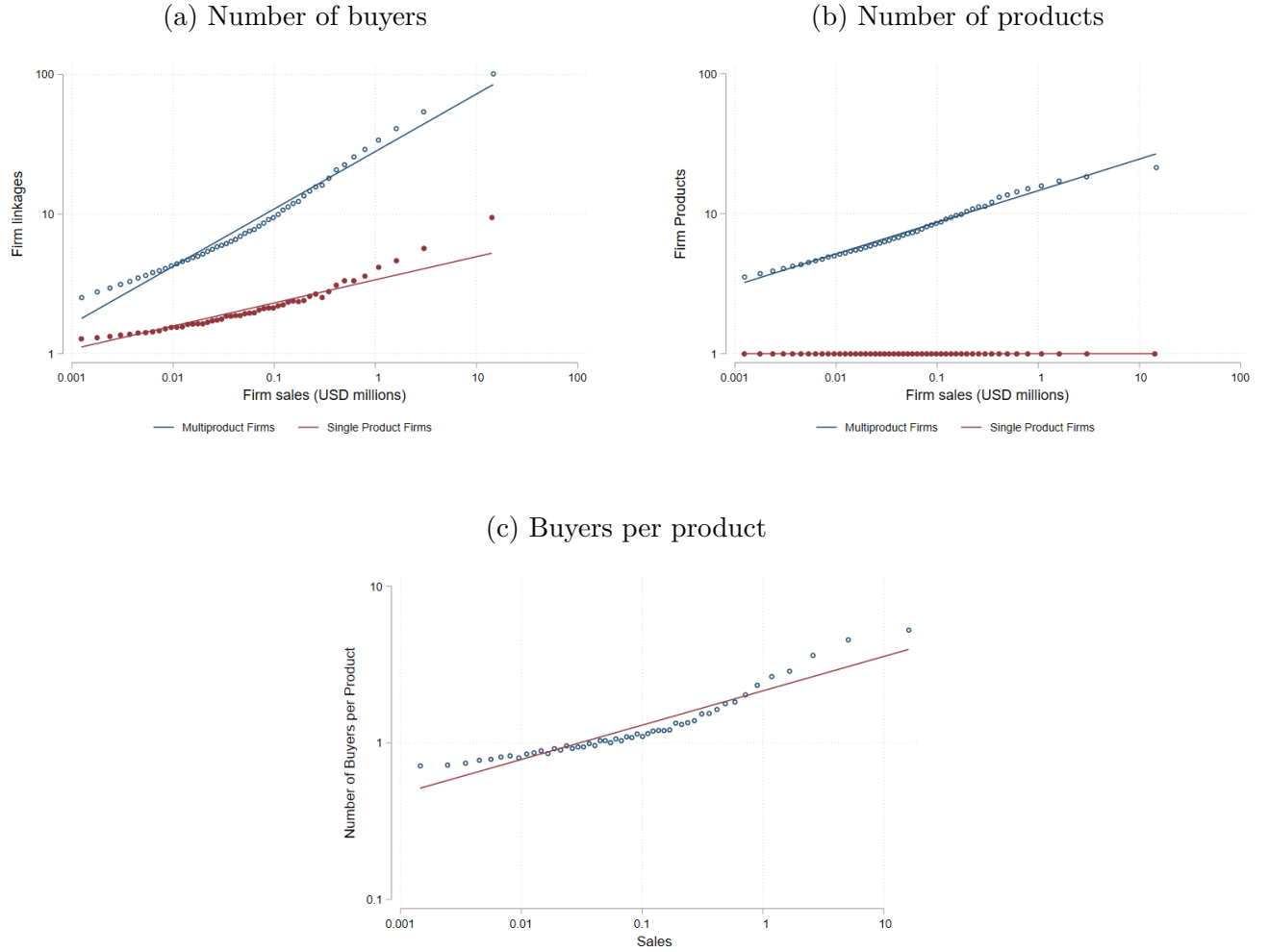
At the same time, a growing production-network literature shows that customer acquisition and buyer–supplier relationships are important determinants of firm growth and the firm-size distribution (Bernard et al., 2022; Eaton et al., 2026; Lim, 2018; Huneus, 2018; Arkolakis et al., 2023; Afrouzi et al., 2023; Einav et al., 2021; Argente et al., 2021; Aekka and Khanna, 2024). This work has developed rich models of buyer–supplier matching and network formation (Bernard et al., 2018; Miyauchi, 2024; Demir et al., 2024; Dhyne et al., 2021); Bernard and Moxnes (2018) provide a survey of this literature. To the best of our knowledge, however, these models typically treat suppliers as single-product firms or subsume product expansion into match formation. In settings in which suppliers sell many products to many buyers, buyer acquisition and product scope are conceptually distinct decisions and need not move one-for-one.

We study these margins using the universe of invoice-level firm-to-firm transactions in Chile (SII Factura Electrónica), matched to administrative firm characteristics, the same network studied

¹Throughout the paper, whenever we refer to firms, we mean suppliers.

by Huneus (2018) and Arkolakis et al. (2023). We focus on manufacturing suppliers, define products at the 5-digit CPC level, and construct the production network from observed supplier–buyer relationships. Figures 1a and 1b show that larger suppliers expand along both extensive margins: they sell to more buyers and offer a broader range of products. Figure 1c further shows that larger suppliers have more buyers per product. Firm size is therefore not associated simply with independently accumulating buyers and products; larger suppliers also deploy their product portfolios across denser buyer networks. These cross-sectional patterns motivate a framework in which buyer acquisition and product scope are separate but interacting firm choices.

Figure 1: Extensive margins and supplier size



Notes: The figure presents binned scatterplots with fitted regression lines relating suppliers’ log total sales to the log number of buyer, products, and buyers per product. Supplier sales are measured at the tax-identifier level and exclude the top 0.1 percent of the sales distribution. We construct the figure at the tax-identifier level, rather than the tax-identifier-by-municipality level, because total sales, including sales to final consumers, are not observed at the latter level. Each dot represents a bin of supplier total sales.

We document three model-facing facts. First, the largest supplier-growth episodes disproportionately combine buyer acquisition and product expansion. Decomposing sales growth between

2017 and 2023 into old-link–old-product, new-link–old-product, old-link–new-product, and new-link–new-product components shows that simultaneous expansion along both margins becomes especially important among the fastest-growing suppliers. This fact motivates modeling buyer acquisition and product scope jointly, since the largest expansions cannot be understood as adjustments along either margin in isolation.

Second, the way firms expand varies systematically with suppliers’ initial conditions. Smaller suppliers rely relatively more on margins involving new products, particularly the joint addition of new buyers and new products, while larger suppliers rely more on existing buyer–product relationships and on forming new buyer links for products already in their portfolios. The composition of growth also varies with supplier age: younger suppliers rely more heavily on margins involving new products and new buyers, whereas older suppliers rely relatively more on acquiring new buyers for products already in their portfolios. The initial-size patterns provide a direct counterpart for the model’s heterogeneous comparative statics, while the age patterns are complementary descriptive evidence because age is not an explicit state variable in the static model.

Third, buyer breadth and product scope co-move strongly within suppliers. In our preferred specification we find that a 10 percent increase in the number of buyers is associated with approximately a 1.3 percent increase in product scope. An alternative specification using new product introductions gives the same qualitative result: a 10 percent increase in buyers is associated with approximately 0.2 additional new products. This complementarity is concentrated among peripheral products rather than core products.² Around the acquisition of a large new buyer, sales of peripheral products rise sharply, while sales of core products change little. This pattern is consistent with the idea that an expansion of the buyer base raises the return to products near the margin of adoption, rather than primarily increasing activity in products that were already central to the supplier.

We also examine whether the positive relationship between the two margins reflects a reallocation of a fixed amount of sales. We find little evidence of such cannibalization. When a supplier introduces a new product within an existing buyer relationship, sales of its existing products do not fall. Likewise, when a supplier acquires a large new buyer, sales to incumbent buyers do not systematically decline. Expansion in products and buyer relationships therefore appears to deepen supplier activity rather than simply reshuffle a fixed amount of revenue across products or customers.

Guided by these facts, we build a static model that adapts the rank-based multi-product structure of [Arkolakis et al. \(2021\)](#)³ to a production network in which destinations are individual

²We define peripheral products as products that account for a relatively small share of the supplier’s revenue, while core products account for a relatively large share.

³In [Arkolakis et al. \(2021\)](#), a firm’s products are ordered by profitability, with a core product at the top of the ranking and progressively less profitable (peripheral) products at higher ranks. Firm product scope is therefore characterized by how far the firm extends down this common product ranking.

buyers. Suppliers differ in productivity and choose their firm-level product scope before deciding which buyer relationships to serve. Developing a product requires a firm-level adoption cost, $F^P(G_s)$, that is paid once by the supplier. Serving a buyer additionally requires a link-formation cost, while selling a particular developed product to that buyer may require a buyer-specific activation cost. Importantly, complementarity between the two margins is not imposed as a primitive of the model. It emerges from the fixed-cost structure. Because the firm-level adoption cost is paid once while its returns accrue across buyer relationships, more buyers raise the summed return to developing an additional product. In the other direction, a broader portfolio creates more opportunities for profitable product–buyer matches, raising the expected surplus from forming a relationship and making it worthwhile to pay the buyer-specific link cost for buyers that would not be profitable with a narrower portfolio.

A second key feature of the model is buyer-specific product matching. A supplier may develop many products, but it does not necessarily sell the same products to every buyer. Products retain a systematic firm-level ranking, with more peripheral products having lower expected profitability, but each developed product also draws an idiosyncratic product–buyer match shock. Different buyers can therefore purchase different, non-nested subsets of the same supplier’s portfolio. This matching structure is disciplined directly by the data. A product’s importance in the supplier’s overall portfolio predicts its importance within individual links, but only partially: the baseline rank-alignment coefficient is approximately 0.27, and the supplier’s top product is not always the top product within every relationship. A rigid common product ladder, in which all buyers move down the same ranking, would imply substantially stronger alignment than we observe.⁴

We calibrate the model by simulated method of moments using the 2017 manufacturing cross-section. The calibration targets the distributions of product scope and buyer reach, buyers per product, sales concentration, and the conditional relationship between buyer breadth and product scope. We also use product specialization and the observed partial alignment between firm-level and within-link product rankings to discipline the product–buyer matching structure. Together, these moments determine how firms expand across buyers and products and how their firm-level portfolios are allocated across individual buyer relationships.

The central quantitative result is that the calibrated model reproduces the positive relationship between product scope and buyer breadth conditional on productivity. The empirical elasticity is 0.13. In the model, the corresponding within-productivity elasticity is approximately 0.18–0.19, depending on how productivity is controlled for. Thus, even among suppliers with similar productivity, those with broader buyer bases optimally carry broader product portfolios.

⁴This common-ladder structure follows the rank-based multiproduct framework of [Arkolakis et al. \(2021\)](#). Our contribution is to allow buyer-specific product matching, so different buyers can purchase different subsets of the supplier’s developed portfolio.

We then show that this co-movement is generated by amortization rather than mechanically by the product-ranking structure. Removing the firm-level adoption cost eliminates the within-productivity relationship between buyer breadth and product scope: the slope falls from approximately 0.19 to zero. Without a firm-level cost, there is nothing to amortize across buyers, and therefore no force linking a larger buyer base to broader firm-level product scope. This distinction is important because a rigid common-ladder structure can also generate a positive relationship between buyers and products, but for a mechanical reason and while imposing much stronger alignment of product rankings across buyers than we observe in the data. For reference, under the deterministic common-ladder structure closest to [Arkolakis et al. \(2021\)](#), the buyer–product relationship survives even when the adoption cost is removed because firm-level scope can be determined by the buyer that reaches farthest down the common product ranking. A supplier with more buyers has more opportunities to reach farther down that ladder, generating buyer–product co-movement even in the absence of amortization. Buyer-specific product matching both accommodates the imperfect alignment of product importance across buyer relationships observed in the data and removes this mechanical channel. As a result, the positive relationship between buyer breadth and product scope in our model arises from the economic mechanism at the center of the paper: a larger buyer base increases the return to paying a firm-level product-adoption cost.

We next use the calibrated model to quantify how the two margins interact and how much each contributes to the firm-size distribution. Comparative statics in the fixed costs show that each cost affects both margins. Lowering the firm-level product-adoption cost increases product scope but also expands the buyer base, because a broader portfolio makes additional relationships profitable. Lowering the buyer-relationship cost has the reverse spillover: it sharply expands buyer reach and also increases product scope because the larger buyer base raises the return to firm-level product adoption. Buyer-relationship policies can therefore affect product expansion even when they do not directly change product-development costs.

The two margins are not equally important quantitatively. When we shut down the buyer margin by replacing each supplier’s buyer network with a single representative buyer, the standard deviation of log sales falls by 37 percent. Shutting down the product margin by restricting suppliers to a single product reduces sales dispersion by only 10 percent. Equivalently, the buyer margin alone reproduces approximately 90 percent of the sales dispersion generated by the full model, compared with approximately 63 percent for the product margin alone. Buyer reach is therefore the dominant source of firm-size dispersion in the calibrated economy, while product scope amplifies the differences generated through the buyer network. This parallels [Bernard et al. \(2022\)](#), who find that downstream factors, and the number of customers in particular, account for most firm-size heterogeneity in the Belgian production network; we show the same ranking holds once suppliers are allowed to be multiproduct, with product scope amplifying rather than

dampening the dispersion.

Finally, we connect the static model back to the growth decompositions that motivate the paper. We subject the calibrated economy to a common productivity shock, re-solve firms' buyer and product decisions, and decompose the resulting change in sales into the same components used in the data. Although the model cannot reproduce an actual time path or relationship exit, it generates the same qualitative heterogeneity in the composition of expansion. Faster-growing, smaller, and less-productive suppliers adjust relatively more through new buyers and new products, whereas larger and more-productive suppliers rely more heavily on sales through existing buyer-product relationships. We interpret this exercise as an untargeted validation of the model's growth mechanics rather than as a structural reproduction of the 2017–2023 transition.

The distinction between buyer and product margins is also supported by their different lifecycle patterns in the data. Sales in new supplier-buyer links peak earlier, decline faster, and exhibit lower survival than supplier-product relationships. These differences reinforce the treatment of buyer reach and product scope as separate margins and are consistent with work viewing the customer base as a slowly accumulated stock (Gourio and Rudanko, 2014; Fitzgerald et al., 2024; Foster et al., 2016). Because the benchmark model is static, however, these lifecycle patterns serve as motivating evidence rather than quantitative targets.

The paper contributes to two related literatures. First, it brings the multi-product firm and production-network perspectives together by treating product scope and buyer acquisition as separate endogenous margins within the same framework. Relative to firm-to-firm matching models in which suppliers are effectively single-product or product expansion is subsumed into match formation (Bernard et al., 2018; Eaton et al., 2026; Bernard et al., 2022; Arkolakis et al., 2023), we keep firm-level product scope as a distinct choice and show it interacts with buyer acquisition through cost amortization. Second, it provides new empirical evidence on how these margins interact: large expansions disproportionately combine new buyers and products, the composition of expansion varies with supplier size and age, and buyer breadth is positively associated with product scope within suppliers. Third, it develops a buyer-specific product-matching structure that allows firm-level product rankings to coexist with substantial heterogeneity in the products sold across individual buyer relationships. This distinction is important because it removes a mechanical source of buyer-product co-movement present in a rigid common-ladder benchmark and makes the amortization mechanism empirically interpretable. Finally, the quantitative exercises show that the interaction between the two margins matters for the firm-size distribution and for policy: reducing buyer-supplier matching frictions can stimulate product expansion indirectly, while lowering product-adoption costs can also encourage network formation.

The remainder of the paper is organized as follows. Section 2 describes the data, sample, and

key definitions. Section 3 presents the lifecycle evidence and the three model-facing empirical facts. Section 4 develops the model. Section 5 describes the calibration strategy and empirical targets. Section 6 presents the quantitative behavior and fit of the calibrated model. Section 7 presents the mechanism diagnostics, fixed-cost comparative statics, margin decompositions, and the comparative-static analog of supplier growth. Section 8 concludes.

2 Data Description

2.1 Administrative data sources

The main dataset combines three administrative sources. Our core data consist of domestic firm-to-firm transactions among Chilean firms, drawn from tax records collected by the Chilean tax authority, the Servicio de Impuestos Internos (SII). These records cover the universe of formal private-sector firms. The SII began collecting these data in 2014, and electronic invoicing became mandatory in 2017. Our sample contains invoices dated from January 2015 through December 2023 and is accessed through the Central Bank of Chile.⁵

Each invoice has two parts. The first contains transaction-level information: anonymized tax identifiers for the supplier and buyer, the value of the transaction, the municipality associated with each party (corresponding to the municipality of the firm’s registered headquarters), and payment terms (cash versus trade credit, including maturity). The second part contains line-item details on the goods and services traded, including a free-text product description, price and quantity (including discounts or surcharges when applicable), and a product identifier. Because the product identifier is assigned at the level of unique text descriptions, the same good is often split into multiple IDs due to minor differences in wording, and the IDs can be extremely granular.⁶ We therefore harmonize products using the Central Product Classification (CPC)⁷.

We merge the transaction records with two additional administrative datasets. The first contains firm balance-sheet and tax information from annual form F22 and monthly form F29, including total sales, intermediate-input expenditures, the wage bill, and fixed assets. The second is the employer–employee dataset from form DJ1887, which reports annual earnings paid by each firm

⁵This paper was developed within the scope of the research agenda conducted by the Central Bank of Chile (CBC) in economic and financial affairs of its competence. The CBC has access to anonymized information from public and private entities through collaboration agreements. To protect the privacy of workers and firms, the CBC requires that the development, extraction, and publication of results not permit the direct or indirect identification of individuals or firms. CBC officials processed the disaggregated data. All analysis was implemented by the authors and does not involve or compromise the SII. Information in the SII databases is of a tax nature and originates in taxpayer self-declarations; its veracity is therefore not the responsibility of the SII.

⁶For example, “BOY SNEAKER” and “GIRL SNEAKER” are classified as different products.

⁷The CPC is an internationally harmonized classification system that organizes goods and services based on standardized concepts, definitions, and coding rules.

to each worker and months of employment. Reported earnings include wages, salaries, bonuses, tips, and other taxable labor-income components. All datasets are anonymized.

2.2 Units of observation and network definitions

Throughout the paper, a firm is defined by a unique anonymized tax identifier. Unless stated otherwise, the term *firm* refers to a supplier. Buyers and suppliers can belong to any sector of the Chilean economy, although the main stylized facts focus on manufacturing suppliers. Suppliers include both upstream (primary) and downstream (secondary) firms in the network, i.e., any firm that sells a good or service to another firm. Transactions are strictly business-to-business because purchases by final consumers are not observed.

Let s denote a supplier, b a buyer, g a 5-digit CPC product, and t a year. Let x_{sbgt} denote sales from supplier s to buyer b of product g in year t . We define

$$\begin{aligned}\mathcal{B}_{st} &= \{b : \exists g \text{ such that } x_{sbgt} > 0\}, & \text{buyers of supplier } s, \\ \mathcal{G}_{st} &= \{g : \exists b \text{ such that } x_{sbgt} > 0\}, & \text{products sold by supplier } s, \\ \mathcal{R}_{st} &= \{(b, g) : x_{sbgt} > 0\}, & \text{active buyer-product relationships of supplier } s.\end{aligned}$$

Supplier sales observed in the transaction network are

$$S_{st}^{B2B} = \sum_b \sum_g x_{sbgt}.$$

When total supplier sales are available from tax filings, we denote them by S_{st}^{total} . The corresponding extensive-margin objects are $|\mathcal{B}_{st}|$, the number of buyers; $|\mathcal{G}_{st}|$, the number of products; and $|\mathcal{R}_{st}|$, the number of active buyer-product relationships. Buyers per product are measured as

$$\frac{|\mathcal{B}_{st}|}{|\mathcal{G}_{st}|}.$$

Sales within a supplier-buyer link are

$$S_{sbt}^{\text{link}} = \sum_g x_{sbgt},$$

while sales associated with a supplier-product relationship are

$$S_{sgt}^{\text{prod}} = \sum_b x_{sbgt}.$$

2.3 Relationship age and cohort measurement

For the lifecycle analysis, we define the age of a supplier–buyer link as the number of years since that pair is first observed trading:

$$\text{Age}_{sbt}^L = t - \min \{ \tau : S_{sb\tau}^{\text{link}} > 0 \}.$$

Analogously, the age of a supplier–product relationship is

$$\text{Age}_{sgt}^P = t - \min \{ \tau : S_{sg\tau}^{\text{prod}} > 0 \}.$$

Supplier age is defined separately as the number of years since the supplier is first observed with positive sales:

$$\text{Age}_{st}^F = t - \min \{ \tau : S_{s\tau}^{B2B} > 0 \}.$$

Because the transaction panel begins in 2015, relationship entry is observed without left-censoring only for links and supplier–product relationships that begin after the start of the sample. Lifecycle regressions therefore use relationships whose first observed activity occurs within the sample and should be interpreted as describing observed relationship age rather than the complete age of relationships that may have existed before 2015.

2.4 Sample construction

We apply several cleaning and sample restrictions. First, we retain firms that report more than one paid worker. Second, we restrict attention to firms that file monthly tax form F29 in all sample years used in the relevant exercise and report positive sales. Following [Burstein et al. \(2024\)](#), we remove transactions that exceed USD 125 million, transactions with non-positive prices or quantities, and invoices in which the supplier and buyer are the same firm. Finally, we use product-level minimum and maximum price filters from the Chilean National Statistics Institute (INE) to remove line items with implausible unit prices. The exact estimation sample varies across exercises because different results require different levels of aggregation and support.

2.5 Descriptive statistics

Table 1 reports descriptive statistics for manufacturing suppliers in 2017 and 2023, separately for multi-product and single-product firms. Multi-product suppliers account for the large majority of firms in both years, although their share declines from 86.2% in 2017 to 83.8% in 2023. They operate at substantially greater scale than single-product suppliers. Their average total sales increase from \$2.35 million to \$4.22 million, compared with an increase from \$0.97 million to \$1.17 million among single-product firms. The large gaps between means and medians, however, reveal

substantial right skewness in both groups. For example, median total sales among multi-product suppliers are only \$0.079 million in 2017 and \$0.100 million in 2023, while the corresponding means are more than \$2 million and \$4 million. The dispersion is similarly pronounced for B2B sales and the number of buyers, indicating that a relatively small number of very large suppliers account for much of the variation.

Multi-product suppliers also maintain much larger buyer networks and broader product portfolios. They serve an average of 68.9 buyers and sell 16.1 products in 2017, compared with 7.6 buyers and one product among single-product suppliers. By 2023, the average number of buyers declines in both groups, to 59.9 for multi-product suppliers and 4.8 for single-product suppliers, while the average number of products among multi-product firms remains relatively stable at 15.8. Median buyer and product counts are considerably lower than their means: the median multi-product supplier has nine buyers and nine products in 2017 and eight of each in 2023. Buyers per product also decline over time, from an average of 3.75 to 3.22 among multi-product suppliers and from 7.64 to 4.77 among single-product suppliers. In both categories, the median is only one buyer per product, again highlighting substantial heterogeneity across firms. Finally, multi-product suppliers have less concentrated buyers than single-product suppliers, but concentration increases modestly over time. The average buyer-concentration ratio rises from 0.51 to 0.54 among multi-product suppliers and from 0.80 to 0.85 among single-product suppliers, while product specialization among multi-product firms increases from 0.54 to 0.55. These patterns indicate that, despite the much broader buyer and product scope of multi-product suppliers, sales remain unevenly distributed across both buyers and products.

3 Empirical Evidence

This section uses the panel structure of the transaction data for two different purposes. We first present lifecycle evidence showing that supplier–buyer links and supplier–product relationships are empirically distinct objects. This evidence motivates the model’s separation of buyer reach and product scope, but it is not itself a quantitative target of the static model. We then document three model-facing facts. The first two concern the margins through which suppliers move between an initial and a final allocation, while the third provides direct evidence for the complementarity mechanism.

3.1 Motivating evidence: Buyer links and products have distinct dynamics

Buyer links and supplier–product relationships follow different lifecycle patterns. This distinction matters because treating a multi-product supplier as a collection of undifferentiated matches

Table 1: Descriptive Statistics for Chilean Manufacturing Suppliers

	2017			2023		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
<i>Panel A: Multiproduct firms (suppliers)</i>						
Total sales (million USD, firm level)	2.35	0.08	43.95	4.22	0.10	84.29
B2B intermediate sales (million USD)	0.96	0.02	29.53	1.67	0.02	53.283
Number of buyers	68.904	9.000	784.554	59.914	8.000	667.048
Number of products	16.061	9.000	20.955	15.764	8.000	21.583
Number of buyers per product	3.749	1.000	22.768	3.220	1.000	20.357
Product specialization ratio	0.541	0.512	0.243	0.553	0.528	0.245
Buyer concentration ratio	0.514	0.475	0.287	0.544	0.514	0.286
Number of firms	37,179	—	—	28,938	—	—
Share of all suppliers (%)	86.2	—	—	83.8	—	—
<i>Panel B: Single-product firms (suppliers)</i>						
Total sales (million USD, firm level)	0.971	0.022	25.378	1.166	0.023	27.427
B2B intermediate sales (million USD)	0.268	0.002	10.445	0.272	0.002	6.572
Number of buyers	7.638	1.000	32.583	4.767	1.000	22.324
Number of products	1.000	1.000	0.000	1.000	1.000	0.000
Number of buyers per product	7.638	1.000	32.583	4.767	1.000	22.324
Product specialization ratio	1.000	1.000	0.000	1.000	1.000	0.000
Buyer concentration ratio	0.801	1.000	0.295	0.845	1.000	0.259
Number of firms	5,935	—	—	5,585	—	—
Share of all suppliers (%)	13.8	—	—	16.2	—	—

Notes: Sales are in million USD. Shares are computed within year among single-product and multiproduct suppliers. A supplier is classified as multi-product in a year if it sells at least two distinct 5-digit CPC products in that year. The product-specialization ratio is the share of supplier sales accounted for by its largest product. The buyer-concentration ratio is defined analogously using the supplier's largest buyer. Buyers per product are calculated at the supplier level before averaging across suppliers.

would conflate two empirically different margins of adjustment.

For supplier–buyer links, we estimate

$$\log S_{sbt}^{\text{link}} = \alpha^L + \sum_{j \geq 1} \beta_j^L \mathbf{1}\{\text{Age}_{sbt}^L = j\} + \mu_{sb} + \gamma_t + \varepsilon_{sbt}^L, \quad (1)$$

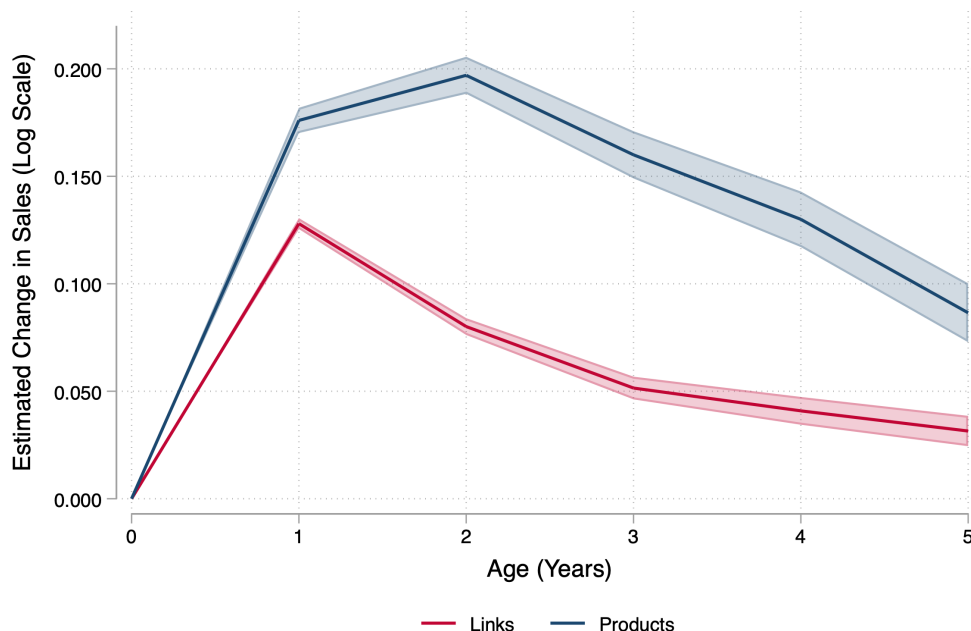
where age zero is the omitted category, μ_{sb} is a supplier–buyer-link fixed effect, and γ_t is a year fixed effect. For supplier–product relationships, we estimate

$$\log S_{sgt}^{\text{prod}} = \alpha^P + \sum_{j \geq 1} \beta_j^P \mathbf{1}\{\text{Age}_{sgt}^P = j\} + \mu_{sg} + \gamma_t + \varepsilon_{sgt}^P, \quad (2)$$

where μ_{sg} is a supplier–product fixed effect. Standard errors are clustered at the relationship level in each specification.

Figure 2 plots the estimated age coefficients. Both relationships display hump-shaped sales profiles, but buyer links peak earlier and decline more quickly. Link sales peak at approximately 0.13 log points above their entry-year level at age one. Supplier–product sales continue rising through age two, reach a peak of approximately 0.20 log points, and remain above the link profile at subsequent ages. By age five, the supplier–product premium remains close to 0.09 log points, whereas the link premium has fallen to approximately 0.03.

Figure 2: Sales over the lifecycle of buyer links and supplier–product relationships



Notes: The figure plots the estimated coefficients from equations (1) and (2). The link specification includes supplier–buyer-link and year fixed effects; the product specification includes supplier–product and year fixed effects. Age zero is the omitted category, so each coefficient measures the average within- relationship change in log sales relative to the first observed year of the relationship. Shaded bands report 95 percent confidence intervals. Standard errors are clustered at the corresponding relationship level.

Appendix A evidence reinforces this distinction. Cohort profiles show that sales associated with older buyer links fall more systematically than sales associated with older supplier–product relationships. Survival also declines more rapidly for buyer links than for products. These results establish that buyer relationships and product scope should be modeled as separate objects. They do not, however, imply that the static model is designed to reproduce the timing of the peaks or the relationship-survival paths. On the other hand, Appendix B shows survival rates for buyer links and supplier–product relationships across entry cohorts. Survival declines sharply for both objects, but buyer links exhibit systematically lower persistence: a substantially smaller share of links remain active one year after entry, and this gap persists at later horizons. Product relationships therefore tend to survive longer than buyer relationships. This survival evidence supports the lifecycle patterns documented above: buyer links not only peak earlier and decline faster in sales, but also exit more quickly. These patterns strengthen the view that the two margins are empirically distinct and should be modeled separately.

3.2 Fact 1: Supplier growth combines buyer and product expansion

We next study how suppliers move between an initial and a final allocation. Although this decomposition is measured over time, its model counterpart is a comparative-static exercise: the static model can compare a baseline equilibrium with a counterfactual equilibrium and decompose the resulting change across buyer and product margins.

Let $t \in \{2017, 2023\}$. Define the sets of buyers and products of supplier s in the baseline year:

$$\mathcal{B}_{s0} = \{b : \exists g \text{ such that } x_{sbg,2017} > 0\}, \quad \mathcal{G}_{s0} = \{g : \exists b \text{ such that } x_{sbg,2017} > 0\}.$$

We classify each unit of 2023 sales according to whether its buyer and product were present in 2017:

$$\begin{aligned} S_{s,2023}^{\text{OLOP}} &= \sum_{b \in \mathcal{B}_{s0}} \sum_{g \in \mathcal{G}_{s0}} x_{sbg,2023}, & \text{old link, old product,} \\ S_{s,2023}^{\text{NLOP}} &= \sum_{b \notin \mathcal{B}_{s0}} \sum_{g \in \mathcal{G}_{s0}} x_{sbg,2023}, & \text{new link, old product,} \\ S_{s,2023}^{\text{OLNP}} &= \sum_{b \in \mathcal{B}_{s0}} \sum_{g \notin \mathcal{G}_{s0}} x_{sbg,2023}, & \text{old link, new product,} \\ S_{s,2023}^{\text{NLNP}} &= \sum_{b \notin \mathcal{B}_{s0}} \sum_{g \notin \mathcal{G}_{s0}} x_{sbg,2023}, & \text{new link, new product.} \end{aligned}$$

Because all 2017 sales occur with buyers and products that are old relative to the 2017 baseline,

$$S_{s,2017} = S_{s,2017}^{\text{OLOP}}.$$

The four components of supplier sales growth are therefore

$$\begin{aligned} \Delta S_s^{\text{OLOP}} &= S_{s,2023}^{\text{OLOP}} - S_{s,2017}, \\ \Delta S_s^{\text{NLOP}} &= S_{s,2023}^{\text{NLOP}}, \\ \Delta S_s^{\text{OLNP}} &= S_{s,2023}^{\text{OLNP}}, \\ \Delta S_s^{\text{NLNP}} &= S_{s,2023}^{\text{NLNP}}. \end{aligned}$$

They satisfy the accounting identity

$$\Delta S_s = S_{s,2023} - S_{s,2017} = \Delta S_s^{\text{OLOP}} + \Delta S_s^{\text{NLOP}} + \Delta S_s^{\text{OLNP}} + \Delta S_s^{\text{NLNP}}. \quad (3)$$

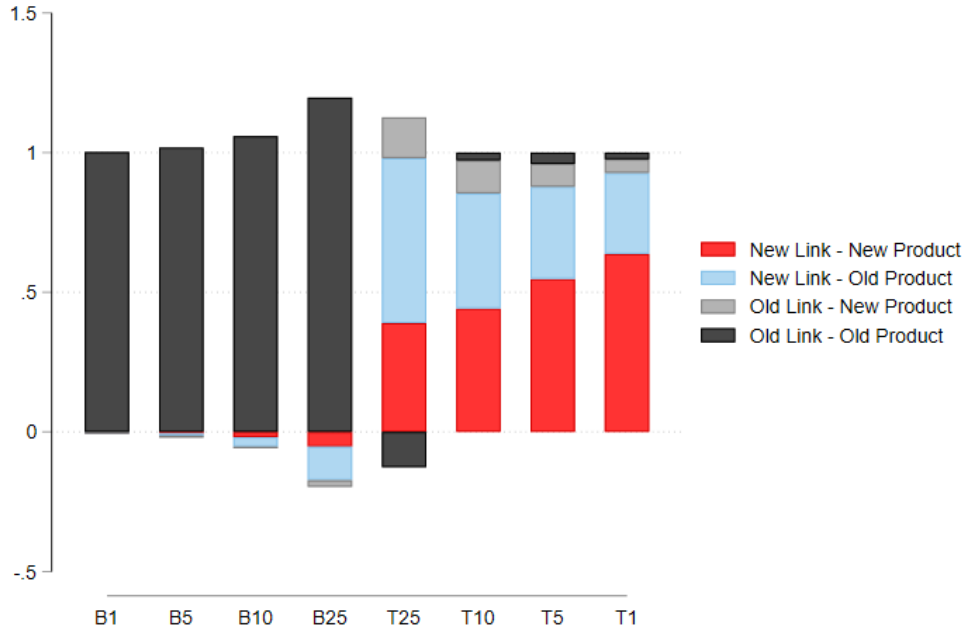
The OLOP component is a net intensive-and-exit term. It includes changes in sales among continuing old-buyer–old-product relationships and losses from old relationships that are no longer active in 2023. The other three components involve at least one extensive-margin change relative

to the baseline.

Decomposition by realized growth. We first group suppliers by their realized 2017–2023 sales growth and compute the average contribution of each component to ΔS_s . Figure 3 asks an ex post accounting question: among suppliers that subsequently shrink or grow by different amounts, which margins account for their realized change in sales?

The lower tail displays little offsetting expansion through new buyers or new products. Contraction is concentrated in OLOP, reflecting lower sales in the continuing portion of the supplier’s baseline buyer–product space and losses from exiting relationships. Because each component is normalized by the supplier’s total change in sales, the OLOP share appears positive for shrinking suppliers: both the OLOP contribution and total sales growth are negative, so their ratio is positive. By contrast, any positive contribution from new buyers or new products is divided by a negative total change and therefore appears as a negative share. In the upper tail, extensive margins account for a much larger share of growth. The NLNP component shows that the fastest-growing suppliers disproportionately combine new buyers and new products rather than expanding along only one margin.

Figure 3: Decomposition of supplier sales growth, 2017–2023



Notes: The figure decomposes supplier sales growth between 2017 and 2023 across intervals of the realized-growth distribution. Each bar reports the average contribution of new link–new product (NLNP), new link–old product (NLOP), old link–new product (OLNP), and old link–old product (OLOP). B denotes intervals in the bottom half and T intervals in the top half. Each category corresponds to the range between adjacent reported cutoffs. For example, B5 denotes the interval from the 1st to the 5th percentile, while T90 denotes the interval from the 90th to the 95th percentile.

Composition of extensive-margin growth. To isolate the composition of expansion involving new buyers or products, define

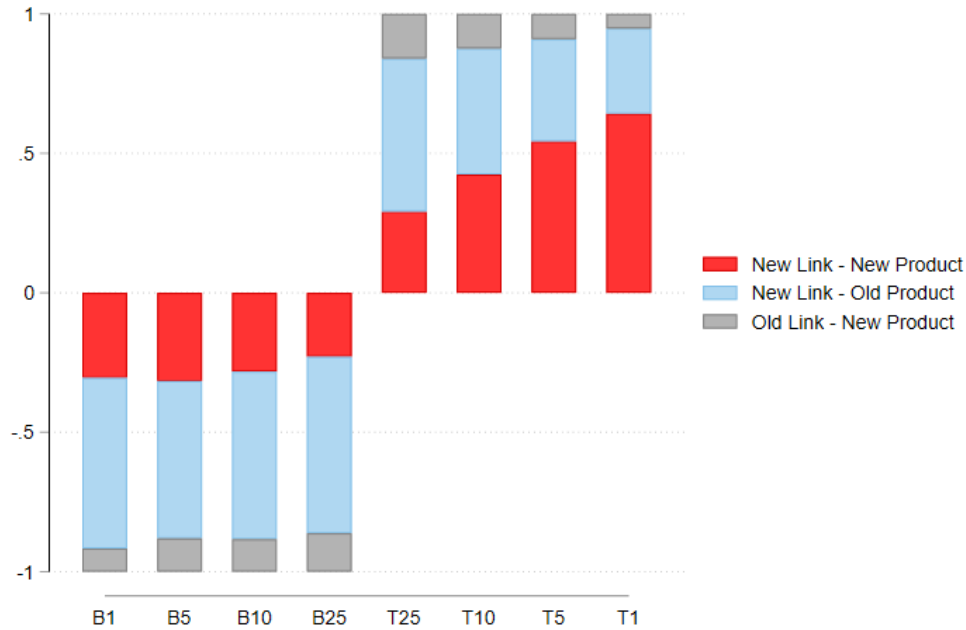
$$\Delta S_s^{\text{EXT}} = \Delta S_s^{\text{NLOP}} + \Delta S_s^{\text{OLNP}} + \Delta S_s^{\text{NLNP}}.$$

For suppliers with positive extensive-margin growth, define

$$\tilde{s}_s^c = \frac{\Delta S_s^c}{\Delta S_s^{\text{EXT}}}, \quad c \in \{\text{NLOP}, \text{OLNP}, \text{NLNP}\}.$$

Figure 4 shows that, conditional on expansion occurring through an extensive margin, NLNP becomes increasingly important among the largest growers. Buyer acquisition and product introduction are therefore not simply two additive sources of growth. The largest expansions are disproportionately associated with activating both margins together.

Figure 4: Composition of extensive-margin growth



Notes: The figure reports the shares of positive extensive-margin growth accounted for by NLNP, NLOP, and OLP across intervals of the realized supplier-growth distribution. The groups are defined using adjacent percentile cutoffs, as in Figure 3.

A further decomposition of OLP separates changes among old buyer–product relationships that remain active through 2023 from losses associated with relationships that exit before 2023. This exercise shows that relationship exit accounts for slightly more than half of the negative growth among suppliers in the lower tail of the realized-growth distribution.

3.3 Fact 2: Expansion shifts from new products and buyers toward established products and new buyers as suppliers become larger and older

The previous figures classify suppliers according to their realized change in sales. The sorting variable is therefore the outcome being decomposed. We now ask a different question: do suppliers that begin from different initial conditions expand through different combinations of buyer acquisition, product introduction, and sales within continuing relationships? We examine two dimensions of initial heterogeneity: suppliers' position in the 2017 sales distribution and their stage in the firm life cycle.

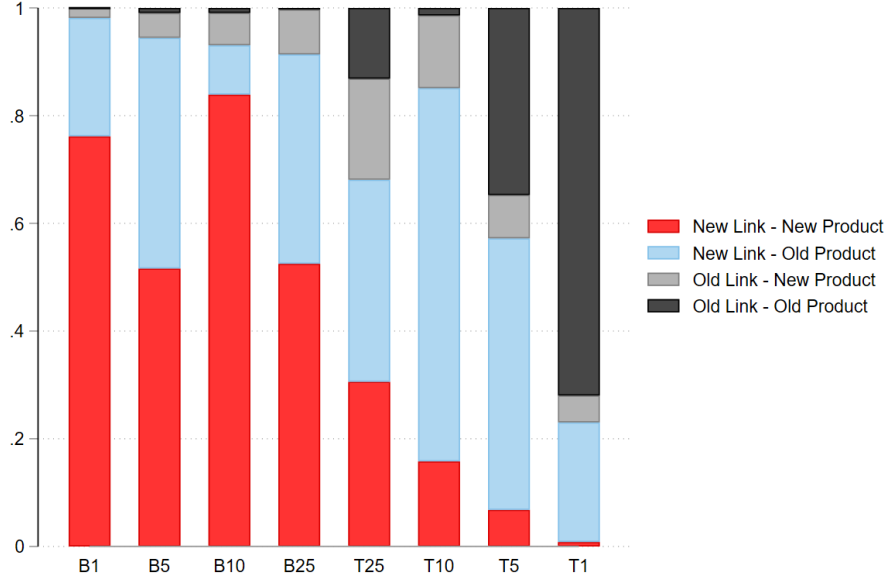
Heterogeneity by initial supplier scale. We first assign suppliers to groups according to their total sales in 2017 and then decompose their sales growth between 2017 and 2023. The current version of this exercise restricts attention to suppliers with positive growth. Accordingly, the figure describes the composition of growth among expanding suppliers; it does not establish which initial-size groups grow faster or are more likely to grow.

Figure 5 reveals a clear pattern across the initial-size distribution. Among suppliers with positive growth, initially smaller suppliers rely relatively more on margins involving new products, particularly the joint addition of new buyers and new products (NLNP). As initial supplier size increases, growth shifts toward existing products: larger suppliers rely relatively more on sales through existing buyer–product relationships (OLOP) and on forming new buyer links for products already in their portfolios (NLOP). Thus, product expansion is relatively more important for initially smaller suppliers, while larger suppliers expand relatively more by deepening existing relationships and extending established products to new buyers. This result has a direct model counterpart. Initial supplier sales can reflect baseline differences in productivity, buyer reach, or product scope. The model can therefore apply a common change in the economic environment and ask whether initially small and large suppliers respond through different combinations of buyer acquisition, product adoption, and changes within continuing relationships.

The distinction between Figures 3 and 5 is important. The first is an ex post accounting exercise: it asks which margins characterize different realized growth outcomes. The second is an ex ante heterogeneity exercise: it asks whether the composition of subsequent expansion varies with a supplier characteristic measured before that expansion occurs. The latter is therefore especially useful for disciplining heterogeneous comparative-static responses in the model.

Heterogeneity over the supplier life cycle. We next ask whether the composition of growth also varies with supplier maturity, while conditioning on suppliers' sales in 2017. This exercise separates differences associated with supplier age from those arising simply because older suppliers

Figure 5: Composition of positive sales growth by initial supplier size



Notes: Suppliers are grouped according to their position in the 2017 total-sales distribution. Within each initial-size group, the figure decomposes positive supplier sales growth between 2017 and 2023 into new link–new product (NLNP), new link–old product (NLOP), old link–new product (OLNP), and old link–old product (OLOP). Because the sample is restricted to suppliers with positive growth, the figure describes how expanding suppliers grow rather than differences in average growth rates or growth probabilities across initial-size groups. Each category corresponds to the interval between adjacent reported percentile cutoffs.

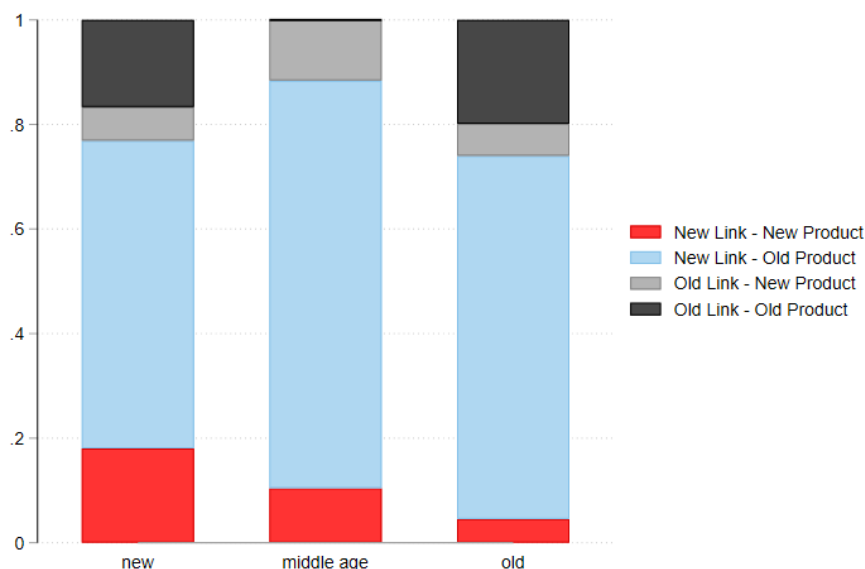
tend to operate at a different initial scale. Figure 6 reports the average decomposition of ΔS_s for young, middle-aged, and old suppliers.

The results suggest that suppliers at different stages of the life cycle expand through different margins. Younger suppliers rely more heavily on margins involving product introduction, particularly the joint NLNP margin in which they add both new buyers and new products. This pattern is consistent with an early stage of firm development in which suppliers are simultaneously building their customer base and developing their product portfolios.

Older suppliers, by contrast, rely relatively more on NLOP growth: they form new buyer relationships for products already included in their existing portfolios. Conditional on initial scale, mature suppliers therefore appear to expand less through additional product scope and more through extending the market reach of established products. The lifecycle evidence suggests a shift from simultaneous experimentation across buyers and products among younger suppliers toward customer acquisition based on an established product portfolio among older suppliers.

The initial-scale and age decompositions show that extensive margins matter across suppliers, but their relative importance varies with firms' initial conditions. Initially smaller suppliers expand relatively more through margins involving new products, especially the joint addition of new buyers and new products, whereas larger suppliers rely relatively more on existing buyer–product relationships and on extending established products to new buyers. These results reinforce the

Figure 6: Composition of sales growth by supplier age



Notes: The figure decomposes supplier sales growth between 2017 and 2023 into NLNP, NLOP, OLNP, and OLOP by supplier age group, conditioning on suppliers' sales in 2017. Young suppliers are between one and five years old, middle-aged suppliers are between six and fifteen years old, and old suppliers are more than fifteen years old. Supplier age is measured as of 2017.

importance of modeling buyer acquisition and product scope as distinct but interacting choices rather than collapsing them into a single match margin.

The initial-size results provide a direct target for the model's heterogeneous comparative statics. The age patterns are more naturally interpreted as complementary descriptive evidence, because supplier age is not an explicit state variable in the static model.

3.4 Fact 3: Buyer breadth and product scope are complementary

The previous facts establish that buyer relationships and product scope are distinct margins. We now ask whether these two margins are systematically related within suppliers. A priori, their relationship could reflect either a trade-off, whereby expanding along one margin comes at the expense of the other, or a complementarity, whereby a larger buyer base raises the return to maintaining a broader product portfolio and, a broader product portfolio, in turn, raises the value of forming additional buyer relationships because it gives the supplier more products through which to generate surplus with a potential buyer, making the fixed cost of forming the relationship easier to cover.

This question is closely related to recent work on the interaction between demand-expansion investments, such as advertising, and innovation. For example, [Cavenaile and Roldan-Blanco \(2021\)](#) provides evidence of substitution between R&D and advertising among U.S. firms, whereas recent models often assume complementarity between customer expansion and innovation, as in

Einav et al. (2021) and Afrouzi et al. (2023). These studies, however, focus on final-consumer settings. In contrast, as discussed above, our setting focuses on supplier–buyer relationships within the production network.

To quantify the relationship between the two margins, we estimate how a supplier’s product scope varies with its contemporaneous number of buyers. Let Buyers_{st} denote the number of distinct buyers served by supplier s in year t , and let Products_{st} denote its number of products (CPCs). Our preferred specification is:

$$\log(\text{Products}_{st}) = \beta_1 \log(\text{Buyers}_{st}) + \beta_2 \log(\text{Products}_{s,t-1}) + \theta \log(\text{Sales}_{st}) + \gamma_t + \delta_s + \varepsilon_{st}, \quad (4)$$

where γ_t are year fixed effects and δ_s are supplier fixed effects. The supplier fixed effects absorb persistent differences across firms, including productivity, managerial ability, product quality, and sectoral specialization, while lagged product scope controls for persistence in the supplier’s portfolio. The coefficient β_1 therefore captures whether, within the same supplier over time, years with a larger buyer base are also years in which the supplier operates a broader product portfolio.

Table 2 reports the estimates. The relationship between buyer breadth and product scope is positive and precisely estimated across all specifications. The coefficient on $\log(\text{Buyers}_{it})$ ranges from 0.13 to 0.25 as controls and fixed effects are added. Column (6) is the most important specification for our purposes. It includes supplier and year fixed effects and controls for contemporaneous sales and lagged product scope. The coefficient falls to 0.13 but remains highly significant. Because both product scope and buyer breadth enter in logs, this coefficient implies that a 10 percent increase in the number of buyers is associated with approximately a 1.3 percent increase in the number of products within the same supplier.

Column (6) is also the empirical moment that maps most closely into the model. The model is static and therefore cannot reproduce within-supplier changes over time literally. In the quantitative exercise, we construct the corresponding model moment by relating product scope to buyer breadth conditional on supplier productivity. The empirical coefficient of 0.13 therefore provides the benchmark against which we evaluate the model’s within-productivity co-movement between the two margins.

Appendix D provides complementary evidence using new product introductions rather than the supplier’s total product scope as the dependent variable. The coefficient on $\log(\text{Buyers}_{it})$ is again positive and precisely estimated across specifications and remains close to 1.9 when supplier fixed effects are included. Thus, within suppliers, years with more buyers are also associated with more product introductions. The magnitude is economically meaningful. In the specification with supplier fixed effects, the coefficient on $\log(\text{Buyers}_{it})$ is approximately 1.9. Because the dependent

variable is the number of new products, this implies that a 10 percent increase in the number of buyers is associated with approximately 0.19 additional product introductions. The appendix results therefore show that the positive relationship between buyer breadth and product scope is also visible at the margin of new product introduction.

These results show a strong positive relationship between the two margins, both in the scope of products a supplier sells and in the introduction of new products. We interpret this evidence as consistent with the complementarity at the center of the model: a broader buyer base raises the returns to product scope, while a broader portfolio of products raises the value of buyer relationships. The data reject a view in which suppliers expand along one margin while holding the other fixed. This motivates a model in which firms choose both (i) investments that expand demand through buyer acquisition and (ii) investments that expand product scope, allowing for interaction between these choices.

Table 2: Relationship between buyer breadth and product scope

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dep. variable: log Num. Products_{it}</i>						
log Buyers _{it}	0.221*** (0.00233)	0.217*** (0.00285)	0.254*** (0.00286)	0.201*** (0.00222)	0.197*** (0.00271)	0.130*** (0.00383)
log Sales _{it}	0.144*** (0.00104)	0.150*** (0.00128)	0.158*** (0.00128)	0.171*** (0.00101)	0.176*** (0.00124)	0.275*** (0.00188)
log Products _{it-1}		0.0759*** (0.00296)	0.0547*** (0.00287)		0.0485*** (0.00246)	0.0202*** (0.00279)
log Sales _{it} ²			-0.00654*** (8.05e-05)			
Observations	186,820	121,609	121,609	186,820	121,609	183,783
R-squared	0.516	0.518	0.542	0.647	0.651	0.916
Firms	34,275	31,784	31,784	34,275	31,784	31,238
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	No	Yes	Yes	No
Supplier FE	No	No	No	No	No	Yes

Notes: The dependent variable is the log of the number of products per supplier of year t . The main regressor is $\log(\text{Buyers}_{it})$. Controls include contemporaneous supplier sales and, in the indicated specifications, lagged product scope and squared log sales. All specifications include year fixed effects; sector and supplier fixed effects depending on the column as indicated. Standard errors are clustered at the supplier level.

4 Model

The stylized facts in the previous section show that firm growth in production networks operates along two distinct extensive margins: expanding product scope and forming or maintaining buyer relationships. They also show that these margins are not only separately important, but systematically linked within firms over time. Reduced-form correlations alone are difficult to interpret because product expansion and buyer accumulation are jointly chosen and shaped by common profitability forces. To discipline interpretation, we develop a tractable framework in which firms choose both firm-level product scope and buyer-level links, and in which the profitability of each margin depends on the other. The model serves three purposes. First, it isolates the mechanism through which the product and buyer margins become complementary. Second, it allows us to quantify how shocks to either product-adoption costs or buyer-linkage costs propagate through both margins and reshape the firm size distribution. Third, it allows us to assess the overall contribution of product scope and buyer scope to the firm size distribution while explicitly accounting for their complementarity.

Our model builds on the logic of firm dynamics in [Hopenhayn \(1992\)](#) and the rank-based multiproduct structure in [Arkolakis et al. \(2021\)](#), but adapts them to a production-network environment in which geographic destinations are individual buyers. The link-formation cost places us in the tradition that models network extensive margins through a fixed cost per relationship ([Lim, 2018](#); [Huneus, 2018](#); [Bernard et al., 2022](#)), as opposed to the search-based approach ([Miyauchi, 2024](#); [Arkolakis et al., 2023](#)). Suppliers are heterogeneous in productivity and choose how many products to develop at the firm level and which buyers to serve through costly relationship formation. Product adoption costs are paid once at the firm level and can therefore be amortized across all buyers served, while link formation requires buyer-specific fixed costs. This distinction is central: because the returns to product adoption accrue across buyer relationships, a larger buyer base raises the payoff to expanding scope; conversely, a broader product portfolio raises the value of forming and maintaining buyer links. The model is designed to rationalize the joint evolution of product scope and buyer networks and to provide a transparent mapping from these decisions to firm-level sales growth.

A separate set of empirical results disciplines how we model demand across products and buyer relationships. In particular, we find little evidence of systematic cannibalization along either margin. [Figure 16](#) follows sales of old and new products to incumbent buyers around the first year in which a supplier introduces a new product within an existing relationship. Sales of existing products do not fall after the introduction; instead, old and new product sales increase together. [Figure 17](#) performs the analogous exercise across buyer relationships, tracing sales to incumbent buyers around the first year in which a supplier acquires a large new buyer. Sales to incumbent

buyers do not decline after the new relationship is formed. These findings motivate a benchmark demand structure that abstracts from systematic displacement of existing sales when suppliers expand their product portfolios or buyer networks. We consider within-link cannibalization separately in the nested-CES extension.

A central modeling choice concerns how a supplier’s products map to its individual buyers. Our baseline allows this mapping to be buyer-specific: a supplier develops a portfolio of products at the firm level, but which of those products are relevant for a given buyer is governed by an idiosyncratic product–buyer match. This is both more realistic and empirically disciplined in our setting. Because we observe product sales at the firm level and within individual supplier–buyer links, we can measure how strongly a product’s importance in the supplier’s overall portfolio predicts its importance within a particular link. If that relationship were near-perfect, a common product ladder shared identically across buyers would be an adequate description; if it is only partial, the data call for buyer-specific product relevance. As we show below, the alignment is positive but far from perfect, which is precisely the pattern a matching structure predicts and a rigid common ladder does not. Allowing the product–buyer mapping to be buyer-specific also connects our setup to two-sided heterogeneity and matching in firm-to-firm trade ([Bernard et al., 2018](#); [Eaton et al., 2026](#); [Demir et al., 2024](#); [Miyauchi, 2024](#)), where which partners transact, and how much, is itself an outcome rather than a primitive.

4.1 Environment, objects, and timing

4.1.1 Agents, products, and the product–buyer match

Suppliers are indexed by $s \in \mathcal{S}$ and buyers by $b \in \mathcal{B}$. Each supplier has productivity $\phi_s > 0$. Products are observed in the data as discrete codes (for example, CPC codes) and indexed by $\omega \in \Omega$; product scope is a discrete choice, while the buyer pool is treated as a continuum approximated in the quantitative model by a large set of draws.

At the firm level, supplier s develops a portfolio of G_s products,

$$\Omega_s = \{\omega_s(1), \dots, \omega_s(G_s)\} \subseteq \Omega, \quad G_s \in \{0, 1, \dots, |\Omega|\},$$

paying the firm-level adoption cost $F^P(G_s)$ described below. Products are ordered from core to peripheral according to their rank $g \in 1, \dots, G_s$, where $g = 1$ denotes the core product. We capture systematic differences in profitability across ranks through an increasing schedule $h(g)$,

$$h(1) = 1, \quad h(g+1) \geq h(g),$$

which scales a product’s marginal cost relative to the supplier’s underlying productivity. Specifi-

cally, as shown in Section 4.3, product-level productivity is $\phi_{s,g} = \phi_s/h(g)$. Thus, absent buyer-specific match heterogeneity, products farther down the ranking are less profitable to supply. Unlike a pure common ladder, however, this ranking governs average profitability and does not determine which products a particular buyer purchases.

Which products are actually traded on a link is determined by an idiosyncratic product–buyer match. Each developed product $g \in \{1, \dots, G_s\}$ draws, for each buyer b , a match shock $\zeta_{sb} > 0$ (i.i.d. across (s, b, g) , with $\mathbb{E}[\zeta_{sb}] = 1$), realized before the buyer-specific activation decision. Product g is made available on link (s, b) if and only if the match is strong enough to cover its activation cost; the set of products traded on the link,

$$\Omega_{sb} = \{\omega \in \Omega_s : \text{product } \omega \text{ matches buyer } b\}, \quad G_{sb} = |\Omega_{sb}| \leq G_s, \quad (5)$$

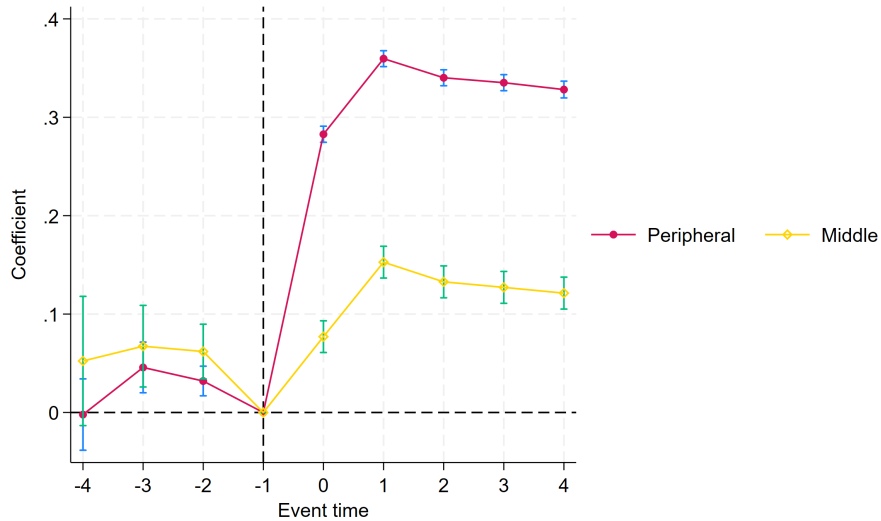
is therefore a buyer-specific subset of the supplier’s portfolio, not a common prefix of a ladder. The link-level scope G_{sb} is an outcome of the match, and different buyers of the same supplier generally hold different, non-nested product sets. The rank schedule $h(g)$ makes core products more likely to match and to sell more when they do, so firm-level and link-level product importance are positively, but imperfectly aligned. The two polar cases are transparent. As the match becomes deterministic, Ω_{sb} collapses to the common subset $\{1, \dots, G_{sb}\}$, reproducing the Arkolakis et al. (2021) rank structure: products share a common profitability ranking across buyers, and each buyer purchases the first G_{sb} products in that ranking, with links differing only in how far down the product ladder they extend. At the other extreme, as the systematic ordering vanishes ($h(g) \equiv 1$), products are ex-ante symmetric and product importance is purely idiosyncratic across buyers.

This matching structure yields a sharp, testable implication for the relationship between firm-level and link-level product hierarchies. A rigid common ladder implies that a product’s rank in the supplier’s overall portfolio should predict its rank within any individual link almost perfectly: every buyer descends the same ordered list. A matching structure instead predicts a positive but partial association where core products tend to be more important within links because they match more often and sell more when matched, but idiosyncratic matches make the within-link hierarchy differ from the firm-level one buyer by buyer. Table 3 shows that the data display exactly this partial alignment. A higher firm-level rank score is associated with only a 0.27–0.29 higher link-level score, and products in the supplier’s top three are 0.08–0.10 more likely to be among a link’s top three; these associations are positive and significant but far from the near-perfect alignment a rigid ladder would imply. Most tellingly, being the supplier’s single top product does not significantly predict being the top product within a given link (column 3). This is difficult to reconcile with a common ladder, under which the core product should be the within-link leader almost everywhere, but is precisely what the matching model predicts: even the firm’s best product fails to match some buyers, and other products lead within those links. We therefore

adopt matching as the baseline and treat the correlation in Table 3 as an identifying moment for the dispersion of the match shock (the parameter θ_ζ introduced in Section 4.3): stronger alignment maps to less dispersed, more deterministic matches, and conversely.

Figure 7 shows that the buyer-product complementarity documented in Table 2 is not uniform across the product ladder: it is concentrated in peripheral products, with core products displaying little response. Specifically, we trace the effect on product-level sales in the years surrounding the addition of a large new buyer, separately for core, middle, and peripheral products (defined by their pre-event rank in the supplier’s portfolio). Core products, which are those already generating the largest share of the supplier’s revenues, exhibit no statistically significant change in sales following a new buyer addition. Peripheral products, by contrast, display a sharp and persistent increase in sales beginning in the year of the event. This asymmetry is exactly what the amortization mechanism in Section 4.8.2 predicts: a new buyer raises the summed return to the firm-level adoption cost $F^P(G_s)$ and thus the firm’s optimal scope G_s^* , so the firm develops additional products at the peripheral (high-rank) margin, while infra-marginal core products, already developed and widely matched, are left essentially unaffected. The evidence therefore validates a central prediction of the model: buyer acquisition operates primarily on the extensive product margin, developing products at the periphery of the firm’s portfolio, rather than intensifying sales of products that were already core.

Figure 7: Buyer-product complementarity across the product ladder



Notes: Event-study estimates of the effect of adding a large new buyer on product-level log sales, plotted separately for core, middle, and peripheral products. Products are classified by their pre-event rank in the supplier’s overall portfolio: core products are those in the top tercile of the firm’s revenue distribution, peripheral products are in the bottom tercile, and middle products are in between. The event is the first year in which a supplier adds a buyer whose size (measured by total purchases) places it in the top decile of the buyer size distribution. The baseline year is $t=1$ (one year before the event) to account for pre-event anticipation effects; all coefficients are measured relative to this baseline. Shaded bands report 95% confidence intervals. Standard errors are clustered at the supplier level.

Table 3: Within-link product-rank correlations

	(1)	(2)	(3)	(4)	(5)	(6)
	Link score	Link rank	Link top-1 dummy	Link top-1 score	Link top-3 dummy	Link top-3 score
<i>Panel A: Link FE + Year FE</i>						
Firm rank score	0.266*** (0.0223)			0.142*** (0.0383)		0.270*** (0.0379)
Firm rank		0.0599*** (0.00482)				
Firm top-1			0.0167 (0.0550)			
Firm top-3					0.105*** (0.0216)	
R-squared	0.012	0.583	0.112	0.115	0.274	0.272
<i>Panel B: Link×Year FE</i>						
Firm score	0.287*** (0.0270)			0.120*** (0.0457)		0.225*** (0.0410)
Firm rank		0.0465*** (0.00625)				
Firm top-1			0.000456 (0.0640)			
Firm top-3					0.0775*** (0.0245)	
R-squared	0.013	0.628	0.131	0.133	0.319	0.320
Observations	56,565,645	56,565,645	56,565,645	56,565,645	48,832,461	48,832,461
Firms	52,334	52,334	52,334	52,334	45,023	45,023
Cluster	Supplier					

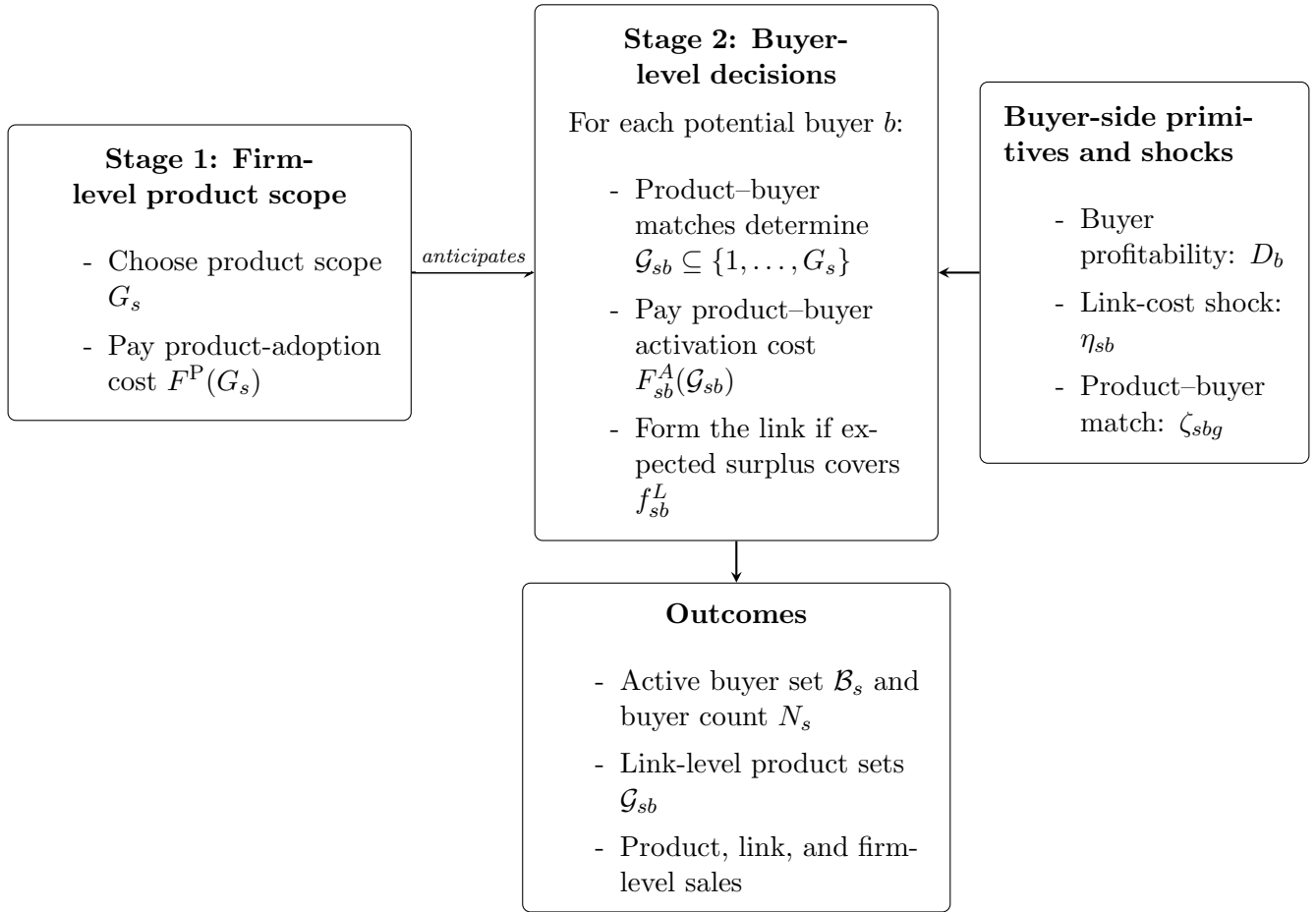
Notes: The unit of observation is a supplier-buyer-product-year. Dependent variables capture the importance of product g within link (s, b) at year t : Link score is the product's sales share within the link, Link rank is its rank position by sales within the link, Link top-1 dummy equals one if the product is the top seller within the link, Link top-1 score is its sales share if it is the top seller zero otherwise, and analogous definitions apply for the top-3 outcomes. The independent variables are the corresponding firm-level analogs computed across all of the supplier's buyers. Panel A uses additive link and year fixed effects; Panel B replaces them with link-by-year fixed effects, which absorb any link-specific shocks that vary over time. Sample sizes and firm counts are identical across panels for a given column. Standard errors clustered at the supplier level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.1.2 Timing and the two-stage structure

The supplier's problem has a recursive two-stage structure. This timing is natural in a production-network environment, where firm growth operates through two distinct extensive margins: expanding the firm's product portfolio and expanding the set of buyers it serves.

Figure 8 summarizes the timing of the supplier's problem and the shocks relevant at each stage. The figure is intended as a roadmap: the supplier first chooses firm-level product scope, and then, conditional on that scope, evaluates buyer relationships and the products sold within each link.

Figure 8: Timing of the supplier's two-stage problem



Notes: The supplier first chooses its firm-level product scope G_s and pays the adoption cost $F^P(G_s)$, anticipating the buyer-level problem. Conditional on G_s , buyer profitability D_b , pair-specific cost heterogeneity η_{sb} , and product-buyer match shocks ζ_{sbg} determine the profitability of serving each potential buyer and which developed products are activated on each link. Activating these products requires the buyer-specific product-activation cost $F_{sb}^A(\mathcal{G}_{sb})$, while forming the relationship requires the link cost f_{sb}^L . The formal problem is solved by backward induction in expected value.

1. **Stage 1: Firm-level product scope.** Supplier s chooses an integer

$$G_s \in \{0, 1, \dots, |\Omega|\},$$

which determines the size of its firm-wide product scope. The supplier pays a fixed cost schedule $F^P(G_s)$ associated with maintaining or adopting this portfolio, anticipating optimal Stage 2 behavior. The choice of G_s should be interpreted as the supplier's current stock of active products in the period, rather than as a once-and-for-all lifetime choice. A dynamic extension would allow firms to adjust this stock over time as buyer opportunities evolve.

2. **Stage 2: Link formation and link-specific scope.** Taking G_s as given, for each potential buyer $b \in \mathcal{B}$, the supplier makes two nested decisions. First, conditional on activating the link, the products made available are those that match the buyer (equation 5); the supplier pays the buyer-specific activation cost $F_{sb}^A(G_{sb}) = \sum_{\omega \in \Omega_{sb}} f_{sb}^A(g_s(\omega))$ on the matched products. This cost is incurred per product per buyer and captures buyer-specific adaptation: customization, certification, packaging, catalog listing, or integration into the buyer's procurement system. Second, the supplier decides whether to activate the link at all by paying the link fixed cost f_{sb}^L , which is incurred once per relationship regardless of G_{sb} and captures the cost of establishing the relationship itself (search, contracting, onboarding, logistics setup). The link is activated whenever the optimized link-level surplus, net of $F_{sb}^A(G_{sb})$, is at least f_{sb}^L .

Roles of the three fixed costs. The key economic distinction is that product adoption is a firm-level investment, whereas link formation and product activation are buyer-level investments. The three fixed costs play conceptually distinct roles:

- $F^P(G_s)$ is paid once at the firm level. Once a product has been adopted, its payoff accrues across multiple buyer relationships. This creates the model's central amortization force.
- f_{sb}^L governs the extensive margin of buyer relationships: whether supplier s forms a link with buyer b .
- $F_{sb}^A(G_{sb})$ governs the intensive margin within an active link: how many of the firm's adopted products are actually made available to that buyer.

The separation between firm-level adoption and buyer-level activation is what generates two-way complementarity between buyer expansion and product scope. Adding buyers raises the return to adopting products because the firm-level adoption cost is amortized across more relationships. A broader product portfolio in turn makes additional buyer relationships profitable, because the supplier has more products with which to generate surplus on each link.

Different from [Arkolakis et al. \(2021\)](#), where product scope is naturally chosen within a firm-destination unit, here in the present production-network setting the central object is instead the

supplier's global product portfolio and its allocation across many buyer relationships. If product fixed costs were purely buyer-specific, with a separate cost $f_{sb}^P(g)$ paid each time product g is sold to buyer b , then adding buyers would mechanically increase $\sum_b G_{sb}$ but would not generate a structural prediction that the supplier's global product scope rises with the number or quality of buyers served. The firm-level adoption cost is therefore the minimal ingredient that turns buyer expansion into an incentive to expand the supplier's product menu.

4.2 Buyer demand and the revenue system

We work in a partial-equilibrium setting in which buyer b has total expenditure $E_b > 0$ on the relevant intermediate-input bundle. In a general-equilibrium closure, E_b can be derived from the buyer's production problem; in the baseline benchmark, we treat it as given.

Let $\mathcal{S}_b \subseteq \mathcal{S}$ denote the set of suppliers active on buyer b . Supplier s develops a firm-level portfolio $\{1, \dots, G_s\}$, but only a buyer-specific subset of that portfolio is traded on link (s, b) . Denote this realized set by

$$\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}, \quad G_{sb} \equiv |\mathcal{G}_{sb}|. \quad (6)$$

Both $\mathcal{S}_b$ and $\mathcal{G}_{sb}$ are endogenous objects determined by the supplier's two-stage problem in Section 4.8. The buyer takes the realized set of available varieties as given when allocating expenditure. Products (s, g) with $s \notin \mathcal{S}_b$ or $g \notin \mathcal{G}_{sb}$ are not offered to buyer b and do not enter the buyer's problem.

For each active product (s, g) with $s \in \mathcal{S}_b$ and $g \in \mathcal{G}_{sb}$, let q_{sbg} denote the quantity purchased by buyer b and let p_{sbg} denote its price. We introduce a buyer–supplier–product match shifter $\zeta_{sbg} > 0$. To preserve a revenue equation that is linear in the match shifter, buyer b allocates expenditure across available products according to CES demand with taste weight $\zeta_{sbg}^{1/\sigma}$ and common elasticity $\sigma > 1$:

$$M_b = \left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg}^{1/\sigma} q_{sbg}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \quad (7)$$

The buyer solves

$$\max_{\{q_{sbg}\}} M_b \quad \text{s.t.} \quad \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} p_{sbg} q_{sbg} \leq E_b. \quad (8)$$

The associated buyer-level CES price index is

$$P_b = \left(\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \quad (9)$$

or equivalently,

$$P_b^{1-\sigma} = \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma}. \quad (10)$$

Standard CES demand then implies that expenditure on variety (s, g) is

$$r_{sbg} \equiv p_{sbg} q_{sbg} = E_b \zeta_{sbg} \left(\frac{p_{sbg}}{P_b} \right)^{1-\sigma}, \quad (11)$$

and hence quantity demand is

$$q_{sbg} = E_b \zeta_{sbg} P_b^{-\sigma} P_b^{\sigma-1}. \quad (12)$$

The match shock is realized after firm-level scope G_s has been chosen and before the product-activation decision on link (s, b) . It therefore plays two roles: it determines whether a developed product is sufficiently relevant for a particular buyer to be activated, and it shifts the realized sales of that product if it is traded. We assume

$$\zeta_{sbg} \sim \text{Pareto}(\theta_\zeta, \zeta_{\min}), \quad \theta_\zeta > 1, \quad \zeta_{\min} = \frac{\theta_\zeta - 1}{\theta_\zeta}, \quad (13)$$

so that $\mathbb{E}[\zeta_{sbg}] = 1$. Draws are i.i.d. across (s, b, g) . The rank schedule $h(g)$ continues to order products by average profitability, but realized product sets are buyer-specific and generally non-nested. A larger θ_ζ reduces match dispersion; as $\theta_\zeta \rightarrow \infty$, ζ_{sbg} converges to one and the model approaches the deterministic common-ladder case. Appendix E shows all the cost minimization problem derivation in detail.

4.3 Supplier technology, ranking, and pricing

4.3.1 Productivity ladder and variable costs

As introduced above, each supplier s draws productivity $\phi_s > 0$. Products are ranked from core ($g = 1$) to peripheral (g large) through a common increasing schedule $h(g)$:

$$\phi_{s,g} \equiv \frac{\phi_s}{h(g)}, \quad h(1) = 1, \quad h(g+1) \geq h(g). \quad (14)$$

Products farther from the core are less efficient to supply and therefore generate lower sales and profits on average.

Producing one unit of product rank g for buyer b uses a linear technology with marginal cost

$$c_{sbg} = \frac{w}{\phi_{s,g}} = \frac{w h(g)}{\phi_s}, \quad (15)$$

where $w > 0$ is the input price. Higher supplier productivity lowers marginal cost for all products,

while higher rank raises marginal cost.

4.3.2 Pricing and product-level sales

Under monopolistic competition with CES demand elasticity $\sigma > 1$, each supplier takes the buyer-level price index P_b as given and chooses prices product by product. Because demand is isoelastic, optimal pricing implies a constant markup over marginal cost:

$$p_{sbq} = \tilde{\sigma} c_{sbq}, \quad \tilde{\sigma} \equiv \frac{\sigma}{\sigma - 1} > 1. \quad (16)$$

Since ζ_{sbq} enters demand multiplicatively, the optimal price does not depend on its realization. Substituting (16) and (15) into (11) yields

$$r_{sbq}(\phi_s, \zeta_{sbq}) \equiv p_{sbq} q_{sbq} = E_b \zeta_{sbq} \left(\frac{\tilde{\sigma} \tau_{sb} w h(g)}{\phi_s P_b} \right)^{1-\sigma}. \quad (17)$$

Rearranging gives

$$r_{sbq}(\phi_s, \zeta_{sbq}) = D_b \phi_s^{\sigma-1} h(g)^{-(\sigma-1)} \zeta_{sbq}, \quad (18)$$

where

$$D_b \equiv E_b \left(\frac{P_b}{\tilde{\sigma} \tau_{sb} w} \right)^{\sigma-1}. \quad (19)$$

The object D_b summarizes buyer size through E_b , competition at buyer b through P_b , and buyer–supplier frictions through τ_{sb} . Equation (18) is the production-network analog of the [Arkolakis et al. \(2021\)](#) product-sales equation: sales rise with supplier productivity, fall with product rank through $h(g)$, and are shifted multiplicatively by the realized product–buyer match.

Under CES pricing, variable operating profits are a constant share of revenue:

$$\pi_{sbq}(\phi_s, \zeta_{sbq}) = \frac{1}{\sigma} r_{sbq}(\phi_s, \zeta_{sbq}) = \frac{D_b}{\sigma} \phi_s^{\sigma-1} h(g)^{-(\sigma-1)} \zeta_{sbq}. \quad (20)$$

Because $h(g)$ is increasing, expected sales and profits decline with rank. The realized match can nevertheless make a peripheral product relevant for one buyer and irrelevant for another.

4.4 Aggregation across products and mapping to observables

Summing (18) over the realized product set $\mathcal{G}_{sb}$ gives total sales from supplier s to buyer b :

$$t_{sb}(\phi_s, \zeta_{sb}) \equiv \sum_{g \in \mathcal{G}_{sb}} r_{sbq}(\phi_s, \zeta_{sbq}) = D_b \phi_s^{\sigma-1} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbq} h(g)^{-(\sigma-1)}. \quad (21)$$

Define the buyer-specific firm efficiency index by

$$H_{sb}(\mathcal{G}_{sb}, \zeta_{sb})^{-(\sigma-1)} \equiv \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} h(g)^{-(\sigma-1)}. \quad (22)$$

Then

$$t_{sb}(\phi_s, \zeta_{sb}) = D_b \phi_s^{\sigma-1} H_{sb}(\mathcal{G}_{sb}, \zeta_{sb})^{-(\sigma-1)}. \quad (23)$$

As in [Arkolakis et al. \(2021\)](#), H_{sb} summarizes how the composition of the realized portfolio—both ranks and matches—maps into effective sales to buyer b .

Structural equations. The structural equations mapping primitives to observables are:

- *Product-level sales.* For a traded product $g \in \mathcal{G}_{sb}$,

$$\log r_{sbg} = \log D_b + (\sigma - 1) \log \phi_s - (\sigma - 1) \log h(g) + \log \zeta_{sbg}. \quad (24)$$

Under $h(g) = g^\alpha$, the conditional rank elasticity of sales is $-\alpha(\sigma - 1)$.

- *Total link sales.* t_{sb} is given by [\(23\)](#).
- *Firm total sales.* Let

$$\mathcal{B}_s \subseteq \mathcal{B} \quad (25)$$

denote the active buyer set of supplier s . It is endogenous and determined by the link-activation rule in [Section 4.8.1](#). Firm s 's total sales are

$$T_s \equiv \sum_{b \in \mathcal{B}_s} t_{sb}. \quad (26)$$

4.5 Fixed costs

4.5.1 Firm-level product adoption cost

Supplier s chooses a portfolio of G_s developed products and pays

$$F^P(G_s) \equiv \sum_{g=1}^{G_s} f^P(g), \quad f^P(g) \geq 0. \quad (27)$$

Adding product g requires paying the incremental firm-level development cost $f^P(g)$. This cost is incurred once by the supplier and is not paid separately for each buyer.

4.5.2 Buyer-specific link setup cost

To serve buyer b , supplier s must incur

$$f_{sb}^L \geq 0, \quad (28)$$

which captures search, contracting, onboarding, logistics, certification, and other relationship-specific investments. This cost is paid once per active relationship.

4.5.3 Buyer-specific product activation cost

A product developed at the firm level may require buyer-specific customization, packaging, certification, or listing before it can be sold to buyer b . Activating developed product g on link (s, b) costs $f_{sb}^A(g) \geq 0$. For any realized product set $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$, total activation costs are

$$F_{sb}^A(\mathcal{G}_{sb}) \equiv \sum_{g \in \mathcal{G}_{sb}} f_{sb}^A(g), \quad F_{sb}^A(\emptyset) = 0. \quad (29)$$

Unlike the deterministic cutoff formulation, $\mathcal{G}_{sb}$ need not be a prefix of the supplier's product ladder.

4.6 Buyer-level primitives, buyer counts, and match heterogeneity

The model contains buyer characteristics, pair-level cost heterogeneity, and product–buyer match heterogeneity. We distinguish these objects before stating the parametric schedules.

4.6.1 Buyer quality versus buyer counts

Buyer quality D_b . The shifter D_b , defined in (19), summarizes buyer expenditure and the competitive environment at buyer b . A larger D_b raises the variable profit generated by every potential product sold to that buyer.

In the quantitative implementation, we discipline buyer-level heterogeneity using measures of buyer scale constructed independently of the focal supplier, including the buyer's total purchases and total sales. These variables provide empirical proxies for the expenditure and market-size components of (D_b) . We additionally use the relationship between these buyer characteristics and supplier–buyer link sales to distinguish persistent buyer-level profitability from pair-specific compatibility, which is captured separately by (η_{sb}) . Section 5 describes this empirical mapping and the associated calibration targets.

Active buyer set $\mathcal{B}_s$ and buyer count N_s . The set $\mathcal{B}_s \subseteq \mathcal{B}$ contains buyers for which supplier s activates the link in Stage 2. The number of active buyers is

$$N_s \equiv |\mathcal{B}_s|. \quad (30)$$

Both $\mathcal{B}_s$ and N_s are endogenous outcomes.

Raw buyer-base aggregator Λ_s . A useful empirical summary of the supplier's buyer base is

$$\Lambda_s \equiv \sum_{b \in \mathcal{B}_s} D_b. \quad (31)$$

It decomposes as

$$\Lambda_s = N_s \bar{D}_s, \quad \bar{D}_s \equiv \frac{1}{N_s} \sum_{b \in \mathcal{B}_s} D_b. \quad (32)$$

Thus Λ_s separates buyer breadth from average buyer quality. In the matching formulation, however, the exact Stage 1 sufficient statistic is the sum of expected product surpluses, introduced below, rather than Λ_s alone. This distinction matters because match probabilities are nonlinear in buyer profitability and activation costs.

4.6.2 Pair-level cost heterogeneity

We decompose link-level costs as

$$f_{sb}^L = \bar{f}_b^L \eta_{sb}, \quad (33)$$

$$f_{sb,0}^A = \bar{f}_b^A \eta_{sb}, \quad (34)$$

where $\bar{f}_b^L, \bar{f}_b^A > 0$ are buyer attributes and $\eta_{sb} > 0$ is an idiosyncratic supplier–buyer compatibility-cost shock with $\mathbb{E}[\eta_{sb}] = 1$. The common factor η_{sb} allows the same pair-specific compatibility to affect both relationship formation and product activation costs.

The pair-level shock η_{sb} and the product-level match ζ_{sbq} play different roles. The former shifts the cost of serving a specific buyer and helps generate dispersion in link formation conditional on supplier productivity. The latter determines which developed products are relevant for that buyer and generates non-nested product sets and realized sales dispersion within links.

4.6.3 Summary of primitives

The model primitives are summarized in Table 4.

We assume

$$\phi_s \perp D_b \perp \eta_{sb} \perp \zeta_{sbq}, \quad (35)$$

Table 4: Model primitives

Class	Object	Role
Supplier	ϕ_s	Supplier productivity
Buyer	D_b	Buyer profitability shifter
Buyer	$\bar{f}_b^L$	Average link-formation difficulty
Buyer	$\bar{f}_b^A$	Average product-activation difficulty
Pair	η_{sb}	Supplier–buyer cost heterogeneity
Product–buyer	ζ_{sbg}	Match quality and realized appeal

subject to the buyer-level heterogeneity encoded in $(D_b, \bar{f}_b^L, \bar{f}_b^A)$.

Endogenous objects. The endogenous outcomes characterized in Section 4.8 are:

- $\mathcal{B}_s$ and N_s : the supplier’s active buyer set and count;
- G_s : firm-level product scope;
- $\mathcal{G}_{sb}$ and G_{sb} : the realized products and number of products traded on link (s, b) ;
- r_{sbg} , t_{sb} , and T_s : realized product-, link-, and firm-level sales.

The empirical analysis observes $(N_s, G_s, G_{sb}, T_s, t_{sb}, r_{sbg})$ in the Chilean firm-to-firm panel and treats $(\phi_s, D_b, \bar{f}_b^L, \bar{f}_b^A, \eta_{sb})$ and the distribution of ζ_{sbg} as primitives to be calibrated.

4.7 Parametric schedules

Following [Arkolakis et al. \(2021\)](#), we impose power-law rank schedules:

$$h(g) = g^\alpha, \quad \alpha \geq 0, \quad (36)$$

$$f^P(g) = f^P(1)g^\delta, \quad \delta \in \mathbb{R}, \quad (37)$$

$$f_{sb}^A(g) = f_{sb,0}^A g^\mu, \quad \mu \in \mathbb{R}. \quad (38)$$

Define

$$\kappa \equiv \mu + \alpha(\sigma - 1). \quad (39)$$

The threshold for a product–buyer match rises with rank when

$$\kappa > 0. \quad (40)$$

In the interior Pareto region, expected per-product surplus declines at rate $g^{\mu - \kappa\theta_\zeta}$. A finite firm-level scope therefore requires

$$\delta + \kappa\theta_\zeta - \mu > 0. \quad (41)$$

Because realized link-level scope is bounded by G_s , no separate unconstrained link-scope cutoff is required in the matching formulation.

4.8 The supplier's two-stage problem

The supplier's problem has a recursive two-stage structure. In Stage 1 the supplier chooses firm-level scope G_s . In Stage 2, conditional on G_s , the supplier evaluates potential links and, after the product–buyer matches are realized, activates the products whose realized operating profits cover their buyer-specific activation costs. We solve the problem by backward induction in expected value.

4.8.1 Stage 2: Link formation and the product–buyer match

Fix firm-level scope G_s . Define the buyer–supplier profitability shifter

$$A_{sb}(\phi_s) \equiv \frac{D_b}{\sigma} \phi_s^{\sigma-1}. \quad (42)$$

Developed product $g \in \{1, \dots, G_s\}$ generates variable profit $A_{sb}h(g)^{-(\sigma-1)}\zeta_{sb}g$ if traded and requires activation cost $f_{sb}^A(g)$. Define

$$\tilde{A}_{sb}(g) \equiv A_{sb}g^{-\alpha(\sigma-1)}, \quad (43)$$

$$\tilde{f}_{sb}(g) \equiv f_{sb,0}^A g^\mu, \quad (44)$$

$$z_{sb}(g) \equiv \frac{\tilde{f}_{sb}(g)}{\tilde{A}_{sb}(g)} = \frac{f_{sb,0}^A}{A_{sb}} g^\kappa. \quad (45)$$

The threshold $z_{sb}(g)$ has a simple economic interpretation: it is the minimum product–buyer match quality required for product g to be profitable on link (s, b) . A lower threshold means that the product is easier to activate for that buyer. The threshold rises with the buyer-specific activation cost and with product rank through g^κ , while it falls with the profitability of the buyer–supplier pair, A_{sb} . Thus, more productive suppliers and more profitable buyers can sustain weaker product–buyer matches, whereas more peripheral products require stronger matches to be traded.

Product g matches buyer b if and only if $\zeta_{sb}g \geq z_{sb}(g)$. The match probability and expected net surplus from product g are

$$p_{sb}(g) \equiv \Pr(\zeta \geq z_{sb}(g)) = \min \left\{ 1, \left(\frac{\zeta_{\min}}{z_{sb}(g)} \right)^{\theta_\zeta} \right\}, \quad (46)$$

$$\begin{aligned} \Psi_{sb}(g) &\equiv \mathbb{E} \left[\left(\tilde{A}_{sb}(g)\zeta - \tilde{f}_{sb}(g) \right)^+ \right] \\ &= \begin{cases} \tilde{A}_{sb}(g) - \tilde{f}_{sb}(g), & \text{if } z_{sb}(g) \leq \zeta_{\min}, \\ \frac{\tilde{f}_{sb}(g)p_{sb}(g)}{\theta_\zeta - 1}, & \text{if } z_{sb}(g) > \zeta_{\min}. \end{cases} \end{aligned} \quad (47)$$

The first branch applies when the product matches with probability one. The second follows from integrating the Pareto density above the activation threshold.

Because product-level profits and activation costs are additively separable, the expected gross value of potential link (s, b) is

$$C_{sb}(G_s; \phi_s) \equiv \sum_{g=1}^{G_s} \Psi_{sb}(g). \quad (48)$$

For compatibility with the gross-link-value notation used elsewhere in the paper, define

$$U_{sb}(G_s; \phi_s) \equiv C_{sb}(G_s; \phi_s). \quad (49)$$

The supplier activates the link ex ante when its expected product surplus covers the link cost:

$$V_{sb}(G_s; \phi_s) = [C_{sb}(G_s; \phi_s) - f_{sb}^L]^+. \quad (50)$$

The corresponding link-activation rule is

$$I_{sb}^*(G_s; \phi_s) = \mathbf{1} \{ C_{sb}(G_s; \phi_s) \geq f_{sb}^L \}. \quad (51)$$

Thus

$$\mathcal{B}_s(G_s; \phi_s) = \{ b \in \mathcal{B} : C_{sb}(G_s; \phi_s) \geq f_{sb}^L \}, \quad N_s(G_s; \phi_s) = |\mathcal{B}_s(G_s; \phi_s)|. \quad (52)$$

Since $\Psi_{sb}(g) > 0$ for every finite threshold, $C_{sb}(G_s)$ is strictly increasing in G_s . Consequently,

$$\mathcal{B}_s(G_s; \phi_s) \subseteq \mathcal{B}_s(G_s + 1; \phi_s). \quad (53)$$

A broader portfolio therefore weakly enlarges the active buyer set. Unlike the deterministic cut-off rule, there is no buyer-specific ceiling beyond which additional firm-level products cease to contribute expected surplus.

Conditional on link activation and the realized matches, the products traded on link (s, b) are

$$\mathcal{G}_{sb}(G_s) = \{g \in \{1, \dots, G_s\} : \zeta_{sb}g \geq z_{sb}(g)\}, \quad G_{sb} = \sum_{g=1}^{G_s} \mathbf{1}\{\zeta_{sb}g \geq z_{sb}(g)\}. \quad (54)$$

Their expected number is

$$\mathbb{E}[G_{sb} \mid G_s] = \sum_{g=1}^{G_s} p_{sb}(g). \quad (55)$$

The firm-level scope G_s remains an upper bound on link-level scope, but the identity and number of products traded are determined by the product–buyer matches rather than by a common prefix.

4.8.2 Stage 1: Firm-level product scope

Anticipating Stage 2, supplier s chooses G_s to maximize expected profit:

$$\Pi_s(G_s; \phi_s) = \sum_{b \in \mathcal{B}} V_{sb}(G_s; \phi_s) - F^P(G_s) = \sum_{b \in \mathcal{B}_s(G_s; \phi_s)} [C_{sb}(G_s; \phi_s) - f_{sb}^L] - F^P(G_s). \quad (56)$$

Optimal scope satisfies

$$G_s^*(\phi_s) \in \arg \max_{G_s \in \{0, 1, \dots, |\Omega|\}} \Pi_s(G_s; \phi_s). \quad (57)$$

Define

$$\Delta \Pi_s(g; \phi_s) \equiv \Pi_s(g; \phi_s) - \Pi_s(g-1; \phi_s). \quad (58)$$

The marginal value of product g decomposes into intensive and extensive channels:

$$\begin{aligned} \Delta \Pi_s(g; \phi_s) = & \underbrace{\sum_{b \in \mathcal{B}_s(g-1; \phi_s)} \Psi_{sb}(g)}_{\text{intensive: deepen active links}} \\ & + \underbrace{\sum_{b \in \mathcal{B}_s(g; \phi_s) \setminus \mathcal{B}_s(g-1; \phi_s)} [C_{sb}(g; \phi_s) - f_{sb}^L] - f^P(g)}_{\text{extensive: newly activated links}}. \end{aligned} \quad (59)$$

If $\Pi_s(G_s; \phi_s)$ is single-peaked, optimal scope is the largest integer g for which $\Delta \Pi_s(g; \phi_s) \geq 0$:

$$G_s^* = \max \{g \in \{0, 1, \dots, |\Omega|\} : \Delta \Pi_s(g; \phi_s) \geq 0\}. \quad (60)$$

The intensive channel differs fundamentally from the deterministic cutoff model. There, the relevant buyer set is $\{b : \tilde{G}_{sb} \geq g\}$, which shrinks with g and can shrink rapidly when buyer-specific cutoffs are dispersed. Here, every buyer active at $g-1$ receives positive expected surplus from developed product g . Heterogeneity changes the size of $\Psi_{sb}(g)$ but does not remove active buyers

from the amortization sum.

4.8.3 Closed-form characterization of $G_s^\star$

To obtain an analytical approximation, hold the active buyer set locally fixed and omit the extensive activation jump. The marginal-product condition becomes

$$\sum_{b \in \mathcal{B}_s} \Psi_{sb}(g) \geq f^P(1)g^\delta. \quad (61)$$

For a marginal product in the interior Pareto region, (47) implies

$$\Psi_{sb}(g) = \Psi_{sb}(1)g^{\mu - \kappa\theta_\zeta}. \quad (62)$$

Define the expected-surplus buyer-base aggregator

$$\Lambda_s^\Psi \equiv \sum_{b \in \mathcal{B}_s} \Psi_{sb}(1) = N_s \bar{\Psi}_s, \quad \bar{\Psi}_s \equiv \frac{1}{N_s} \sum_{b \in \mathcal{B}_s} \Psi_{sb}(1). \quad (63)$$

The power-law last-product condition is therefore

$$\Lambda_s^\Psi g^{\mu - \kappa\theta_\zeta} \geq f^P(1)g^\delta. \quad (64)$$

The approximate closed-form scope is

$$G_s^\star \approx \left[\frac{\Lambda_s^\Psi}{f^P(1)} \right]^{\frac{1}{\delta + \kappa\theta_\zeta - \mu}}. \quad (65)$$

In the interior Pareto region, the rank-one surplus is

$$\Psi_{sb}(1) = \frac{\zeta_{\min}^{\theta_\zeta}}{\theta_\zeta - 1} A_{sb}^{\theta_\zeta} (f_{sb,0}^A)^{1 - \theta_\zeta}. \quad (66)$$

Using (42), define the match-adjusted buyer-base aggregator

$$\Lambda_s^M \equiv \sum_{b \in \mathcal{B}_s} D_b^{\theta_\zeta} (f_{sb,0}^A)^{1 - \theta_\zeta}. \quad (67)$$

Then

$$\Lambda_s^\Psi = \frac{\zeta_{\min}^{\theta_\zeta}}{(\theta_\zeta - 1)\sigma^{\theta_\zeta}} \phi_s^{\theta_\zeta(\sigma - 1)} \Lambda_s^M, \quad (68)$$

and therefore

$$G_s^* \propto \left[\frac{\phi_s^{\theta_\zeta(\sigma-1)} \Lambda_s^M}{f^P(1)} \right]^{\frac{1}{\delta + \kappa \theta_\zeta - \mu}}. \quad (69)$$

This expression preserves the amortization mechanism: the firm-level development cost is paid once, while the expected return is summed across all active buyers. It also shows that, under Pareto matching, the exact quality aggregator is match-adjusted rather than simply $\Lambda_s = \sum_b D_b$. When activation costs are homogeneous, the distinction reduces to replacing the sum of D_b with a power mean of buyer quality.

4.8.4 Joint determination and numerical solution

The equilibrium objects $(G_s^*, \mathcal{B}_s^*)$ satisfy

$$\mathcal{B}_s^* = \{b \in \mathcal{B} : C_{sb}(G_s^*; \phi_s) \geq f_{sb}^L\}, \quad (70)$$

$$G_s^* \in \arg \max_{G_s \in \{0, 1, \dots, |\Omega|\}} \left\{ \sum_{b \in \mathcal{B}} [C_{sb}(G_s; \phi_s) - f_{sb}^L]^+ - F^P(G_s) \right\}. \quad (71)$$

These equations jointly characterize buyer breadth and product scope, but they do not require treating the active buyer set as fixed when solving the exact problem: the objective in (71) already embeds the Stage 2 activation rule.

Existence and uniqueness. Existence follows immediately because G_s is chosen from a finite set and the Stage 2 policy is well defined for every candidate G_s . Uniqueness is not guaranteed without additional restrictions on the discrete profit sequence. When $\Pi_s(G_s; \phi_s)$ is single-peaked, the last-product rule (60) delivers a unique optimum up to ties.

Exact numerical solution. For each supplier and each candidate $G \in \{0, 1, \dots, |\Omega|\}$, compute $\Psi_{sb}(g)$, $C_{sb}(G)$, $V_{sb}(G)$, and $\Pi_s(G)$. The exact solution is the value of G that maximizes $\Pi_s(G)$. The closed form (65) can be used as a starting value for a local search when $|\Omega|$ is large, but all reported numerical solutions can be verified against the exact discrete objective.

4.9 Properties of the equilibrium

4.9.1 Two-way complementarity

The matching model generates complementarity in both directions: more buyers imply weakly more products, and more products imply weakly more buyers.

More buyers imply weakly more products. Partition buyers into types $k \in \{1, \dots, K\}$ and let $\mathcal{B}_{s,k}$ be the set of active type- k buyers, with

$$N_{s,k} \equiv |\mathcal{B}_{s,k}|, \quad N_s = \sum_{k=1}^K N_{s,k}. \quad (72)$$

Let $\bar{\Psi}_{g,k}(\phi_s) \geq 0$ denote the expected surplus from product g for an active buyer of type k . Holding the active set locally fixed and abstracting from activation jumps, the adoption condition is

$$\sum_{k=1}^K N_{s,k} \bar{\Psi}_{g,k}(\phi_s) \geq f^P(g). \quad (73)$$

[Monotonicity in buyers conditional on productivity] Fix supplier productivity ϕ_s . Suppose $\bar{\Psi}_{g,k}(\phi_s) \geq 0$ for every buyer type k and product rank g , and suppose the adoption rule is characterized by (73). Then optimal firm-level scope is weakly increasing in every $N_{s,k}$.

Proof. Increasing any $N_{s,k}$ weakly raises the left-hand side of (73) at every rank g and leaves the firm-level development cost unchanged. Hence every rank profitable before the increase remains profitable afterward, so the largest adopted rank cannot fall. $\square$

The exact condition (59) reinforces this result: an additional active buyer adds a nonnegative intensive surplus and may also raise the value of future link activation.

More products imply weakly more buyers. For every potential buyer b ,

$$C_{sb}(G_s + 1; \phi_s) = C_{sb}(G_s; \phi_s) + \Psi_{sb}(G_s + 1) \geq C_{sb}(G_s; \phi_s). \quad (74)$$

Therefore,

$$I_{sb}^*(G_s + 1; \phi_s) \geq I_{sb}^*(G_s; \phi_s), \quad (75)$$

and the active buyer set satisfies the nesting property in (53).

Joint determination. More buyers increase the number of relationships over which the supplier can amortize firm-level product-development costs. More developed products raise the expected value of each potential link and can activate additional buyers. These two forces jointly generate the positive co-movement between buyer breadth and product scope observed in the data.

4.9.2 Marginal-link representation

For a fixed G_s , define the expected net surplus of potential link (s, b) :

$$S_{sb}(G_s; \phi_s) \equiv C_{sb}(G_s; \phi_s) - f_{sb}^L. \quad (76)$$

Order potential buyers by $S_{sb}(G_s; \phi_s)$ from highest to lowest and let $b(n; G_s)$ denote the buyer in position n . The active buyer count is the number of nonnegative link surpluses:

$$N_s(G_s; \phi_s) = \max \{n : S_{s,b(n;G_s)}(G_s; \phi_s) \geq 0\}, \quad (77)$$

with the convention that the maximum is zero when no link has nonnegative surplus. Equivalently, when an interior marginal buyer exists,

$$C_{s,b(N_s;G_s)}(G_s; \phi_s) \geq f_{s,b(N_s;G_s)}^L, \quad C_{s,b(N_s+1;G_s)}(G_s; \phi_s) < f_{s,b(N_s+1;G_s)}^L. \quad (78)$$

Since $C_{sb}(G_s; \phi_s)$ is non-decreasing in G_s for every buyer, the number of active links is weakly increasing in firm-level product scope. This is the marginal-link representation of the “more products $\Rightarrow$ more buyers” channel.

5 Model Calibration

5.1 Calibration strategy and empirical targets

We calibrate the model to the cross-sectional distribution of manufacturing suppliers in 2017. We use 2017 as the baseline year because electronic invoicing was mandatory by that year and because it provides a common cross section in which buyer relationships, product scope, and supplier sales can all be measured consistently. Using 2017 also allows us to avoid contamination from large aggregate shocks that affected the Chilean economy in subsequent years, including the social unrest that began in late 2019 and the COVID-19 pandemic. These events generated unusually large disruptions to firm sales, buyer–supplier relationships, and production decisions that are outside the mechanisms captured by our static model.

The calibration is designed to discipline the joint distribution of the two extensive margins at the center of the model, buyer base and product scope, as well as the allocation of products and sales across those margins.

The quantitative exercise compares two versions of the model. M1 is the deterministic rank model, in which all buyers draw from the same ordered product hierarchy and therefore receive the first subset of the supplier’s product ranking. M2 is the matching model developed in Section 4, in which the product–buyer shock ζ_{sbj} generates buyer-specific, non-nested product sets. The two models share the same basic productivity, buyer, and cost structure but differ in how the supplier’s developed product portfolio maps into the products traded with each buyer. This comparison allows us to assess which features of the data can be reproduced by the deterministic rank structure alone and which require buyer-specific product matching.

Calibration method. Let Θ_m denote the vector of parameters calibrated in model $m \in \{\text{M1}, \text{M2}\}$, and let $\mathcal{M}_m$ denote the corresponding set of targeted moments. We estimate Θ_m by simulated method of moments (SMM), minimizing an equally weighted sum of squared relative deviations between simulated and empirical moments:

$$\hat{\Theta}_m = \arg \min_{\Theta_m} \sum_{j \in \mathcal{M}_m} \left(\frac{m_j^{\text{model}}(\Theta_m) - m_j^{\text{data}}}{m_j^{\text{data}}} \right)^2. \quad (79)$$

Using relative deviations places moments measured in products, buyers, slopes, and shares on a comparable scale.

Table 6 summarizes the empirical moments and their model counterparts. It reports the moments used in the SMM objective, additional untargeted moments used to evaluate model fit, and model predictions for objects for which we do not yet have a direct empirical counterpart. M1 is calibrated to nine moments. M2 targets the same moments and additionally uses the product-specialization ratio and within-link rank alignment to discipline the product–buyer matching mechanism, for a total of eleven targeted moments. Moments that are not used in the objective function are reported as validation rather than being mechanically fit.

The simulated moments are constructed from realized trade using definitions that mirror those in the data. In particular, a developed product counts toward observed firm scope only if it is sold to at least one buyer, and a buyer counts toward observed buyer reach only if positive trade occurs on the supplier–buyer relationship. This distinction is largely immaterial in the deterministic rank model but matters in the matching model, where a developed product may fail to match any buyer and different buyers may trade different subsets of the supplier’s portfolio.

Common empirical targets. The first group of moments in Table 6 disciplines the distribution of product scope and buyer reach. We target the median, mean, and standard deviation of the number of products, G_s , together with the median and mean number of buyers, N_s . In the 2017 manufacturing sample, multiproduct suppliers sell a median of 9 and an average of 16.1 products, while serving a median of 9 and an average of 68.9 buyers. The large gaps between the means and medians are important because they require the model to reproduce not only the typical supplier but also the strong right skewness in both extensive margins.

We additionally target the median and mean number of buyers per product, N_s/G_s . These moments discipline the joint behavior of buyer reach and product scope rather than either marginal distribution separately. A model could reproduce the average number of buyers and products while generating the wrong way in which the two margins are combined within firms. Buyers per product helps rule out this possibility.

Sales concentration provides a further restriction on the composition of the buyer network. We target the mean buyer-concentration ratio, defined as the share of supplier sales accounted for by its largest buyer. This moment is informative about buyer-quality heterogeneity, relationship costs, and selection into the supplier’s active buyer set.

Finally, we target the conditional relationship between buyer breadth and product scope. Its empirical counterpart is the supplier-fixed-effect specification in Table 2, in which the coefficient on $\log N_{st}$ is approximately 0.13. Because the model is static, it has no within-supplier time variation and therefore cannot reproduce the empirical fixed-effect regression literally. We construct its closest model counterpart by estimating

$$\log G_s = \beta^{\text{model}} \log N_s + \Gamma(\phi_s) + \varepsilon_s, \quad (80)$$

where $\Gamma(\phi_s)$ denotes a flexible set of productivity-bin fixed effects. The resulting coefficient captures the relationship between buyer reach and product scope among suppliers with similar productivity. We use this moment to discipline the strength of the buyer–product co-movement that is central to the amortization mechanism.

Moments specific to product–buyer matching. M2 uses two additional moments to discipline how the supplier’s firm-level product hierarchy maps into individual buyer relationships. The first is the mean product-specialization ratio, defined as the share of supplier sales accounted for by its largest product. The second is the within-link rank-alignment coefficient documented in Table 3. In the data, the relationship between a product’s importance in the supplier’s overall portfolio and its importance within an individual supplier–buyer link is positive but far from one, with a baseline coefficient of approximately 0.27.

This moment is particularly informative about the dispersion of ζ_{sbq} . A larger θ_ζ implies less dispersed product–buyer matches and therefore a closer correspondence between the firm-level and within-link product hierarchies. A smaller θ_ζ generates greater buyer-specific variation and weaker rank alignment. Product specialization also helps distinguish systematic rank decay, governed by α , from idiosyncratic product–buyer matching, governed by θ_ζ .

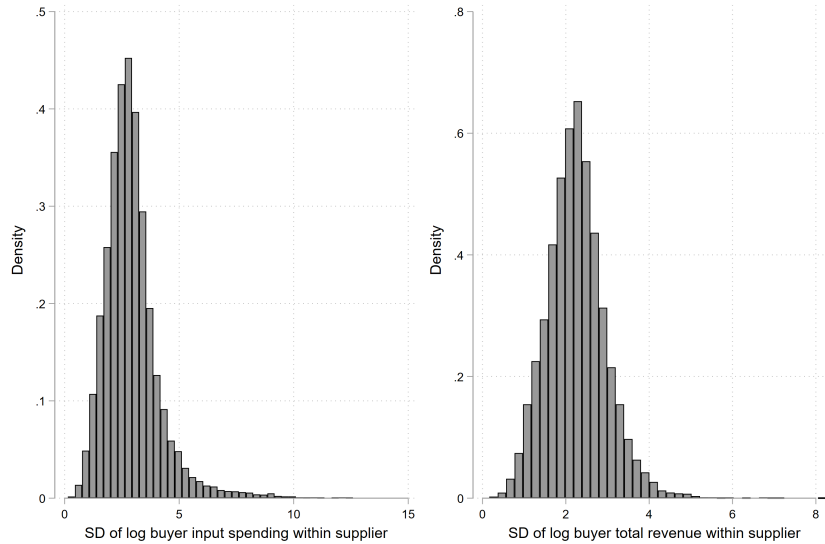
We do not use these two moments to calibrate M1. In the deterministic common-ladder structure, the mapping between firm-level and link-level product rankings is largely imposed by construction, while α is fixed externally. We therefore report M1’s performance on these dimensions as an out-of-sample diagnostic rather than forcing the model to match them.

Buyer heterogeneity and the calibration of D_b . The model also allows buyers to differ in profitability through D_b . Buyer total purchases and total sales provide observable measures of buyer scale that are constructed independently of the focal supplier. Figure 9 shows sub-

stantial heterogeneity in these measures even among buyers served by the same supplier. This within-supplier dispersion motivates allowing buyer quality to vary substantially in the quantitative model.

In the current calibration, however, we do not map these observed buyer-size measures one-for-one into D_b . Instead, D_b is represented by a reduced-form distribution whose dispersion, σ_B , is estimated jointly with the remaining parameters using the firm-level moments in Table 6. The buyer-size evidence in Figure 9 therefore provides empirical motivation for this heterogeneity, while the exact dispersion of D_b is determined jointly in the SMM calibration.

Figure 9: Within-supplier dispersion in buyer size, 2017



Notes: The figure reports the distribution across manufacturing suppliers of within-supplier heterogeneity in buyer size in 2017. For each supplier, we calculate the standard deviation across its active buyers of the logarithm of buyer input spending (left panel) and the logarithm of buyer total revenue (right panel). Buyer input spending is measured excluding purchases from the focal supplier, so that the measure does not mechanically reflect the size of the supplier–buyer relationship. Buyer total revenue is obtained from firms’ administrative tax filings. The sample is restricted to suppliers with at least five buyers and nonmissing information for the corresponding buyer-size measure. Each observation in the histograms is one supplier, and the vertical axes report density.

5.2 Parameter discipline

Table 5 reports the parameter values, their sources, and the moments that provide the main discipline for each free parameter. Table 6 reports the resulting fit of both models to the targeted and untargeted moments. Identification is joint: changing any cost or heterogeneity parameter generally affects several moments simultaneously. The moment assignments in Table 5 should therefore be interpreted as the primary sources of discipline rather than as one-to-one identification statements.

The level of the buyer-specific product-activation cost, $\bar{f}_0^A$, governs how easily developed products are activated across buyer relationships and is therefore primarily disciplined by product scope, buyers per product, and sales concentration. The link-cost level, $\bar{f}^L$, acts directly on the extensive buyer margin and is primarily disciplined by the buyer-count and buyers-per-product moments. The level of the firm-level product-adoption cost, f_0^P , determines the overall cost of expanding the supplier’s global portfolio and therefore affects both product scope and, through amortization, the buyer base that can support that scope.

The parameter δ governs how the incremental firm-level adoption cost changes as the supplier moves toward more peripheral products. It is therefore particularly important for the dispersion and upper tail of the product-scope distribution. Supplier-level cost heterogeneity, summarized by σ_c , generates common variation in relationship costs across all buyers of a supplier. It therefore creates variation in buyer reach that survives aggregation across links and, through the amortization mechanism, generates co-movement between buyer reach and product scope conditional on productivity. For this reason, the within- ϕ slope β in equation (80) is an informative moment for σ_c . Buyer-quality dispersion, σ_B , instead governs heterogeneity in the profitability of potential buyers and contributes to dispersion in both buyer reach and product scope.

In M2, θ_ζ governs product–buyer match dispersion, while α governs systematic sales decay along the supplier’s product ranking. The two parameters are estimated jointly because both affect the relationship between firm-level and link-level product importance. Within-link rank alignment is particularly informative about θ_ζ , while product specialization provides additional discipline on α and helps separate systematic rank decay from idiosyncratic matching. In M1, α is fixed externally because the deterministic product ladder provides substantially less independent variation with which to identify these two forces separately.

We also fix the activation-cost rank parameter μ across specifications. The currently available aggregate moments provide little independent information about μ once the remaining rank and cost parameters are free to adjust. A direct empirical measure of the distribution of product scope within supplier–buyer links would provide substantially sharper discipline for this parameter.

The CES elasticity, the parameters of the supplier-productivity distribution, and the remaining idiosyncratic cost and sales shocks are fixed externally or normalized rather than estimated from the same SMM moment set. Table 5 reports these values and their sources.

Targeted versus validation moments. Not every moment reported in Table 6 enters the SMM objective. We report the standard deviation of buyer counts, the standard deviation of buyers per product, and the median concentration measures as over-identification checks. These moments are strongly influenced by the upper tail of the supplier distribution and are not assigned to a separate structural parameter in the current calibration.

We also report the share of single-product suppliers as a validation moment for both models. Neither M1 nor M2 contains a separate force designed to generate the empirical mass at one product, so targeting this share would require the calibration to use the existing mechanisms to fit an object they were not designed to explain. Finally, the mean and standard deviation of products per link are reported as model predictions because they currently have no direct data counterpart in the calibration. The comparison of targeted moments, untargeted moments, and model-only predictions in Table 6 therefore provides a transparent assessment of both model fit and the dimensions along which additional structure or empirical discipline may be useful.

Table 5: Full parameter list, calibration, and source across models

Parameter	M1	M2	Source	Identified / set by
<i>Fixed or normalized parameters</i>				
σ (CES elasticity)	4.0	4.0	Literature	Standard value; not estimated.
θ_ϕ (productivity tail)	4.5	4.5	Literature	Standard value; not estimated.
$\phi_{\min}$ (productivity floor)	1.0	1.0	Normalization	Units of productivity.
σ_η (pair-cost shock)	0.7	0.7	Assumption	Link-level cost heterogeneity; not fit to aggregate firm moments.
σ_ξ (appeal shock)	0.8	0.8	Assumption	Ex-post dispersion in realized link sales.
σ_{fL}, σ_{fA} (buyer-cost draws)	0.3	0.3	Assumption	Cross-buyer dispersion in link and activation costs.
μ (activation convexity)	1.00	1.00	Assumption	Not identified by the reported moments. Working on link-level scope measure. AGM's analog is -1.17 .
<i>Calibrated and model-specific parameters</i>				
$\bar{f}_0^A$ (activation cost)	0.403	0.002	Data (SMM)	Level of the per-product, per-buyer activation cost: sets how many products a link carries and hence firm scope. Matched: Mean G , Buyer concentration ratio (mean), Buyers per product (mean).
$\bar{f}^L$ (link cost)	90.28	349.6	Data (SMM)	Level of the per-relationship cost: sets the extensive margin of buyers. Matched: Median N (buyers), Buyers per product (med.), Buyers per product (mean).
f_0^P (adoption cost)	0.052	0.013	Data (SMM)	Level of the firm-level adoption cost: shifts scope and, through amortization, the buyer count that supports it. Matched: Mean N , Buyers per product (mean), Mean G .
δ (adoption convexity)	0.717	4.241	Data (SMM)	Curvature of the adoption cost: sets the upper tail and the mean-median gap of the scope distribution. Matched: SD of G , Mean G , Median G (products).
θ_ζ (match dispersion)	–	1.230	M1 imposed; M2 SMM	Dispersion of the product-buyer match: sets how tightly the firm-level product hierarchy predicts the within-link one. Matched: Within-link rank alignment, Product specialization ratio (mean), Buyer concentration ratio (mean).
σ_c (capability dispersion)	0.719	0.084	Data (SMM)	Supplier-level cost dispersion: generates dispersion in buyer base. Matched: Within- ϕ amortization slope β , Median N , Mean N , Buyers per product (mean).
σ_B (buyer-quality dispersion)	1.009	1.811	Data (SMM)	Dispersion of buyer quality: spreads the scope and buyer distributions across suppliers. Matched: Median G (products), SD of G , Mean N .
α (rank decay)	0.300	0.044	M1 imposed; M2 SMM	M1: fixed externally. M2: systematic rate at which sales decline across product ranks. Jointly with θ_ζ , disciplined by Product specialization ratio (mean), Within-link rank alignment and Mean N .

Notes: Each model is calibrated on its own moment set: a parameter's identifying moments (last column) are targeted in every model in which that parameter is free, and only there. M1 imposes α and μ externally and has neither a match shock nor a menu cost, so it does not target the product-specialization ratio, the within-link rank alignment, or the single-product share; those are reported for M1 as validation (Table 6). Every moment named in this column appears as a row of Table 6.

σ_B does not follow mechanically from the buyer-side objects in equation (19): the model uses a reduced-form distribution of buyer quality in place of the composite E_b/P_b and τ_{sb} heterogeneity, so σ_B replaces that source rather than being derived from separate estimates.

Table 6: Identifying moments: data vs. model

Moment	Data	M1 ladder	M2 matching
<i>Product scope</i>			
Median G (products)	9.000	12.00	8.000
Mean G	16.06	14.96	9.694
SD of G	20.95	11.52	5.273
<i>Buyer base</i>			
Median N (buyers)	9.000	9.000	7.000
Mean N	68.90	87.01	66.52
<i>Link breadth</i>			
Buyers per product (med.)	1.000	0.727	0.875
Buyers per product (mean)	3.749	2.579	3.144
<i>Sales composition and product hierarchy</i>			
Product specialization ratio (mean)	0.541	0.314	0.343
Buyer concentration ratio (mean)	0.514	0.356	0.470
Within-link rank alignment (Table 3)	0.266	0.910	0.143
<i>Extensive margin and complementarity</i>			
Single-product share	0.138	0.000	0.000
Within- ϕ amortization slope β	0.130	0.185	0.19
<i>Reported, targeted by no model</i>			
SD of N	784.6	376.8	273.3
Buyers per product (SD)	22.77	5.389	7.537
Product specialization ratio (med.)	0.512	0.291	0.305
Buyer concentration ratio (med.)	0.475	0.249	0.412
<i>Model prediction with no data counterpart</i>			
Products per link (mean G_{sb})	n.a.	16.32	26.72
Products per link (SD)	n.a.	12.43	11.73
Targeted moments		9	11
Free parameters		6	8
SMM criterion		0.833	1.488

Notes: Each row reports an empirical moment and its counterpart in M1 and M2. M1 targets nine moments, while M2 targets eleven; the two additional M2 targets are the mean product-specialization ratio and within-link rank alignment. The single-product share and the moments reported under “Reported, targeted by no model” are validation moments and do not enter the SMM objective. Product specialization is the share of supplier sales accounted for by its largest product. Products per link are currently reported only as model predictions because they have no direct empirical target in the calibration. The SMM criterion reports the value of equation (79) evaluated at the calibrated parameters.

6 Quantitative Results

This section presents the quantitative behavior of the calibrated matching model (M2 of Section 5). We solve the model at the simulated-method-of-moments optimum on an economy of $S = 16,000$ suppliers facing a potential-buyer pool of $B = 8,000$, with the product ceiling set at $G_{\max} = 300$ so that no firm is bound by the enumeration grid. Productivity is Pareto ($\theta_\phi = 4.5$, $\phi_{\min} = 1$), the CES elasticity is $\sigma = 4$, and the remaining primitives take the calibrated values in Table 5: an activation cost $\bar{f}_0^A = 0.002$, a link cost $\bar{f}^L = 349.6$, a firm-level adoption cost with base $f_0^P = 0.013$ and convexity $\delta = 4.241$, a product-buyer match dispersion $\theta_\zeta = 1.230$, a supplier-level capability dispersion $\sigma_c = 0.084$, a buyer-quality dispersion $\sigma_B = 1.811$, and a rank-decay parameter $\alpha = 0.044$. Every firm-level moment is computed from realized transactions: a supplier's products are the distinct codes it actually sells and its buyers are those that actually receive at least one product.

The section is organized around the model's central testable prediction: conditional on productivity, firms with more buyers should have more products, and then inspects the model's other objects to assess where the calibration succeeds and where it is strained.

6.1 The central prediction: within-productivity co-movement of scope and buyer base

Figure 10 plots the relationship between product scope G_s and buyer base n_s in two ways. Panel (a) is a binscatter of $\log G_s$ against $\log n_s$ computed separately within each productivity quintile: firms are sorted into five bins by ϕ_s , and within each bin their scope is averaged across cells of buyer count. Panel (b) strips out productivity entirely by residualizing both $\log G_s$ and $\log n_s$ on twenty $\log \phi_s$ bins and plotting the residuals against one another, so that what remains is the co-movement of product scope and buyer breadth among firms of essentially identical productivity.

In panel (a) the five quintile lines are ordered from lower-left (low ϕ) to upper-right (high ϕ), but each individual line slopes up: within a productivity band, firms that reach more buyers carry more products. Panel (b) makes this the whole picture. Once productivity sorting is removed, the residual relationship is clean and positive with a slope of $\beta = 0.194$. Table 7 traces the same object through regressions: the unconditional elasticity of scope to buyer count is 0.229 (column 1), falls to 0.181 once productivity enters linearly (column 3), and is 0.194 with twenty productivity-bin fixed effects (column 4). Column (3), which conditions on productivity linearly, is the specification that maps most directly to column (6) of the empirical Fact 3 regression (Table 2): the model's 0.181 is of the same order and sign as the data's 0.130, somewhat above it.

Adding the capability shock as a direct control (column 5) barely moves the model coefficient, from 0.181 to 0.185, because the calibrated dispersion of c_s is small, so the within- ϕ relationship

is not an artifact of unobserved capability.

The variance decomposition in Table 8 sharpens the point: productivity alone explains 91.5% of the variance of $\log G_s$, and of the remaining within- ϕ variation, buyer count absorbs nearly three quarters (72.2%). Table 9 shows the within-quintile correlation between $\log G_s$ and $\log n_s$ rising from 0.83 in the lowest productivity quintile to 0.99 in the highest.

This is the amortization mechanism made visible and is the single prediction the model exists to deliver. The firm-level adoption cost $F^P(G_s)$ is paid once and its return accrues across the buyer base, so a firm with more buyers earns a higher summed return to each developed product and optimally develops more of them, even holding productivity fixed. The unconditional slope alone would be uninformative, because more productive firms mechanically have both more buyers and more products; it is the within- ϕ slope, isolated in panel (b), that identifies amortization as opposed to productivity sorting.

Match with the data. The empirical counterpart is the Fact 3 regression of Table 2, column 6: a within-supplier elasticity of product scope to buyer count of 0.130, estimated with supplier and year fixed effects. The model’s within- ϕ slope is of the same order and sign, 0.194 with productivity-bin fixed effects (column 4) and 0.181 with a linear productivity control (column 3), running somewhat above the data value. Because the matching model’s simulated moments carry sampling noise, β varies across simulation seeds between roughly 0.15 and 0.22. The calibration reproduces the sign, the economic magnitude, and the robustness of the relationship to controlling for productivity, while modestly overstating its strength. The mapping is not exact, the data estimand is identified from variation within a supplier over time and the model is static, so the productivity controls stand in for the supplier fixed effect, but the amortization channel the regression isolates is clearly present and economically large.⁸

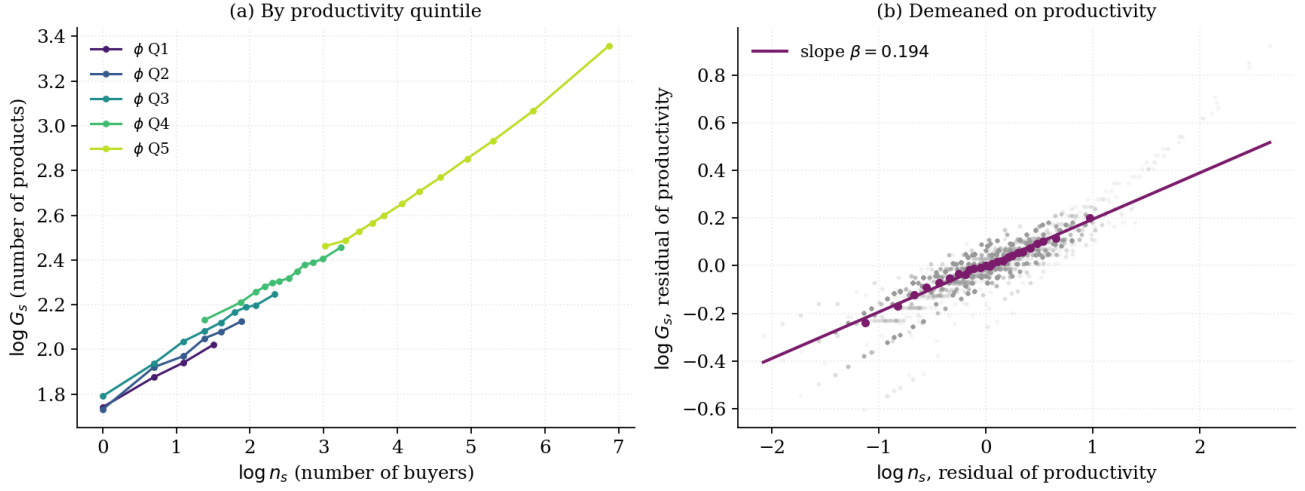
6.2 Marginal distributions and calibration fit

Figure 11 shows three panels, which are histograms of the firm-level product scope G_s (left), the number of buyers N_s (middle), and log total sales $\log T_s$ (right), across the simulated population of suppliers. The dashed lines mark the medians.

Product scope is right-skewed with a median of 11; the buyer count is strongly right-skewed with a median near 10 and a long upper tail reaching several hundred buyers; and total sales are approximately log-normal. Corner masses at the enumeration ceilings are negligible, so the optima are interior and the calibration is not distorted by the grid.

⁸The within- ϕ slope is the most seed-sensitive moment of the calibrated matching model: across four independent draws it takes the values 0.15–0.22 (mean ≈ 0.19). The figures and tables in this section are reported at a representative seed ($\beta = 0.194$). The deterministic-ladder model M1, by contrast, delivers $\beta = 0.185$ essentially regardless of seed.

Figure 10: Within-productivity co-movement of product scope and buyer breadth



Notes: Calibrated matching model (M2). Panel (a): binscatter of $\log G_s$ against $\log n_s$ within each productivity quintile; the within-line slope is the within- ϕ amortization channel and the ordering of the lines is productivity sorting. Panel (b): the same two variables after residualizing each on twenty $\log \phi_s$ bins, which removes the sorting; the residual slope $\beta = 0.154$ is the model analog of the within-firm elasticity of scope to buyer breadth, against an empirical target of 0.130 (Table 2, column 6).

These three distributions are the primitives of firm size in the model, the two extensive margins and their sales consequence, and the calibration must reproduce not only their central tendency but their skewness, since the paper's motivating fact is that a small number of large, broad-scope suppliers account for most activity.

Match with the data. The model reproduces the features of the firm-size distribution that the exercise targets. The medians line up closely with the 2017 manufacturing data, median scope of 11 against 9 in the data, and median buyer count of 10 against 9, and the log-normal shape of sales matches the heavy-tailed firm-size distribution in Table 1. Two dimensions are left untargeted by design and are exactly where a static, multiproduct benchmark is not meant to bind. The dispersion of scope is somewhat below the data (the standard deviation of G_s falls short of the empirical 21), and the model concentrates on multiproduct suppliers, so the 13.8% single-product mass in the data sits outside its scope. Neither bears on the mechanism at issue: both are governed by forces the benchmark deliberately leaves out, the extreme upper tail of firm size and the discrete margin at a single product, rather than by any parameter the calibration uses to fit the buyer-product co-movement. We therefore report them as boundaries of the current specification rather than as tension with the central result.

Table 7: Regressions of $\log G_s$ on $\log n_s$, $\log \phi_s$, and $\log c_s$ (calibrated M2)

	(1) $\log G_s$	(2) $\log G_s$	(3) $\log G_s$	(4) $\log G_s$	(5) $\log G_s$
$\log n_s$	0.229 (0.000)		0.181 (0.001)	0.194 (0.001)	0.185 (0.001)
$\log \phi_s$		1.489 (0.005)	0.342 (0.008)		0.315 (0.009)
$\log c_s$					0.075 (0.007)
ϕ -bin FE	No	No	No	Yes	No
R^2	0.971	0.915	0.976	0.976	0.977

Table 8: Variance decomposition of $\log G_s$ (calibrated M2)

	Share of Var($\log G_s$)
Explained by $\log \phi_s$ alone	0.915
Explained by $\log \phi_s + \log n_s$	0.976
Explained by $\log \phi_s + \log c_s$	0.919
Within- ϕ residual share	0.085
Additional share explained by $\log n_s$ as % of within- ϕ residual	0.061 72.2%

6.3 Sales across margins

Figure 12 shows three panels. Each panel scatters log total sales $\log T_s$ against one firm-level object, the buyer count $\log n_s$ (left), product scope $\log G_s$ (middle), and productivity $\log \phi_s$ (right), with points colored by productivity quintile. The informative feature is the relative tightness of each relationship and how mixed the productivity colors are within it.

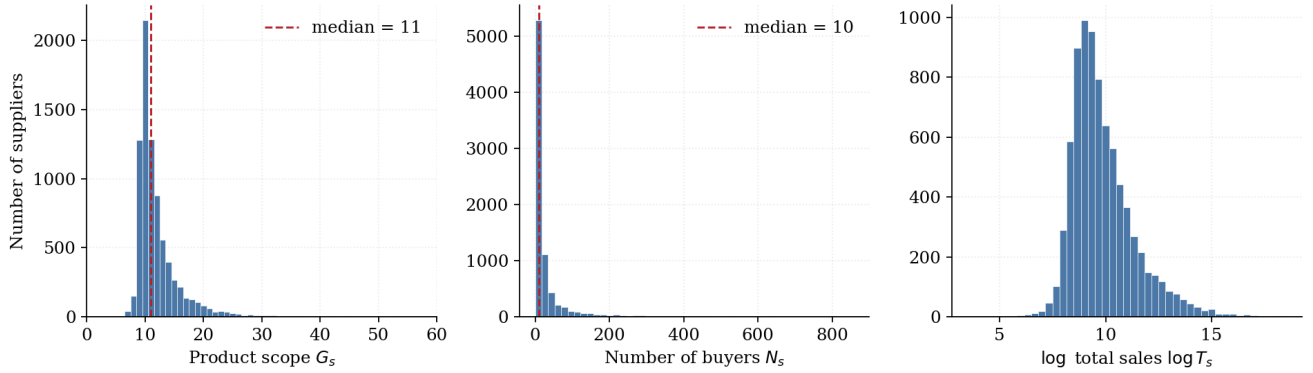
Sales rise most tightly with product scope: in the middle panel the productivity colors are nearly mixed, so once a firm's scope is known, its sales are largely pinned down. Sales rise less tightly with the buyer count: in the left panel the colors fan out, because at a given number of buyers a more productive firm earns more per relationship. The relationship with productivity is the loosest of the three, at any given productivity level, sales span a wide range, reflecting variation in the buyer base, buyer quality, and the appeal shock.

The ordering identifies G_s as the most aggregated endogenous summary of the firm: scope internalizes productivity, capability, and the buyer base simultaneously, which is why it predicts sales better than any single primitive. This is an internal-consistency check on the mechanism, which confirms that scope is an endogenous outcome that bundles the firm's advantages rather than an exogenous characteristic.

Table 9: Within- ϕ -quintile correlations between G_s and n_s (calibrated M2)

Quintile	$\rho(G_s, n_s)$	$\rho(\log G_s, \log n_s)$	# firms
Q1	0.823	0.828	1600
Q2	0.846	0.883	1600
Q3	0.835	0.888	1600
Q4	0.875	0.904	1600
Q5	0.897	0.991	1600
Pooled	0.747	0.986	8000

Figure 11: Marginal distributions of firm-level outcomes (calibrated M2)



Notes: Left: firm-level product scope G_s . Middle: number of buyers N_s (top percentile trimmed for display). Right: log total sales $\log T_s$. Medians (dashed) align with the 2017 calibration targets of 9 products and 9 buyers; the scope distribution is more concentrated than in the data, and the model generates no single-product suppliers.

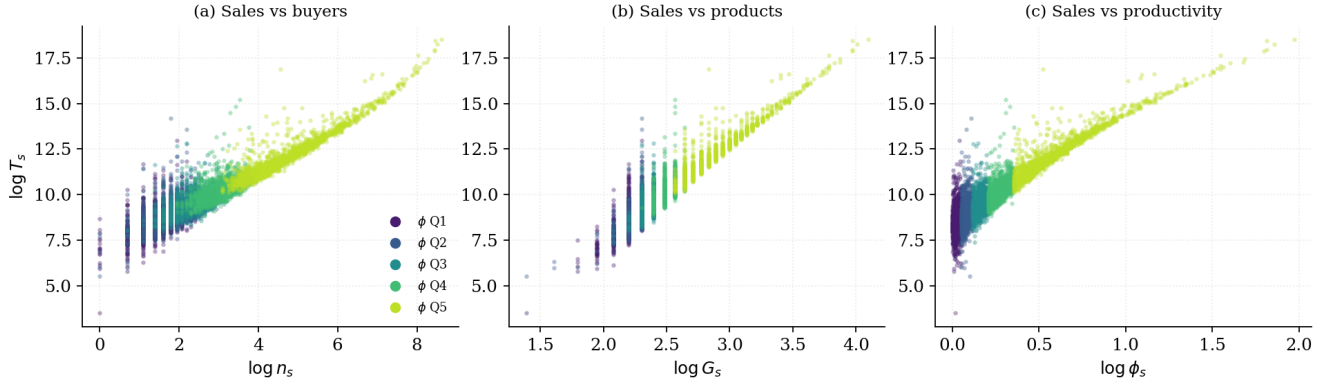
Match with the data. The pattern reproduces the cross-sectional fact in Figure 12 that larger suppliers expand along both extensive margins at once, more buyers and more products, and that firm size is associated with a denser use of the product portfolio across the buyer network. That larger firms are more productive on average, while sales also reflect scope and buyer acquisition, is exactly the loosening of the sales–productivity relationship visible in the right panel.

7 Counterfactuals

7.1 Shutting down the firm-level adoption cost

The calibrated model lets us ask what the firm-level adoption cost F^P , the source of the amortization and complementarity, contributes. By setting it to zero and re-solving the economy at otherwise unchanged parameters we discover the exercise has two payoffs: it confirms that the central within- ϕ result is amortization rather than a mechanical relationship, and it shows that the same complementarity is what shapes the firm-size distribution.

Figure 12: Total sales across margins (calibrated M2)



Notes: Log total sales $\log T_s$ against $\log n_s$ (left), $\log G_s$ (middle), and $\log \phi_s$ (right), colored by productivity quintile. Sales co-move most tightly with product scope, because G_s is the most aggregated endogenous outcome; the sales–productivity relation is loosest because sales also reflect the buyer base and demand-side heterogeneity.

Amortization is what ties product scope and the buyer base. Table 10 sets the matching model against the deterministic-ladder alternative (M1 of Section 5) with and without the adoption cost. In the matching model, removing F^P sends the within- ϕ slope from 0.194 to zero and product scope runs to the enumeration ceiling (median $11 \rightarrow 300$): with no firm-level cost nothing bounds scope, $\log G_s$ becomes constant across firms, and the slope vanishes by construction. This is the sign of amortization, the slope exists only because the once-paid cost is spread across the buyer base. The ladder behaves oppositely: its slope not only survives but rises, from 0.185 to 0.228, while scope stays bounded (median $14 \rightarrow 17$). The reason is that in the ladder, once the adoption cost is removed, firm scope is pinned by the deepest active buyer’s desired scope, an order statistic that increases mechanically with the number of buyers, so the co-movement persists with no cost to amortize. This contrast is the sharpest statement of why the matching model is the better vehicle for the mechanism: in M2 the within- ϕ co-movement is genuine amortization, verified by its disappearance when F^P is removed, whereas in M1 the same co-movement is a mechanical artifact of the ladder that no adoption cost is needed to produce. Scaling F^P around its calibrated value in M2, rather than removing it, leaves the slope essentially unchanged, between 0.15 and 0.17 for costs from a quarter to four times the calibrated level, so the slope collapses only at exactly zero.

Table 10: Mechanism diagnostic: the within- ϕ slope with and without the firm-level adoption cost

	M1 (ladder)	M2 (matching)
Within- ϕ slope β , calibrated	0.185	0.194
Within- ϕ slope β , $F^P = 0$	0.228	0.000
Median product scope, calibrated	14	9
Median product scope, $F^P = 0$	17	300

Amortization shapes the firm-size distribution. The complementarity does more than link the two extensive margins; it also concentrates sales. Because large firms have both more buyers and more products, and the two margins reinforce one another through amortization, sales pile up in the largest firms. Table 11 quantifies this by comparing the sales distribution of the full model to the $F^P = 0$ economy. Removing the adoption cost compresses the distribution: the standard deviation of log sales falls by 53% and the share of aggregate sales held by the top 1% of firms falls by 66%. The mechanism is transparent, with no firm-level cost every supplier expands to the ceiling on both margins, so firms homogenize and the cross-firm dispersion that the complementarity generated disappears. The firm-level adoption cost is therefore not only what makes the within- ϕ co-movement structural; it is also a quantitatively important source of the firm-size dispersion and sales concentration observed in the data. A milder version of the experiment that holds the marginal distributions of scope and buyer breadth fixed but decouples the two margins delivers the same message in smaller magnitude, confirming that the concentration is driven by the joint determination of the two margins rather than by either margin alone.

Table 11: The firm-level adoption cost and the sales distribution (calibrated M2)

	Full model (both mechanisms)	No F^P (adoption cost removed)
Sales dispersion, $SD(\log T_s)$	1.95	0.91 (−53%)
Top 1% sales share	0.71	0.24 (−66%)

7.2 Comparative statics in the fixed costs

Having established that the firm-level product adoption cost is what ties the two margins together, we now ask how the calibrated economy responds when the fixed costs that govern those margins are moved rather than removed. We scale each cost up and down and trace the effect on both extensive margins and on the sales distribution.

Table 12 halves and doubles the firm-level product-adoption cost F^P (the product margin) and the buyer-relationship, or matching, cost $\bar{f}^L$ (the buyer margin), holding everything else at the calibration. Three results stand out.

First, and most tellingly, each cost moves both margins. Lowering the product-adoption cost raises median scope from 9 to 10 but also raises the median buyer count from 6 to 7, because cheaper products make each buyer relationship more valuable and so worth forming; lowering the matching cost raises the median buyer count sharply, from 6 to 18, and also raises median scope from 9 to 10, because a larger buyer base raises the return to adopting products through amortization. The symmetry is the two-way complementarity made quantitative: neither fixed

cost can be moved without the other margin responding, so the product and buyer margins cannot be targeted in isolation.

Second, the two costs are nonetheless specialized levers. The matching cost $\bar{f}^L$ is the dominant instrument for the buyer margin, halving it triples the median buyer count and doubling it halves it, and correspondingly drives buyers per product, which ranges from 5.0 when matching is cheap to 1.4 when it is dear. The product-adoption cost moves scope more than it moves buyers. The matching cost is also the dominant lever for the shape of the sales distribution: cheaper matching spreads the distribution, lowering both the dispersion of log sales (from 1.94 to 1.69) and the top-1% sales share, because many more firms reach broad buyer bases; dearer matching concentrates it. In this sense the buyer margin, governed by $\bar{f}^L$, does more of the work in shaping firm-size dispersion than the product margin does, consistent with the buyer base being the object that carries the within- ϕ variation in the model.

Third, the amortization mechanism itself is robust to the level of the fixed costs. The within- ϕ slope β stays between 0.18 and 0.20 across all four perturbations, so the co-movement of scope and buyer breadth is a structural feature of the model rather than an artifact of the particular calibrated cost levels.

These comparative statics carry a direct policy reading. A subsidy that lowers the buyer-matching cost $\bar{f}^L$ raises not only the buyer base but also product scope, because the complementarity transmits the shock across margins: reducing buyer-supplier matching frictions stimulates product innovation through amortization. This is the quantitative counterpart of the policy argument in the introduction, and it is a channel that a single-margin model, in which product scope and buyer acquisition are chosen independently, cannot deliver. The third fixed cost, the per-product activation cost $\bar{f}^A$, is by contrast quantitatively inert at the firm level, which is why we vary only F^P and $\bar{f}^L$ above. Scaling $\bar{f}^A$ from half to ten times its calibrated value leaves median scope (9), the median buyer count (6), buyers per product (≈ 2.3), the dispersion of log sales (≈ 1.93), and the within- ϕ slope (0.195) essentially unchanged. The reason is that the calibrated activation cost is very small, so the product-buyer match threshold sits below the support of ζ and activation almost never binds; a large increase eventually begins to trim products from individual links, the selection-on-sets margin discussed in Section 6, but that operates strictly within links and leaves firm-level scope and the buyer base untouched. The two extensive margins are therefore shaped by the adoption and matching costs, not by the activation cost.

Shutting down one extensive margin at a time. The comparative statics before move the two margins jointly; a complementary exercise shuts each margin down entirely and asks which one the sales distribution depends on. In the products-only economy every supplier faces a single representative buyer of average quality, so the buyer margin is switched off and cross-firm sales variation comes only from productivity and product scope. In the buyers-only economy every

Table 12: Comparative statics in the two fixed costs (calibrated M2)

	Baseline	Product cost F^P		Matching cost $\bar{f}^L$	
		−50%	+100%	−50%	+100%
Median products G_s	9	10	7	10	8
Median buyers N_s	6	7	5	18	3
Buyers per product N_s/G_s	2.3	2.2	2.3	5.0	1.4
Sales dispersion $SD(\log T_s)$	1.94	1.87	1.96	1.69	2.06
Top 1% sales share	0.71	0.59	0.64	0.63	0.71
Within- ϕ slope β	0.195	0.194	0.200	0.183	0.201

supplier is restricted to a single product, so the product margin is switched off and variation comes only from productivity and the buyer base. Both are computed on the calibrated economy, holding each firm’s realized objects fixed, so productivity remains active in both and neither distribution collapses to a point.

Table 13 reports the result, and it is asymmetric. Removing the buyer margin (products only) cuts the dispersion of log sales by 37% and the top-1% sales share by a third; removing the product margin (buyers only) cuts dispersion by only 10%. Equivalently, the buyer margin on its own reproduces about 90% of the full sales dispersion while the product margin on its own reproduces only about 63%. The buyer margin, not the product margin, is doing most of the work in generating firm-size dispersion and sales concentration. This is the demand-side counterpart of the calibration result that within- ϕ dispersion in scope is carried by the buyer base: firm size in this model is primarily a story about how many, and how good, a supplier’s buyers are, with product scope amplifying rather than originating the dispersion.

Table 13: Shutting down one extensive margin at a time (calibrated M2)

	Full model (both margins)	Products only (representative buyer)	Buyers only (single product)
Sales dispersion, $SD(\log T_s)$	1.92	1.21 (−37%)	1.73 (−10%)
Top 1% sales share	0.68	0.45 (−33%)	0.51 (−25%)

7.3 A comparative-static analog of the growth decomposition

The empirical facts of Section 2 decompose realized firm growth into four buyer×product margins: old-link–old-product (OLOP), new-link–old-product (NLOP), old-link–new-product (OLNP), and new-link–new-product (NLNP). The model is static and cannot reproduce a time path, but it admits a comparative-static analog: we shock the calibrated economy, re-solve, and split each supplier’s change in sales into the same four quadrants, using its baseline buyer set and product

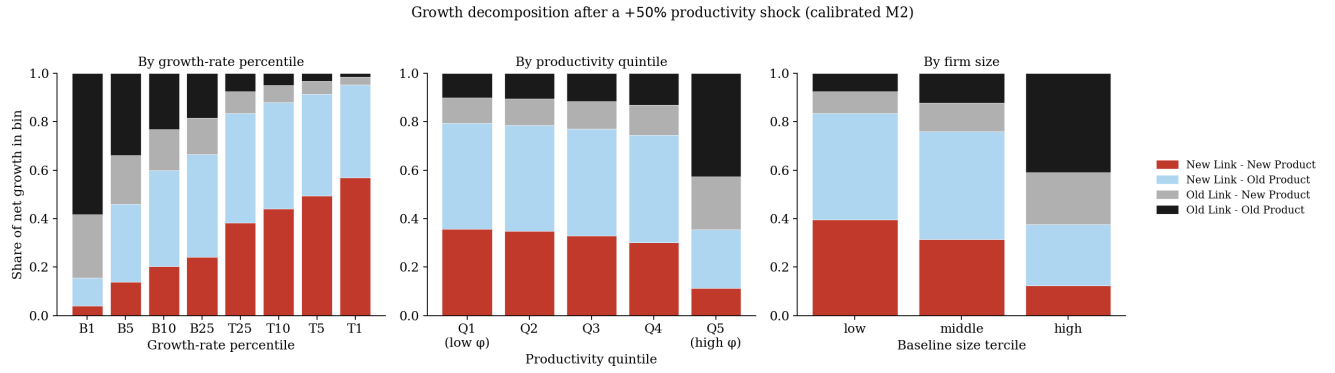
ladder as the "old" reference. We report the analog under a uniform +50% productivity shock, which raises every ϕ_s by half; a demand shock delivers the same qualitative picture.

Because a positive shock weakly enlarges every supplier's active buyer set and product ladder, the baseline sets nest inside the post-shock sets and the decomposition is exact. OLOP is the additional revenue earned on baseline buyer-product cells as their sales rise; NLOP and OLNP are the revenue on newly activated buyers (selling old products) and on newly developed products (sold to old buyers); and NLNP is the revenue on cells that pair a new buyer with a new product. Within each bin of Figure 13 we sum each component across suppliers and divide by the bin's total net growth, so the four shares sum to one.

The exercise reproduces the qualitative shape of the empirical decomposition (Figures 3 to 6) without ever having targeted it. Growth among the fastest-growing, smallest, and least-productive suppliers is dominated by the extensive margins: the top growth-rate percentile draws 57% of its net growth from new-buyer-new-product cells and a further 38% from new buyers selling existing products, and the lowest productivity quintile and smallest size tercile show the same extensive tilt. Growth among the slowest-growing, largest, and most-productive suppliers is instead intensive, concentrated in OLOP: the bottom growth percentile draws 58% of its growth from deepening existing buyer-product relationships, and the top productivity quintile and largest size tercile echo this. The gradient from OLOP to NLNP across the growth distribution, and from extensive to intensive as productivity and size rise, is exactly the pattern the data display, which we read as a validation of the model's growth mechanics rather than as a fitted result.

Two caveats fix the interpretation. The static model has no product/link exit and applies a common shock, so the bottom growth-percentile bins collect suppliers that grow least, and therefore intensively, rather than suppliers that contract, as in the data; and because a uniform shock makes every supplier grow, the growth rates themselves are not informative, only their composition is. The figure should therefore be read as a comparative-static illustration of how the composition of growth varies across the firm distribution, not as a structural reproduction of the 2017–2023 transition.

Figure 13: Growth decomposition after a +50% productivity shock (calibrated M2)



Notes: Comparative-static analog of the growth decomposition. Every supplier's productivity ϕ_s is raised by 50% and the economy re-solved; the change in each supplier's sales is split into new-link-new-product (NLNP), new-link-old-product (NLOP), old-link-new-product (OLNP), and old-link-old-product (OLOP) relative to its baseline buyer set and product ladder. Within each bin the four components are summed across suppliers and normalized to sum to one. Left: bins of the realized growth-rate distribution. Middle: baseline productivity quintiles. Right: baseline size terciles. The model is static, so the exercise captures the composition of growth, not a time path (see text).

8 Conclusion

This paper studies the joint role of buyer relationships and product scope in shaping firm size inside production networks. Using the universe of firm-to-firm transactions in Chile, we show that these are empirically distinct but closely connected margins. Buyer relationships have shorter and less persistent life cycles than supplier–product relationships, and large supplier expansions disproportionately combine new buyers with new products. More importantly for the model, buyer breadth and product scope co-move strongly within suppliers. Conditional on sales and supplier fixed effects, a 10 percent increase in the number of buyers is associated with a 1.3 percent increase in product scope. The same relationship appears at the product-entry margin: years in which a supplier serves more buyers are also years in which it introduces more products. The response is concentrated among peripheral products, consistent with buyer expansion raising the return to products near the margin of adoption.

We develop a static model in which suppliers jointly choose their firm-level product portfolio and the buyers they serve. The central mechanism is a firm-level product-adoption cost that is paid once but generates returns across multiple buyer relationships. A larger buyer base therefore raises the return to adopting additional products, while a broader product portfolio raises the value of forming additional buyer relationships. We embed this mechanism in a rank-based multiproduct structure and allow for idiosyncratic product–buyer matches. The matching structure is important in a production network because the data reject a rigid common product ladder: firm-level product importance predicts within-link importance only partially, and different buyers of the same supplier purchase different, non-nested subsets of its portfolio.

The calibrated model reproduces the positive relationship between buyer breadth and product scope conditional on productivity. The corresponding elasticity is about 0.19 in the model, compared with 0.13 in the data. Shutting down the firm-level adoption cost provides a direct test of the mechanism. In the matching model, the within-productivity relationship between buyers and products disappears when the adoption cost is removed. In the deterministic rank model, by contrast, the relationship survives because firms with more buyers mechanically have a greater chance of reaching farther down the common product ranking. This comparison shows why allowing for buyer-specific product matching matters for interpreting the observed co-movement as evidence of amortization rather than as a mechanical consequence of the product ladder.

The quantitative exercises also show that the interaction between the two margins matters for the distribution of firm sales. Changes in either the product-adoption cost or the buyer-relationship cost affect both product scope and buyer reach. In particular, lowering buyer-relationship costs expands the buyer base and, through amortization, also increases product scope. The two margins are not equally important quantitatively, however. Removing the buyer margin reduces the

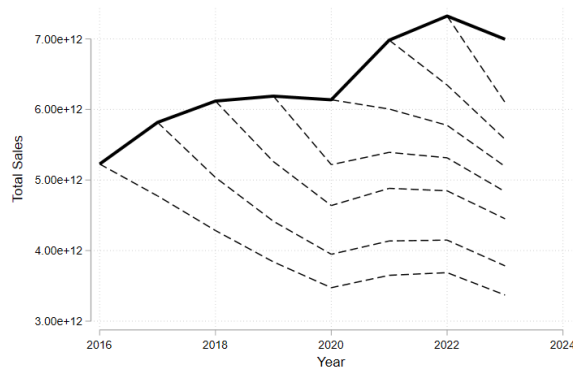
dispersion of log sales by 37 percent, compared with 10 percent when the product margin is removed. Equivalently, the buyer margin alone reproduces about 90 percent of the sales dispersion generated by the full model, while the product margin alone reproduces about 63 percent. Buyer reach is therefore the main source of firm-size dispersion in the calibrated economy, with product scope amplifying differences generated through the buyer network.

Although several of our empirical facts are measured over time, the benchmark model is static. We therefore interpret its growth exercises as comparative statics rather than as a model of firm dynamics. Nevertheless, when the calibrated economy is perturbed, the model reproduces the qualitative reallocation across old and new buyers and products observed in the data: smaller and less productive suppliers adjust relatively more through extensive margins, whereas larger and more productive suppliers rely more heavily on existing buyer–product relationships. A natural next step is to introduce buyer accumulation and product adjustment dynamically, allowing the model to speak directly to the life-cycle patterns documented in the data.

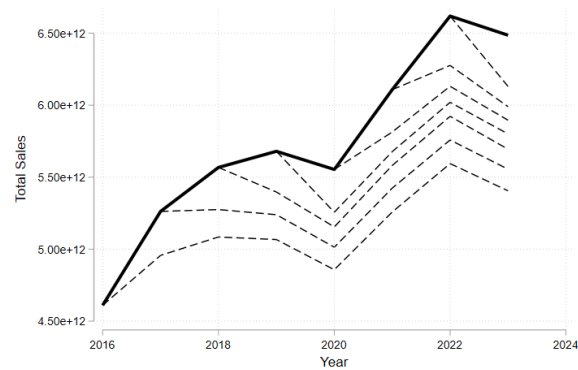
More broadly, the results show that product scope cannot be studied separately from the network of buyers over which products are sold. Policies that reduce frictions to forming supplier–buyer relationships may affect not only the number of relationships firms maintain but also their incentives to expand their product portfolios. Conversely, policies that lower the cost of product adoption can increase the value of forming new buyer relationships. Accounting for these interactions is therefore important for understanding how buyer–supplier networks and multiproduct production jointly shape the distribution of firm activity.

Appendix

A Evolution of total sales by cohort and year

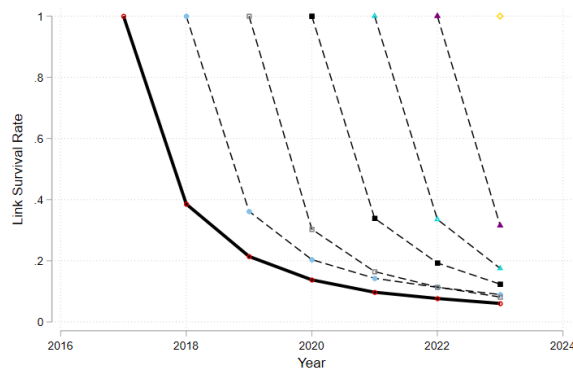


(a) Link total sales per cohort and year

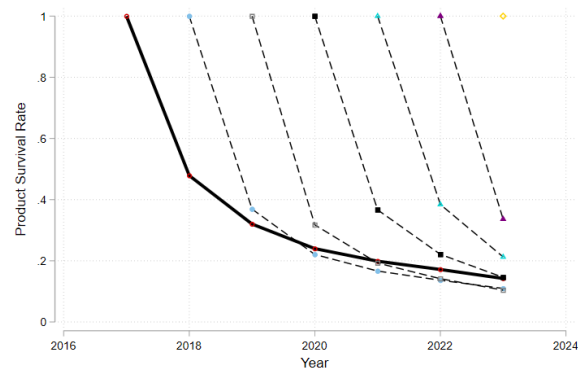


(b) Product total sales per cohort and year

B Survival rate of links and products



(a) Link survival rate



(b) Product survival rate

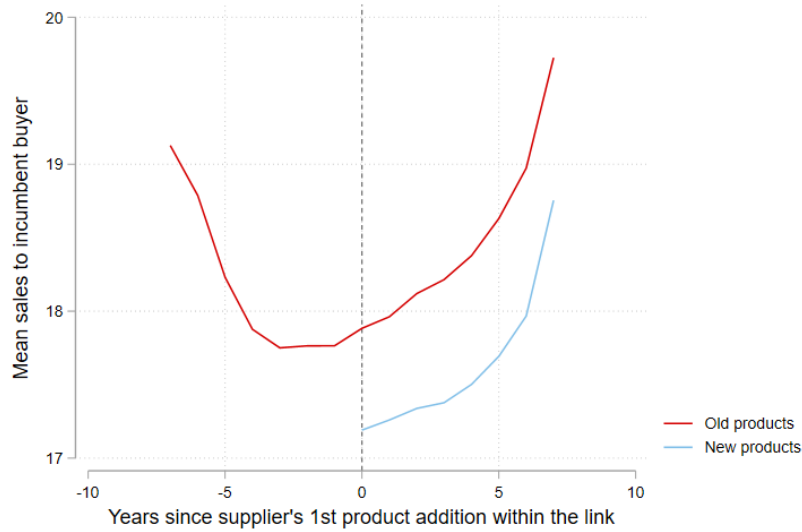
C Ruling out cannibalization

This appendix reports the two cannibalization exercises discussed in Section 4. The first examines whether introducing a new product within an existing buyer relationship reduces sales of the supplier’s existing products. Figure 16 shows no such decline. Instead, sales of old products rise after the new product is introduced, while sales of the new product grow alongside them. This rules out within-link cannibalization: expanding the product menu offered to a buyer deepens the overall relationship rather than reshuffling revenue across products. This pattern is consistent with the separability assumption maintained in the benchmark and, in the nested-CES extension of Appendix F, with the case in which the within-supplier and across-supplier elasticities are similar, $\varepsilon \approx \nu$.

The second exercise asks whether acquiring a large new buyer diverts sales away from existing buyer relationships. Figure 17 shows that sales to incumbent buyers do not decline after the new buyer is added. Incumbent sales recover around the event and return to their baseline level, with no systematic post-event reduction. We therefore find no evidence of link-level cannibalization.

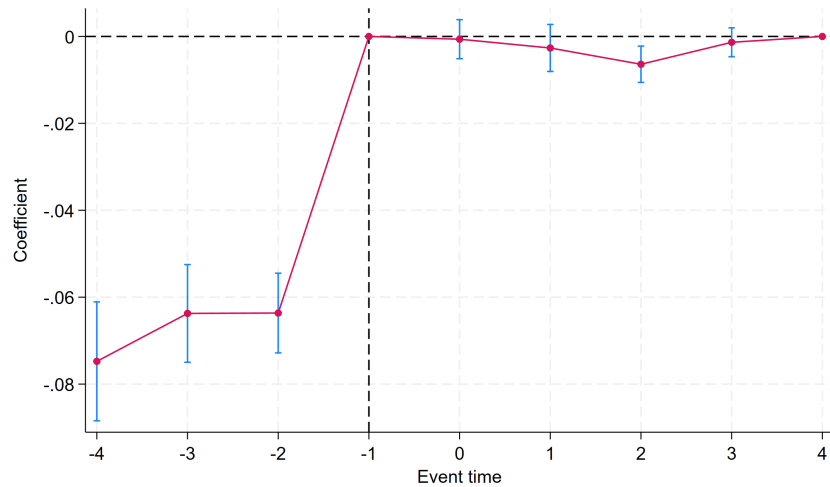
Taken together, the two exercises suggest that cannibalization is not a first-order force along either margin in our setting. Product expansion does not systematically reduce sales of existing products within a link, and buyer expansion does not systematically reduce sales to incumbent buyers.

Figure 16: No within-link product cannibalization



Notes: Mean log sales to the incumbent buyer for old products (red) and new products (blue) in years surrounding the first addition of a new product within an existing supplier–buyer link ($t=0$). Old products are those sold on the link before the new product is introduced; new products are those added at $t=0$. The figure includes only supplier–buyer links that survive throughout the window shown. The upward trajectory of old-product sales after $t=0$ indicates that introducing a new product deepens the existing relationship rather than cannibalizing incumbent product revenues.

Figure 17: No link-level cannibalization from new large buyer



Notes: Event-study estimates of the effect on log sales to incumbent buyers of the supplier's first addition of a large new buyer ($t=0$), defined as a buyer in the top decile of the buyer size distribution. Coefficients are measured relative to $t=1$ to account for pre-event anticipation. The pre-event decline reflects selection: suppliers that are about to add a large buyer tend to have temporarily depressed sales to incumbents in prior years. Sales recover to baseline immediately after the event and show no post-event decline, ruling out diversion of revenue from existing links to the new large buyer. Shaded bands report 95% confidence intervals; standard errors are clustered at the supplier level.

D New Product Introduction and Buyer Breadth

This appendix provides complementary evidence on the relationship between buyer breadth and product introduction. While the main analysis in Section 3.4 studies the relationship between the number of buyers and a supplier’s total product scope, here we ask whether suppliers also introduce more new products in years in which they serve more buyers.

Let NewProducts_{st} denote the number of products sold by supplier s in year t that were not sold by the supplier in year $t - 1$. We estimate

$$\text{NewProducts}_{st} = \beta_1 \log(\text{Buyers}_{st}) + \theta_1 \log(\text{Sales}_{st}) + \gamma_t + \delta_s + \varepsilon_{st}, \quad (81)$$

where γ_t denotes year fixed effects and δ_s denotes supplier fixed effects. Some specifications omit the quadratic sales term or replace supplier fixed effects with sector fixed effects, as indicated in the table.

Table 14 reports the results. The coefficient on $\log(\text{Buyers}_{st})$ is positive and precisely estimated across all specifications. Most importantly, the relationship remains strong when supplier fixed effects are included, so the result is identified from variation within the same supplier over time. In column (5), the coefficient is 1.934, implying that a 10 percent increase in the number of buyers is associated with approximately 0.19 additional new product introductions.⁹

The results therefore complement the product-scope evidence in the main text. Suppliers with a broader buyer base not only sell a larger number of products, but also tend to introduce more products in years in which they serve more buyers.

⁹More precisely, a 10 percent increase implies $1.934 \times \log(1.10) \simeq 0.18$ additional new products. The 0.19 figure uses the standard log-change approximation.

Table 14: Relationship between buyer breadth and new product introduction

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dep. variable: New Products_{it}</i>						
log Buyers _{it}	1.955*** (0.0173)	1.958*** (0.0213)	1.938*** (0.0176)	1.952*** (0.0217)	1.934*** (0.0308)	1.927*** (0.0423)
log Sales _{it}	0.0622*** (0.00869)	-0.275*** (0.0157)	0.0629*** (0.00876)	-0.273*** (0.0164)	0.538*** (0.0210)	1.000*** (0.0822)
log Products _{it-1}		0.318*** (0.0254)		0.296*** (0.0254)		0.00642 (0.0527)
log Sales _{it} ²		-0.0839*** (0.00208)		-0.0805*** (0.00214)		0.0583*** (0.00784)
Observations	186,439	119,517	186,426	119,726	183,401	112,946
R-squared	0.190	0.183	0.200	0.201	0.609	0.641
Firms	34,263	31,792	34,263	31,808	31,225	25,013
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	No	No
Supplier FE	No	No	No	No	Yes	Yes

Notes: The dependent variable is the number of new products introduced by supplier i in year t , defined as products sold in year t that were not sold by the supplier in year $t - 1$. The main regressor is $\log(\text{Buyers}_{it})$. Controls include contemporaneous supplier sales and, in the indicated specifications, lagged product scope and squared log sales. All specifications include year fixed effects; sector and supplier fixed effects depending on the column as indicated. Standard errors are clustered at the supplier level.

E Buyer Cost Minimization Problem (discrete formulation)

This appendix derives the buyer's cost-minimization problem in discrete form, consistent with the endogenous active sets generated by the supplier's two-stage problem in Section 4. The derivation proceeds in two steps: the top nest across suppliers (Section E), and the inner nest across products within each supplier (Section F).

C.1 Top nest: across suppliers

Buyer b chooses labor L_b and intermediate-input quantities $\{m_{b,s}\}_{s \in \mathcal{S}_b}$ to minimize total cost. Let $\mathcal{S}_b \subseteq \mathcal{S}$ denote the endogenous set of suppliers active on buyer b , determined by the supplier's two-stage problem in Section 4. The buyer takes $\mathcal{S}_b$ as given and solves

$$\min_{L_b, \{m_{b,s}\}_{s \in \mathcal{S}_b}} wL_b + \sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s}$$

subject to the production constraint and the intermediate-input bundle:

$$1 = Y_b = \varphi_b L_b^\alpha M_b^{1-\alpha}, \quad 0 < \alpha < 1, \quad (82)$$

$$M_b = \left[\sum_{s \in \mathcal{S}_b} m_{b,s}^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}, \quad \nu > 1. \quad (83)$$

Varieties $s \notin \mathcal{S}_b$ are not offered to b and do not enter the buyer's choice set; equivalently, $m_{b,s} = 0$ for $s \notin \mathcal{S}_b$ by construction.

Minimizing total cost subject to the production constraint gives the dual unit cost

$$\eta_b(w, \{P_{b,s}\}_{s \in \mathcal{S}_b}) = \frac{1}{\tilde{\varphi}_b} w^\alpha P_b^{1-\alpha}, \quad \tilde{\varphi}_b := \varphi_b \alpha^\alpha (1-\alpha)^{1-\alpha}, \quad (84)$$

and the across-supplier CES price index

$$P_b = \left[\sum_{s \in \mathcal{S}_b} P_{b,s}^{1-\nu} \right]^{\frac{1}{1-\nu}}. \quad (85)$$

Let μ_b be the multiplier on (82). Substituting (83) into (82), the Lagrangian is

$$\mathcal{L} = wL_b + \sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s} + \mu_b \left(1 - \varphi_b L_b^\alpha \left(\underbrace{\left[\sum_{s \in \mathcal{S}_b} m_{b,s}^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}}_{M_b} \right)^{1-\alpha} \right).$$

The first-order condition with respect to labor is

$$\frac{\partial \mathcal{L}}{\partial L_b} = w - \mu_b \varphi_b \alpha L_b^{\alpha-1} M_b^{1-\alpha} = 0. \quad (\text{L})$$

Write $K = \sum_{s \in \mathcal{S}_b} m_{b,s}^{\frac{\nu-1}{\nu}}$, so $M_b = K^{\frac{\nu}{\nu-1}}$. Then

$$\frac{\partial M_b}{\partial m_{b,s}} = K^{\frac{1}{\nu-1}} m_{b,s}^{-\frac{1}{\nu}} = M_b^{1/\nu} m_{b,s}^{-1/\nu}. \quad (86)$$

Using the chain rule, for each $s \in \mathcal{S}_b$,

$$\frac{\partial \mathcal{L}}{\partial m_{b,s}} = P_{b,s} - \mu_b \varphi_b (1 - \alpha) L_b^\alpha M_b^{-\alpha} \left(M_b^{1/\nu} m_{b,s}^{-1/\nu} \right) = 0.$$

Define the composite shadow price

$$\tau := \mu_b \varphi_b (1 - \alpha) L_b^\alpha M_b^{-\alpha} > 0, \quad \Rightarrow \quad P_{b,s} = \tau M_b^{1/\nu} m_{b,s}^{-1/\nu}. \quad (m_s)$$

Equation (87) implies

$$m_{b,s} = \left(\frac{\tau}{P_{b,s}} \right)^\nu M_b, \quad s \in \mathcal{S}_b. \quad (87)$$

The only object that matters for $\{m_{b,s}\}_{s \in \mathcal{S}_b}$ is the single shadow price τ . Total spending on the supplier bundle is, using (87) and (85),

$$\sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s} = M_b \tau^\nu \sum_{s \in \mathcal{S}_b} P_{b,s}^{1-\nu} = M_b \tau^\nu P_b^{1-\nu}.$$

By duality, $\sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s} = M_b P_b$, so the shadow price must equal the dual index,

$$\tau = P_b,$$

and (87) becomes the standard top-nest Hicksian demand across suppliers

$$\boxed{m_{b,s} = M_b \left(\frac{P_{b,s}}{P_b} \right)^{-\nu}, \quad s \in \mathcal{S}_b,} \quad (88)$$

with dual price relation

$$\boxed{\sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s} = M_b P_b.} \quad (89)$$

Labor–bundle cost shares and unit cost

From (L) and the definition of τ ,

$$\frac{w}{\tau} = \frac{\alpha}{1-\alpha} \cdot \frac{M_b}{L_b} \xrightarrow{\tau=P_b} wL_b = \alpha(wL_b + P_b M_b), \quad P_b M_b = (1-\alpha)(wL_b + P_b M_b).$$

Let the minimized cost be $\eta_b := wL_b + \sum_{s \in \mathcal{S}_b} P_{b,s} m_{b,s} = wL_b + P_b M_b$. Using (82) with these shares,

$$1 = \varphi_b \left(\frac{\alpha \eta_b}{w} \right)^\alpha \left(\frac{(1-\alpha) \eta_b}{P_b} \right)^{1-\alpha} = \varphi_b \alpha^\alpha (1-\alpha)^{1-\alpha} \frac{\eta_b}{w^\alpha P_b^{1-\alpha}}.$$

Thus the unit cost coincides with (84):

$$\eta_b(w, \{P_{b,s}\}_{s \in \mathcal{S}_b}) = \frac{1}{\tilde{\varphi}_b} w^\alpha P_b^{1-\alpha}, \quad \tilde{\varphi}_b := \varphi_b \alpha^\alpha (1-\alpha)^{1-\alpha}. \quad (90)$$

C.2 Inner nest: across products within a supplier

Given total spending $R_{sb} \equiv P_{b,s} m_{b,s}$ on supplier s from the top nest, buyer b allocates this spending across the products that supplier s activates on the link (s, b) . Let $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$ denote the endogenous set of products activated by supplier s on link (s, b) , with cardinality $G_{sb} = |\mathcal{G}_{sb}|$, determined by the supplier's link-level scope choice in Section 4. The buyer takes $\mathcal{G}_{sb}$ as given.

The within-supplier product-line composite is

$$m_{b,s} = \left[\sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad \varepsilon > 1, \quad (91)$$

where x_{sbg} is the quantity of product g that buyer b purchases from supplier s , and ε is the elasticity of substitution across products within a supplier.¹⁰ Products $g \notin \mathcal{G}_{sb}$ are not offered on link (s, b) and satisfy $x_{sbg} = 0$ by construction.

Given prices $\{p_{sbg}\}_{g \in \mathcal{G}_{sb}}$ and target composite $m_{b,s}$, the buyer's expenditure-minimization prob-

¹⁰The benchmark of Section 4 corresponds to $\varepsilon = \nu$ (single CES nest, no within-supplier cannibalization). The nested-CES extension of Section 5 allows $\varepsilon \neq \nu$ and delivers within-link cannibalization when $\varepsilon > \nu$.

lem at the inner nest is

$$\min_{\{x_{sbg}\}_{g \in \mathcal{G}_{sb}}} \sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} \quad \text{s.t.} \quad \left[\sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \geq m_{b,s}.$$

Let λ_{sb} be the multiplier on the constraint. The Lagrangian is

$$\mathcal{L}_{sb} = \sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} + \lambda_{sb} \left(m_{b,s} - \left[\sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \right).$$

Write $\tilde{K} = \sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}}$, so $m_{b,s} = \tilde{K}^{\frac{\varepsilon}{\varepsilon-1}}$ and

$$\frac{\partial m_{b,s}}{\partial x_{sbg}} = \tilde{K}^{\frac{1}{\varepsilon-1}} x_{sbg}^{-\frac{1}{\varepsilon}} = m_{b,s}^{1/\varepsilon} x_{sbg}^{-1/\varepsilon}. \quad (92)$$

The first-order condition with respect to x_{sbg} for each $g \in \mathcal{G}_{sb}$ is

$$\frac{\partial \mathcal{L}_{sb}}{\partial x_{sbg}} = p_{sbg} - \lambda_{sb} m_{b,s}^{1/\varepsilon} x_{sbg}^{-1/\varepsilon} = 0, \quad (93)$$

which implies

$$p_{sbg} = \lambda_{sb} m_{b,s}^{1/\varepsilon} x_{sbg}^{-1/\varepsilon} \implies x_{sbg} = \left(\frac{\lambda_{sb}}{p_{sbg}} \right)^{\varepsilon} m_{b,s}, \quad g \in \mathcal{G}_{sb}. \quad (94)$$

Total spending on supplier s , using (94), is

$$\sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} = m_{b,s} \lambda_{sb}^{\varepsilon} \sum_{g \in \mathcal{G}_{sb}} p_{sbg}^{1-\varepsilon}.$$

Defining the within-supplier CES price index

$$P_{b,s} = \left[\sum_{g \in \mathcal{G}_{sb}} p_{sbg}^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \quad (95)$$

the spending identity becomes

$$\sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} = m_{b,s} \lambda_{sb}^{\varepsilon} P_{b,s}^{1-\varepsilon}.$$

By duality, $\sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} = P_{b,s} m_{b,s}$, so the shadow price equals the within-supplier CES price index,

$$\lambda_{sb} = P_{b,s},$$

and (94) becomes the standard within-supplier Hicksian demand across products

$$\boxed{x_{sbg} = m_{b,s} \left(\frac{p_{sbg}}{P_{b,s}} \right)^{-\varepsilon}, \quad g \in \mathcal{G}_{sb},} \quad (96)$$

with dual price relation

$$\boxed{\sum_{g \in \mathcal{G}_{sb}} p_{sbg} x_{sbg} = P_{b,s} m_{b,s}.} \quad (97)$$

Product-level expenditure shares

Combining the top-nest Hicksian (88) with the inner-nest Hicksian (96) gives the full demand for product g from supplier s by buyer b in terms of price indices and the buyer's composite M_b :

$$x_{sbg} = M_b \left(\frac{P_{b,s}}{P_b} \right)^{-\nu} \left(\frac{p_{sbg}}{P_{b,s}} \right)^{-\varepsilon}, \quad s \in \mathcal{S}_b, g \in \mathcal{G}_{sb}. \quad (98)$$

Equivalently, product-level expenditure shares decompose as

$$r_{sbg} \equiv p_{sbg} x_{sbg} = \underbrace{P_b M_b}_{\text{total expenditure}} \cdot \underbrace{\frac{P_{b,s}^{1-\nu}}{P_b^{1-\nu}}}_{\text{share on supplier } s} \cdot \underbrace{\frac{p_{sbg}^{1-\varepsilon}}{P_{b,s}^{1-\varepsilon}}}_{\text{share on product } g \text{ within } s}, \quad (99)$$

which is the standard two-stage budgeting representation: total expenditure $P_b M_b$ is allocated across active suppliers $s \in \mathcal{S}_b$ according to relative price indices with elasticity ν , and then across active products $g \in \mathcal{G}_{sb}$ within each supplier according to relative prices with elasticity ε . The benchmark of Section 4 collapses both stages into a single CES by setting $\varepsilon = \nu$, in which case $r_{sbg} = P_b M_b \cdot p_{sbg}^{1-\nu} / P_b^{1-\nu}$ and within-supplier cannibalization is absent.

F Extension 1: Nested CES and within-link cannibalization

The benchmark in Section 4 treats product-level profits as additively separable within a link: adding a product on link (s, b) does not change the profitability of the supplier's other products sold to the same buyer. This separability comes from assuming a single CES nest in which a supplier's products and other suppliers' products are equally substitutable. In this section we relax that assumption by specifying demand as a *nested* CES system, with an outer nest across suppliers (elasticity ν) and an inner nest across products within each supplier (elasticity ε). The two stages of the supplier's problem (firm-level scope G_s in Stage 1; link activation and link-level scope G_{sb} in Stage 2) are preserved, as are the active sets $\mathcal{S}_b$ and $\mathcal{G}_{sb}$. The new ingredient is

that products sold by the same supplier to the same buyer now interact, generating within-link cannibalization whenever $\varepsilon \neq \nu$.

F.1 Demand: nested CES across suppliers and products

Let $\mathcal{S}_b \subseteq \mathcal{S}$ denote the endogenous set of suppliers active on buyer b , and let $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$ denote the endogenous set of products that supplier $s \in \mathcal{S}_b$ activates on link (s, b) , with cardinality $G_{sb} = |\mathcal{G}_{sb}|$. Both objects are determined by the supplier's two-stage problem (Section 4) and taken as given by the buyer. Products in the supplier's adopted portfolio that are not activated on the link, as well as suppliers not in $\mathcal{S}_b$, do not enter the buyer's choice set.

Outer nest: across suppliers. Let X_{sb} denote the composite input that buyer b purchases from supplier $s \in \mathcal{S}_b$ (the supplier's “variety” inside buyer b). The buyer's overall intermediate-input bundle is

$$M_b = \left[\sum_{s \in \mathcal{S}_b} X_{sb}^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}, \quad \nu > 1, \quad (100)$$

where ν governs substitution *across* suppliers: a higher ν means buyer b more easily switches spending between suppliers.

Inner nest: across products within a supplier. For each active supplier $s \in \mathcal{S}_b$, the within-supplier product-line composite is

$$X_{sb} = \left[\sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad \varepsilon > 1, \quad (101)$$

where x_{sbg} is the quantity of product g that supplier s sells to buyer b , and ε governs substitution *within* a supplier's product line. The benchmark of Section 4 is the special case $\varepsilon = \nu$, in which the two nests collapse and cannibalization vanishes.

Price indices. Let p_{sbg} denote the delivered price of product g sold by supplier s to buyer b . The within-supplier (inner-nest) price index is

$$P_{sb} = \left[\sum_{g \in \mathcal{G}_{sb}} p_{sbg}^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \quad (102)$$

the dual to (101) and the minimum unit cost of one unit of the supplier composite X_{sb} given prices $\{p_{sbg}\}_{g \in \mathcal{G}_{sb}}$. The buyer-level (outer-nest) price index across suppliers is

$$P_b = \left[\sum_{s \in \mathcal{S}_b} P_{sb}^{1-\nu} \right]^{\frac{1}{1-\nu}}, \quad (103)$$

the dual to (100). The full derivation of these objects from buyer cost minimization is in Appendix C.

Two-stage budgeting. Given total expenditure E_b , the buyer first allocates spending across suppliers and then across products within each supplier:

$$R_{sb} \equiv P_{sb} X_{sb} = E_b \left(\frac{P_{sb}}{P_b} \right)^{1-\nu}, \quad (\text{outer}) \quad (104)$$

$$r_{sbg} \equiv p_{sbg} x_{sbg} = R_{sb} \left(\frac{p_{sbg}}{P_{sb}} \right)^{1-\varepsilon}, \quad g \in \mathcal{G}_{sb}. \quad (\text{inner}) \quad (105)$$

Equations (104)–(105) make the two opposing effects of adding a product explicit. Adding a product g to $\mathcal{G}_{sb}$ lowers P_{sb} (the inner-nest price index becomes smaller), which raises supplier s 's share of buyer b 's total spending through the outer nest. At the same time, the inner-nest allocation reshuffles revenue across the supplier's own products. This reshuffling is the within-link cannibalization channel. Its strength relative to the demand-stealing channel is governed by ε versus ν : cannibalization is strongest when products within a supplier are closer substitutes than suppliers' product lines are across firms, i.e., when $\varepsilon > \nu$.

F.2 Technology, pricing, and link-level revenue

The marginal-cost schedule is the same as in the benchmark:

$$c_{sbg} = \frac{w h(g)}{\phi_s}, \quad (106)$$

where w is the input price, ϕ_s supplier productivity, and $h(g)$ the rank schedule (increasing in g). Optimal pricing under monopolistic competition with outer-nest elasticity ν across suppliers gives

a constant markup over marginal cost:¹¹

$$p_{sb} = \tilde{\nu} c_{sb} = \tilde{\nu} \frac{w h(g)}{\phi_s}, \quad \tilde{\nu} \equiv \frac{\nu}{\nu - 1} > 1. \quad (107)$$

If supplier s activates exactly G products on link (s, b) (i.e., $G_{sb} = G$, with products ranked $g = 1, \dots, G$), the within-supplier price index becomes

$$P_{sb}(G) = \tilde{\nu} \frac{w}{\phi_s} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}. \quad (108)$$

Substituting (108) into (104) gives total link-level expenditure as a function of G :

$$R_{sb}(G) = E_b \left(\frac{P_b \phi_s}{\tilde{\nu} w} \right)^{\nu-1} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{\nu-1}{\varepsilon-1}}. \quad (109)$$

Product-level revenue within the link, by the inner-nest allocation (105), is

$$r_{sb}(g; G) = R_{sb}(G) \frac{h(g)^{1-\varepsilon}}{\sum_{j=1}^G h(j)^{1-\varepsilon}}, \quad g = 1, \dots, G. \quad (110)$$

Equation (110) shows cannibalization at the product level: when the supplier adds a new product, the denominator $\sum_{j=1}^G h(j)^{1-\varepsilon}$ changes, so the revenue of every infra-marginal product on the link is reallocated. Whether $R_{sb}(G)$ itself rises with G depends on the sign of $(\nu - 1)/(\varepsilon - 1)$ and on the marginal contribution $h(G)^{1-\varepsilon}$, which is positive whenever $\varepsilon > 1$ and $h(\cdot)$ is finite.

Under the constant-markup rule, variable profits are a constant fraction of revenues:

$$\sum_{g=1}^G \pi_{sb}^V(g; G) = \frac{1}{\nu} \sum_{g=1}^G r_{sb}(g; G) = \frac{1}{\nu} R_{sb}(G). \quad (111)$$

This is the key analytical simplification: although product-level revenues are jointly determined by cannibalization, total variable profits on the link depend on G only through the scalar $R_{sb}(G)$. The buyer-level problem therefore reduces to a one-dimensional choice over G , exactly as in the benchmark.

¹¹We maintain the constant-markup rule throughout. This is an analytical convenience: under nested CES with $\varepsilon \neq \nu$, fully strategic multiproduct pricing would deliver markups that depend on the inner-nest elasticity (see the discussion in [Arkolakis et al. \(2021\)](#), Appendix E, and [Hottman et al. \(2016\)](#)). The constant-markup approximation keeps variable profits proportional to revenues and isolates the scope cannibalization mechanism on the demand side. It can be defended as a benchmark in which suppliers price product-by-product taking the supplier-level price index as given.

F.3 The supplier's two-stage problem under nested CES

We retain the two-stage structure of Section 4. In Stage 1, supplier s chooses firm-level adopted scope G_s and pays the firm-level adoption cost $F^P(G_s)$. In Stage 2, for each potential buyer b , the supplier decides whether to activate the link (paying f_{sb}^L) and how many products $G_{sb} \in \{0, 1, \dots, G_s\}$ to activate on the link (paying $F_{sb}^A(G_{sb})$). The only thing that changes relative to the benchmark is the within-link revenue function $R_{sb}(G)$ in (109), which now embeds the inner-nest cannibalization.

Stage 2: link-level value function. Given firm-level scope G_s , the maximum operating value supplier s can extract from buyer b is

$$V_{sb}(G_s; \phi_s) = \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{1}{\nu} R_{sb}(G) - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}, \quad (112)$$

or, substituting (109) explicitly,

$$V_{sb}(G_s; \phi_s) = \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{E_b}{\nu} \left(\frac{P_b \phi_s}{\tilde{\nu} w} \right)^{\nu-1} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{\nu-1}{\varepsilon-1}} - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}. \quad (113)$$

The first term is total variable profit from activating G products on the link, the second is the link fixed cost (paid only if at least one product is activated), and the third is the cumulative product-activation cost. The optimal link-level scope and link activation are characterized jointly by

$$G_{sb}^*(G_s; \phi_s) \in \arg \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{1}{\nu} R_{sb}(G) - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}, \quad (114)$$

with the link activated if and only if $G_{sb}^*(G_s; \phi_s) \geq 1$. The active buyer set at firm-level scope G_s is

$$\mathcal{B}_s(G_s) = \{b \in \mathcal{B} : G_{sb}^*(G_s; \phi_s) \geq 1\}. \quad (115)$$

Two features distinguish (114) from the benchmark Stage-2 problem. First, the marginal gain from adding product g on the link is no longer simply $A_{sb}(\phi_s) h(g)^{-(\nu-1)} - f_{sb}^A(g)$, because the inner-nest term $[\sum_j h(j)^{1-\varepsilon}]^{(\nu-1)/(\varepsilon-1)}$ in $R_{sb}(G)$ is not additive across ranks when $\varepsilon \neq \nu$. Second, the marginal gain depends on *which* products are already active through the cannibalization channel: adding rank G reshuffles revenue from infra-marginal ranks $g < G$, so the incremental profit from rank G accounts for both the new revenue it generates and the revenue it absorbs from existing products. The closed-form scope cutoff $\tilde{G}_{sb}(\phi_s)$ in equation (39) of the benchmark no longer obtains in general; the link-level problem must be solved as a discrete choice over $G \in \{0, 1, \dots, G_s\}$. Different buyers can optimally choose different scopes because E_b , f_{sb}^L , and $F_{sb}^A(\cdot)$ all vary across buyers.

Stage 1: firm-level scope. Substituting the Stage 2 value functions into the supplier's outer problem,

$$\Pi_s(G_s; \phi_s) = \sum_{b \in \mathcal{B}} V_{sb}(G_s; \phi_s) - F^P(G_s), \quad (116)$$

and firm-level scope solves

$$G_s^*(\phi_s) \in \arg \max_{G_s \in \{0, 1, \dots, |\Omega|\}} \Pi_s(G_s; \phi_s). \quad (117)$$

The structure is identical to the benchmark (cf. equations (44)–(45)): for each candidate G_s , the firm earns $V_{sb}(G_s; \phi_s)$ from every potential buyer b (zero if the link is inactive), aggregates across the buyer pool, and subtracts the firm-level adoption cost $F^P(G_s)$. The amortization mechanism is preserved: the firm-level cost is paid once but the payoff accrues across all active relationships, so a larger and more valuable buyer base raises the return to expanding firm-level scope.

Fixed-point structure. As in the benchmark (Section 4.8.4), the joint solution $(G_s^*, \mathcal{B}_s^*)$ is a fixed point: the active buyer set $\mathcal{B}_s$ depends on G_s through the link-activation rule embedded in (114)–(115), and the optimal G_s depends on $\mathcal{B}_s$ through the sum in (116). The monotonicity arguments of the benchmark carry over: the map $G_s \mapsto \mathcal{B}_s(G_s)$ is weakly increasing in G_s (more firm-level products weakly raise link surplus), and the map $\mathcal{B}_s \mapsto G_s^*(\mathcal{B}_s)$ is weakly increasing (more buyers raise the marginal return to scope through amortization). Tarski's theorem again guarantees existence on the finite lattice. The same iterative algorithm in Section 4.8.4 applies, with the only change being that step 3 (the inner Stage 2 solve) requires numerical optimization over $G \in \{0, 1, \dots, G_s\}$ rather than the closed-form rule of the benchmark.

Comparison with the benchmark. Conditional on G_s , each buyer-level problem in (113) is still a scalar discrete choice. What is lost relative to the benchmark is not existence or tractability of the supplier's problem, but the simple closed-form rank-cutoff characterization $\tilde{G}_{sb} = (B_b \phi_s^{\nu-1} / \nu f_{sb,0}^A)^{1/(\mu+\alpha(\nu-1))}$ of the benchmark (equation 39). When $\varepsilon = \nu$, the inner aggregator in (109) reduces to $\sum_{g=1}^G h(g)^{-(\nu-1)}$ and the benchmark's analytical structure is recovered exactly. The nested-CES extension is therefore a generalization that nests the benchmark at $\varepsilon = \nu$ and introduces cannibalization for $\varepsilon \neq \nu$, with $\varepsilon > \nu$ being the empirically relevant case in which a supplier's own products are closer substitutes than products across suppliers.

F.4 Complementarity in both directions: buyers $\leftrightarrow$ products

The two-way complementarity established in Section 4.9.1 carries over to the nested-CES extension, though through slightly different channels.

More buyers $\Rightarrow$ more products. Holding productivity ϕ_s fixed, expanding the active buyer set raises the return to firm-level adoption because the cost $F^P(G_s)$ is amortized across more

relationships. Formally, the marginal gain from raising G_s from $g - 1$ to g is

$$\Delta\Pi_s(g; \phi_s) = \sum_{b \in \mathcal{B}} [V_{sb}(g; \phi_s) - V_{sb}(g - 1; \phi_s)] - f^P(g), \quad (118)$$

which is weakly increasing in the size and quality of the active buyer set: each additional active buyer contributes a non-negative term $V_{sb}(g; \phi_s) - V_{sb}(g - 1; \phi_s) \geq 0$ to the gross gain, while $f^P(g)$ is fixed at the firm level. The monotonicity proof of Proposition 1 in the benchmark applies unchanged: holding ϕ_s fixed, the optimal G_s^* is weakly increasing in the buyer base.

More products $\Rightarrow$ more buyers. Holding ϕ_s fixed, raising G_s weakly relaxes the link-level constraint $G_{sb} \leq G_s$ in (114), so the maximized link value $V_{sb}(G_s; \phi_s)$ is weakly increasing in G_s . Under nested CES, this channel operates through two reinforcing mechanisms relative to the benchmark. First, a higher G_s permits a wider link-level scope G_{sb} on links where the firm-level cap was binding, which lowers the inner-nest price index P_{sb} and increases supplier s 's share of buyer b 's total expenditure through the outer nest (104). Second, the wider product line lets buyers substitute across more of the supplier's products, raising buyer-level utility for given total spending. Both forces push link surplus up, so previously unprofitable links can become profitable as G_s rises. The active-set inclusion $\mathcal{B}_s(G_s) \subseteq \mathcal{B}_s(G_s + 1)$ from the benchmark therefore continues to hold.

Joint determination. The two directions of complementarity again pin down the equilibrium $(G_s^*, \mathcal{B}_s^*)$ jointly. The economic content is the same as in the benchmark: amortization on the cost side (more buyers raise the return to firm-level adoption) and surplus expansion on the demand side (more products raise the value of each active link) reinforce each other. The nested-CES specification strengthens the second channel through the inner-nest substitution effect, but does not alter the qualitative complementarity that drives the co-movement of buyer breadth and product scope observed in Stylized Facts 2 and 3.

G Extension 2: Dynamic buyer accumulation over the life cycle

The benchmark of Section 4 and the nested-CES extension of Section F are static within-period environments. They clarify how a given buyer base shapes firm-level scope G_s , link-level scope G_{sb} , and within-link cannibalization, but they do not by themselves explain why young firms begin life with few buyers and only gradually build their buyer base over time. To capture that life-cycle margin, we add a simple dynamic buyer-accumulation block on top of the within-period problem.

The extension is designed to capture three additional facts. First, young firms have fewer buyers than mature firms even after conditioning on standard firm-level controls. Second, buyer accumulation is gradual rather than instantaneous. Third, if link survival is relatively stable over the life cycle, the main source of low buyer base among young firms is slow acquisition of new buyers rather than unusually high buyer churn. These patterns motivate a specification in which the active buyer set $\mathcal{B}_s$ of Section 4 is itself a state variable that the firm builds up through costly investment, while the within-period scope and link-activation decisions are inherited from the benchmark. Treating the buyer base as a costly-to-build state variable connects this extension to models of customer accumulation (Arkolakis, 2010; Gourio and Rudanko, 2014; Fitzgerald et al., 2024), adapted here to firm-to-firm relationships rather than final consumers.

G.1 Buyer-base state variable

Let $a \in \{0, 1, 2, \dots\}$ denote firm age, measured as periods since entry. Let

$$\Lambda_{s,a} \geq 0 \tag{119}$$

denote the buyer-base aggregator of supplier s at age a , defined exactly as in 31 of the benchmark, $\Lambda_s = \sum_{b \in \mathcal{B}_s} B_b$, but now indexed by age to reflect the gradual accumulation of relationships. As in the static benchmark, $\Lambda_{s,a}$ admits the decomposition $\Lambda_{s,a} = N_{s,a} \bar{B}_{s,a}$ into the number of active buyers $N_{s,a} = |\mathcal{B}_{s,a}|$ and their average quality $\bar{B}_{s,a}$, so the state captures both how many buyers the supplier has accumulated and how good those buyers are on average.

In each period the supplier chooses a buyer-acquisition effort

$$n_{s,a}^B \geq 0, \tag{120}$$

which determines the expected mass of new buyers reached at age a . Successful acquisition depends on aggregate matching conditions in the buyer market: let $m^B > 0$ denote matching efficiency, and let $\bar{B}^{\text{new}} > 0$ denote the average buyer-quality contribution of a newly acquired relationship. Let $\delta^B \in (0, 1)$ denote the per-period survival probability of an existing buyer relationship. The expected law of motion for the buyer-base aggregator is

$$\mathbb{E}[\Lambda_{s,a+1} \mid \Lambda_{s,a}, n_{s,a}^B] = \delta^B \Lambda_{s,a} + m^B \bar{B}^{\text{new}} n_{s,a}^B. \tag{121}$$

The buyer base next period is the sum of surviving current relationships and newly acquired buyers. The specification deliberately keeps δ^B age-invariant; this is the assumption that isolates differences in acquisition (rather than churn) as the source of life-cycle differences in buyer-base growth, consistent with the survival evidence in Appendix B.

G.2 Cost of buyer accumulation

To build its buyer base, supplier s incurs a convex networking cost

$$C_a^B(n_{s,a}^B) \equiv \xi_a^B \frac{f^B}{\gamma_B} (n_{s,a}^B)^{\gamma_B}, \quad \gamma_B > 1, \quad f^B > 0, \quad (122)$$

where $\xi_a^B \geq 1$ is an optional age-specific networking wedge. The convexity parameter $\gamma_B > 1$ ensures that it is costly to acquire many buyers all at once, which gives firms an incentive to smooth buyer accumulation over the life cycle. If ξ_a^B is declining in a , then young firms face an additional handicap in acquiring buyers early in life, which strengthens the gradual-accumulation pattern.

The cost C_a^B can be interpreted as buyer-side networking, sales, promotion, or prospecting effort. The substantive content is minimal: acquiring new buyers is costly, the marginal cost of rapid acquisition rises with effort, and these frictions are what slow down young firms.

G.3 Dynamic supplier problem

The within-period block is inherited unchanged from one of the static specifications:

- In the benchmark (Section 4), the within-period profit is determined by the two-stage problem in Section 4.8, which delivers a flow profit

$$\Pi_s^{\text{flow,bench}}(G_s; \phi_s, \Lambda_{s,a}) = \sum_{b \in \mathcal{B}_{s,a}} V_{sb}(G_s; \phi_s) - F^P(G_s),$$

where the link-level value functions $V_{sb}(G_s; \phi_s)$ are given by equation (43) and the buyer-base aggregator $\Lambda_{s,a} = \sum_{b \in \mathcal{B}_{s,a}} B_b$ summarizes the relevant component of the buyer base for Stage 1.

- In the nested-CES extension (Section F), within-period profit takes the same outer form but uses the cannibalization-adjusted link values from equation (113).

To keep notation compact, let

$$\Pi_s^{\text{flow}}(G_s; \phi_s, \Lambda_{s,a}) \quad (123)$$

denote the current-period profit block inherited from whichever within-period version is being used. The supplier's dynamic problem is then

$$W_s(\phi_s, a, \Lambda_{s,a}) = \max_{G_s, n_{s,a}^B} \{ \Pi_s^{\text{flow}}(G_s; \phi_s, \Lambda_{s,a}) - C_a^B(n_{s,a}^B) + \beta \mathbb{E}[W_s(\phi_s, a+1, \Lambda_{s,a+1})] \}, \quad (124)$$

subject to the law of motion (121), where $\beta \in (0, 1)$ is the discount factor. The firm therefore chooses, each period, both its current product scope G_s (the Stage-1 decision of the within-period

problem) and its current buyer-acquisition effort $n_{s,a}^B$ (the new dynamic margin), internalizing how acquisition today affects future profits through $\Lambda_{s,a+1}$.

Decomposition of decisions. It is useful to be explicit about which decisions are static versus dynamic in this extension. Conditional on $(G_s, \Lambda_{s,a})$, all link-level decisions remain static and within-period: which links to activate (the Stage-2 problem of Section 4.8.1) and how many products to activate on each link (the link-level scope G_{sb}) are determined exactly as in the benchmark, using the link-activation rule (114) and the Stage-2 closed form (37). What the dynamic block adds is a layer above: the firm now also chooses $n_{s,a}^B$ to shape the future evolution of the buyer base, taking into account that a larger $\Lambda_{s,a+1}$ will raise next period's Stage-1 return to firm-level scope G_s through amortization.

Euler-style condition for buyer acquisition. The first-order condition with respect to $n_{s,a}^B$ takes the standard intertemporal form. Differentiating (124) and using (121),

$$\underbrace{\xi_a^B f^B (n_{s,a}^B)^{\gamma_B-1}}_{\text{marginal acquisition cost}} = \beta m^B \bar{B}^{\text{new}} \cdot \underbrace{\mathbb{E} \left[\frac{\partial W_s(\phi_s, a+1, \Lambda_{s,a+1})}{\partial \Lambda_{s,a+1}} \right]}_{\text{marginal value of the buyer base}}. \quad (125)$$

The marginal value of the buyer base on the right-hand side is what links the dynamic and static blocks: by the envelope theorem applied to (124), it equals the discounted sum of all future increments to Π_s^{flow} from carrying an extra unit of Λ forward, which through amortization translates into a higher optimal G_s in future periods.

G.4 Connection to the benchmark scope rule

In the benchmark, the approximate Stage-1 scope rule (equation 55) ties optimal firm-level scope to the buyer-base aggregator. Embedding that rule in the dynamic environment with $\Lambda_{s,a}$ in place of the static aggregator yields a life-cycle scope rule

$$G_s^*(\phi_s, \Lambda_{s,a}) \approx \text{integer} \left\{ \left[\frac{\Lambda_{s,a} \phi_s^{\sigma-1}}{\sigma f^P(1)} \right]^{\frac{1}{\delta + \alpha(\sigma-1)}} \right\}, \quad (126)$$

where the parameter σ is the benchmark single-CES elasticity (which corresponds to ν in the nested-CES notation of Section ??). Even holding productivity fixed, gradual buyer accumulation generates gradual product-scope expansion over the life cycle, because $\Lambda_{s,a}$ enters (126) with a positive exponent under the finite-scope condition $\delta + \alpha(\sigma - 1) > 0$.

The same intuition carries over to the nested-CES extension. A higher $\Lambda_{s,a}$ increases the payoff to paying the firm-level adoption cost $F^P(G_s)$, raises the value of activating broader link-level scopes G_{sb} on inframarginal relationships, and pushes link-surplus above the activation threshold

f_{sb}^L on the extensive margin (cf. equation 52). Within-link cannibalization in the nested-CES case prevents the same closed form as in (126), but the qualitative monotonicity is preserved: G_s^* is weakly increasing in $\Lambda_{s,a}$, so the life-cycle pattern of scope expansion is the same.

G.5 Equivalent truncated-history representation

For comparison with the dynamic production-network literature (e.g., Einav et al. (2021), Aekka and Khanna (2024)), the buyer-base state can equivalently be written as a truncated history of past acquisition decisions. Iterating (121) forward and taking expectations,

$$\Lambda_{s,a} \approx m^B \bar{B}^{\text{new}} \sum_{\tau=0}^{A_B} (\delta^B)^\tau n_{s,a-1-\tau}^B, \quad (127)$$

where A_B is a truncation horizon chosen so that contributions from cohorts older than $a - A_B$ are negligible. The truncation is accurate whenever buyer relationships do not survive indefinitely, since the contribution of past cohorts decays geometrically at rate δ^B . This representation makes the dynamic block tractable: rather than tracking the entire history of buyer relationships, the firm can summarize the dynamic buyer side either with the scalar state $\Lambda_{s,a}$ in (121) or with the finite-dimensional vector of recent acquisition choices in (127).

G.6 Interpretation and mapping to the stylized facts

This extension adds a dynamic production-network margin without discarding the multiproduct structure central to the paper. The four versions of the model deliver a unified picture. The benchmark (Section 4) explains why a larger buyer base raises firm-level scope through amortization of $F^P(G_s)$. The link-level scope mechanism within the benchmark (Section 4.8.1) explains why different buyers can absorb different numbers of products. The nested-CES extension (Section F) explains how products interact within a link via cannibalization; and the present extension explains how the buyer base itself $\Lambda_{s,a}$ is accumulated over time.

Together, these blocks rationalize the life-cycle composition of growth documented in Fact 2 (Section 3) and Figure 5. Young firms begin life with a small $\Lambda_{s,a}$, and because building the buyer base is costly, and possibly especially costly when ξ_a^B is high at low a , buyer accumulation is gradual. Since firm-level scope G_s^* is increasing in $\Lambda_{s,a}$ via (126), product scope expands gradually alongside the buyer base. The composition of growth shifts predictably with age: at low a , the marginal value of Λ in (125) is high (because amortization gains are large when current scope and buyer base are small), so firms invest heavily in both new buyers and new products, consistent with the elevated NLNP and OLNP shares in the youngest age group in Figure 5. As $\Lambda_{s,a}$ grows toward its steady-state level, the marginal value of additional buyers falls and the composition of

growth tilts toward expanding an existing product portfolio across new buyers (NLOP), consistent with the older age groups in Figure 5.

The extension thus links dynamic production-network growth to dynamic multiproduct growth within a single framework, while leaving open natural quantitative exercises: estimating γ_B and the age profile of ξ_a^B from the observed gradient of buyer accumulation, and using the model to decompose the contribution of slow buyer accumulation versus slow product adoption to the early-life-cycle growth gap.

References

- Aekka, Anuraag and Gaurav Khanna (2024) “Endogenous Production Networks and Firm Dynamics,” Technical report, JMP.
- Afrouzi, Hassan, Andres Drenik, and Ryan Kim (2023) “Concentration, Market Power, and Misallocation: The Role of Endogenous Customer Acquisition,” Working Paper 31415, National Bureau of Economic Research, [10.3386/w31415](https://doi.org/10.3386/w31415).
- Argente, David, Doireann Fitzgerald, Sara Moreira, and Anthony Priolo (2021) “How Do Firms Build Market Share?” Technical Report 31415, [10.2139/ssrn.3831706](https://ssrn.com/abstract=3831706).
- Argente, David, Munseob Lee, and Sara Moreira (2018) “How do Firms Grow? The Life Cycle of Products Matters,” 2018 Meeting Papers 1174, Society for Economic Dynamics, <https://EconPapers.repec.org/RePEc:red:sed018:1174>.
- Arkolakis, Costas (2010) “Market Penetration Costs and the New Consumers Margin in International Trade,” *Journal of Political Economy*, 118 (6), 1151–1199, <http://www.jstor.org/stable/10.1086/657949>.
- Arkolakis, Costas, Sharat Ganapati, and Marc-Andreas Muendler (2021) “The Extensive Margin of Exporting Products: A Firm-Level Analysis,” *American Economic Journal: Macroeconomics*, 13 (4), 182–245, [10.1257/mac.20150370](https://doi.org/10.1257/mac.20150370).
- Arkolakis, Costas, Federico Huneus, and Yuhei Miyauchi (2023) “Production Network Formation, Trade, and Welfare,” NBER Working Paper 30954, National Bureau of Economic Research.
- Bernard, Andrew B., Emmanuel Dhyne, Glenn Magerman, Kalina Manova, and Andreas Moxnes (2022) “The Origins of Firm Heterogeneity: A Production Network Approach,” *Journal of Political Economy*, 130 (7), 1765–1804.
- Bernard, Andrew B. and Andreas Moxnes (2018) “Networks and Trade,” *Annual Review of Economics*, 10, 65–85.
- Bernard, Andrew B., Andreas Moxnes, and Karen Helene Ulltveit-Moe (2018) “Two-Sided Heterogeneity and Trade,” *Review of Economics and Statistics*, 100 (3), 424–439.
- Bernard, Andrew B., Stephen J. Redding, and Peter K. Schott (2010) “Multiple-Product Firms and Product Switching,” *American Economic Review*, 100 (1), 70–97.
- (2011) “Multiproduct Firms and Trade Liberalization,” *Quarterly Journal of Economics*, 126 (3), 1271–1318.

- Burstein, Ariel, Javier Cravino, and Marco Rojas (2024) “Input price dispersion across buyers and misallocation*,” *Working Paper*.
- Cavenaile, Laurent and Pau Roldan-Blanco (2021) “Advertising, Innovation, and Economic Growth,” *American Economic Journal: Macroeconomics*, 13 (3), 251–303, [10.1257/mac.20180461](https://doi.org/10.1257/mac.20180461).
- Copeland, Adam and Adam Hale Shapiro (2016) “Price Setting and Rapid Technology Adoption: The Case of the PC Industry,” *The Review of Economics and Statistics*, 98 (3), 601–616, [10.1162/REST_a.00539](https://doi.org/10.1162/REST_a.00539).
- Demir, Banu, Ana Cecília Fielér, Daniel Yi Xu, and Kelly Kaili Yang (2024) “O-Ring Production Networks,” *Journal of Political Economy*, 132 (1), 200–247.
- Dhyne, Emmanuel, Ayumu Ken Kikkawa, Magne Mogstad, and Felix Tintelnot (2021) “Trade and Domestic Production Networks,” *Review of Economic Studies*, 88 (2), 643–668.
- Eaton, Jonathan, Samuel S. Kortum, and Francis Kramarz (2026) “Firm-to-Firm Trade: Imports, Exports, and the Labor Market,” *Econometrica*, 94 (4), 1135–1170.
- Einav, Liran, Peter J. Klenow, Jonathan D. Levin, and Raviv Murciano-Goroff (2021) “Customers and Retail Growth,” NBER Working Papers 29561, National Bureau of Economic Research, Inc, <https://ideas.repec.org/p/nbr/nberwo/29561.html>.
- Fitzgerald, Doireann, Stefanie Haller, and Yaniv Yedid-Levi (2024) “How Exporters Grow,” *Review of Economic Studies*.
- Foster, Lucia, John Haltiwanger, and Chad Syverson (2016) “The Slow Growth of New Plants: Learning about Demand?” *Economica*, 83 (329), 91–129.
- Gourio, François and Leena Rudanko (2014) “Customer Capital,” *Review of Economic Studies*, 81 (3), 1102–1136.
- Hopenhayn, Hugo A. (1992) “Entry, Exit, and firm Dynamics in Long Run Equilibrium,” *Econometrica*, 60 (5), 1127–1150, <http://www.jstor.org/stable/2951541>.
- Hottman, Colin J., Stephen J. Redding, and David E. Weinstein (2016) “Quantifying the Sources of Firm Heterogeneity,” *The Quarterly Journal of Economics*, 131 (3), 1291–1364, <https://doi.org/10.1093/qje/qjw012>.
- Huneus, Federico (2018) “Production Network Dynamics and the Propagation of Shocks,” Working paper.
- Lim, Kevin (2018) “Endogenous Production Networks and the Business Cycle,” Working paper.

Mayer, Thierry, Marc J. Melitz, and Gianmarco I. P. Ottaviano (2014) “Market Size, Competition, and the Product Mix of Exporters,” *American Economic Review*, 104 (2), 495–536.

Miyauchi, Yuhei (2024) “Matching and Agglomeration: Theory and Evidence from Japanese Firm-to-Firm Trade,” *Econometrica*, 92 (6), 1869–1905.